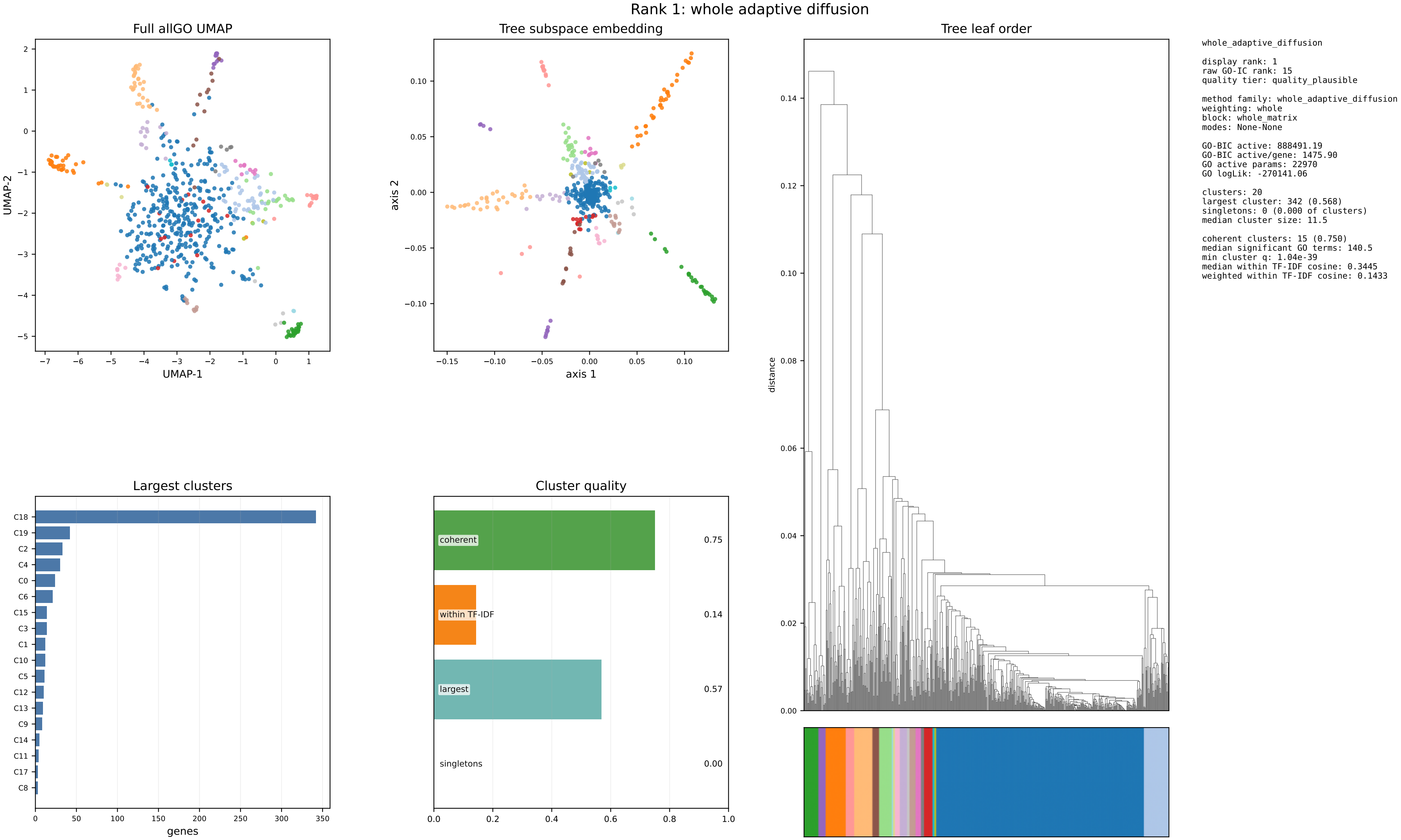
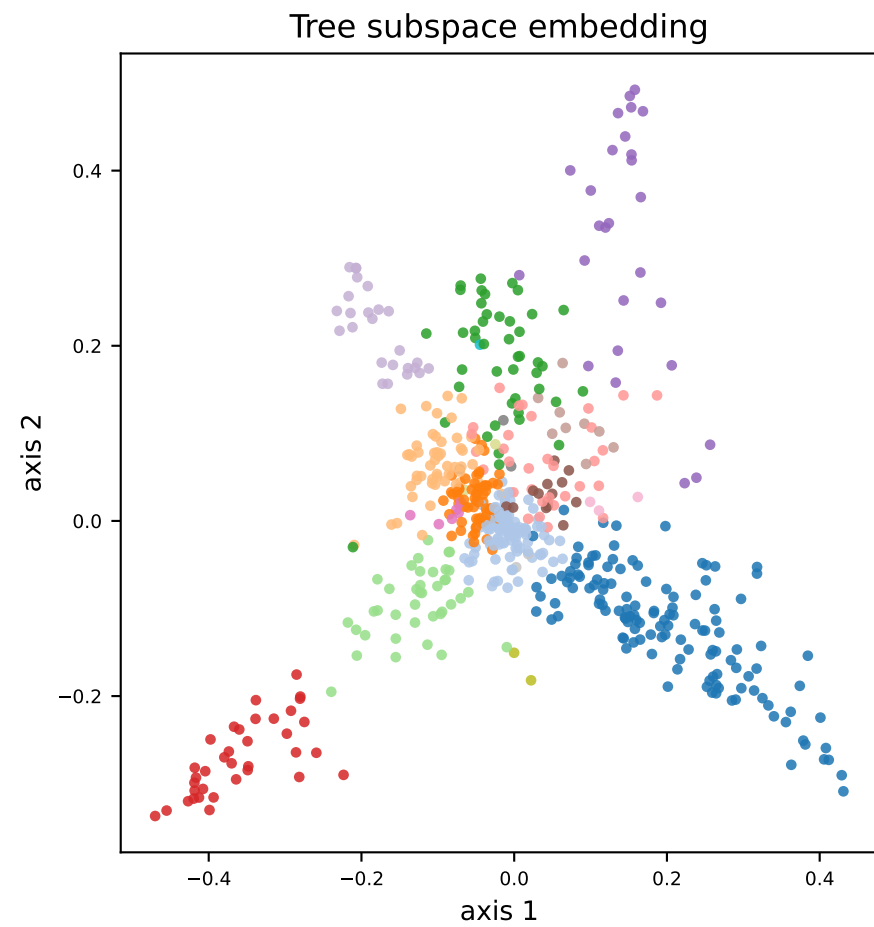
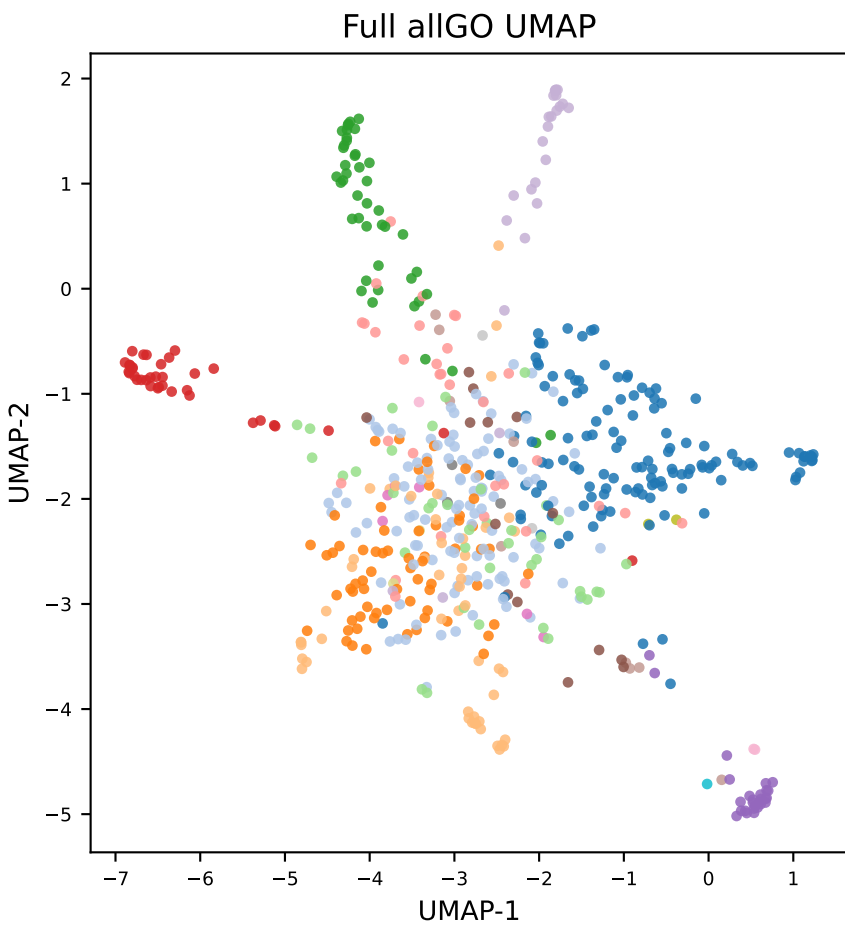




Rank 2: legacy c2ef, TF-IDF modes 02-05



legacy_c2ef_tfidf_components_02_05

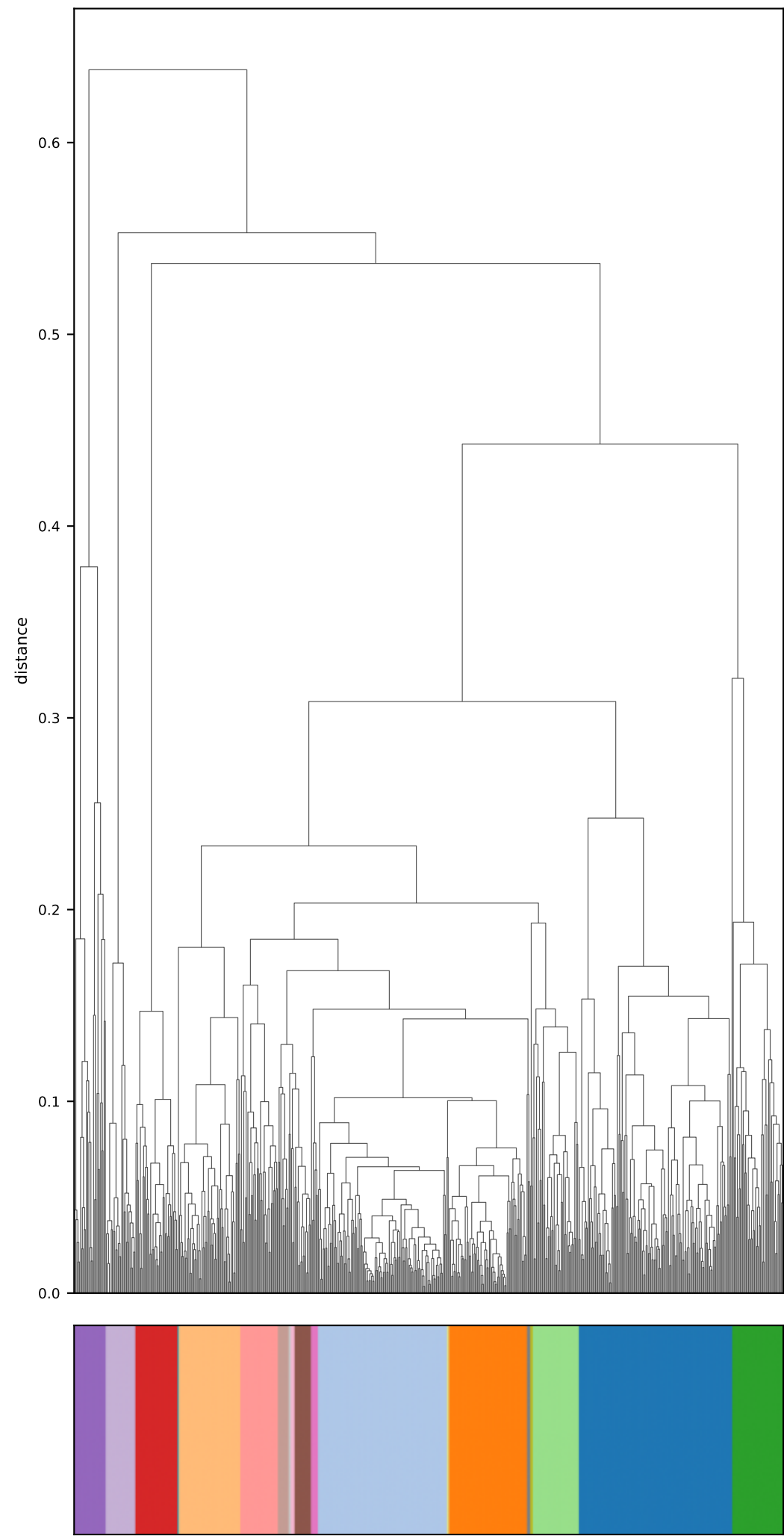
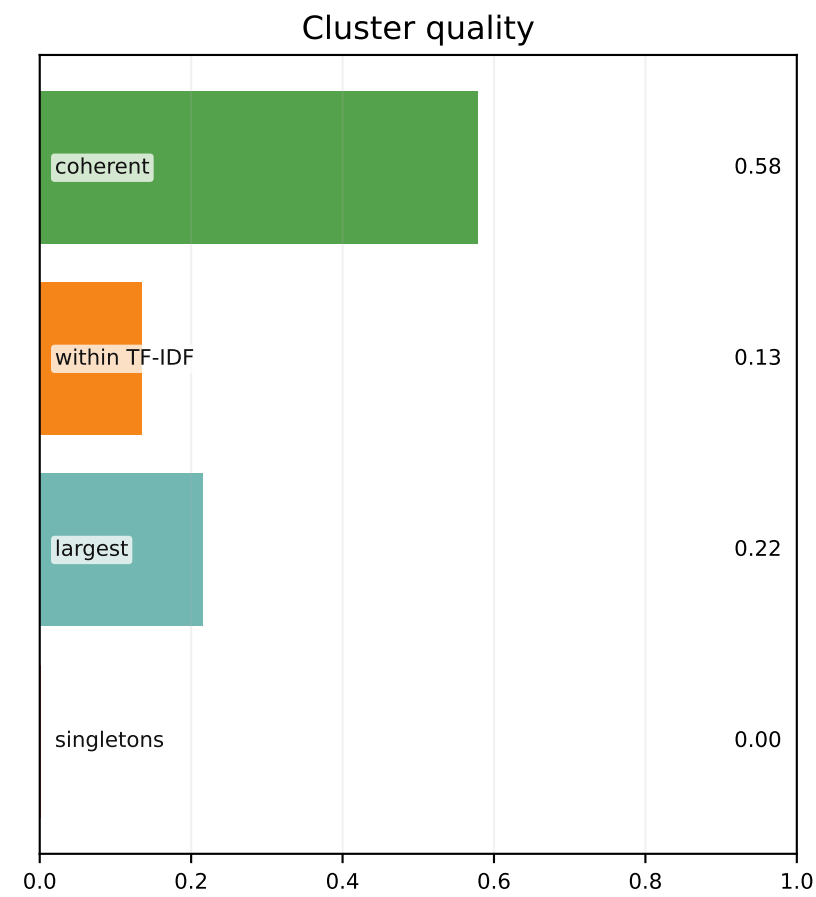
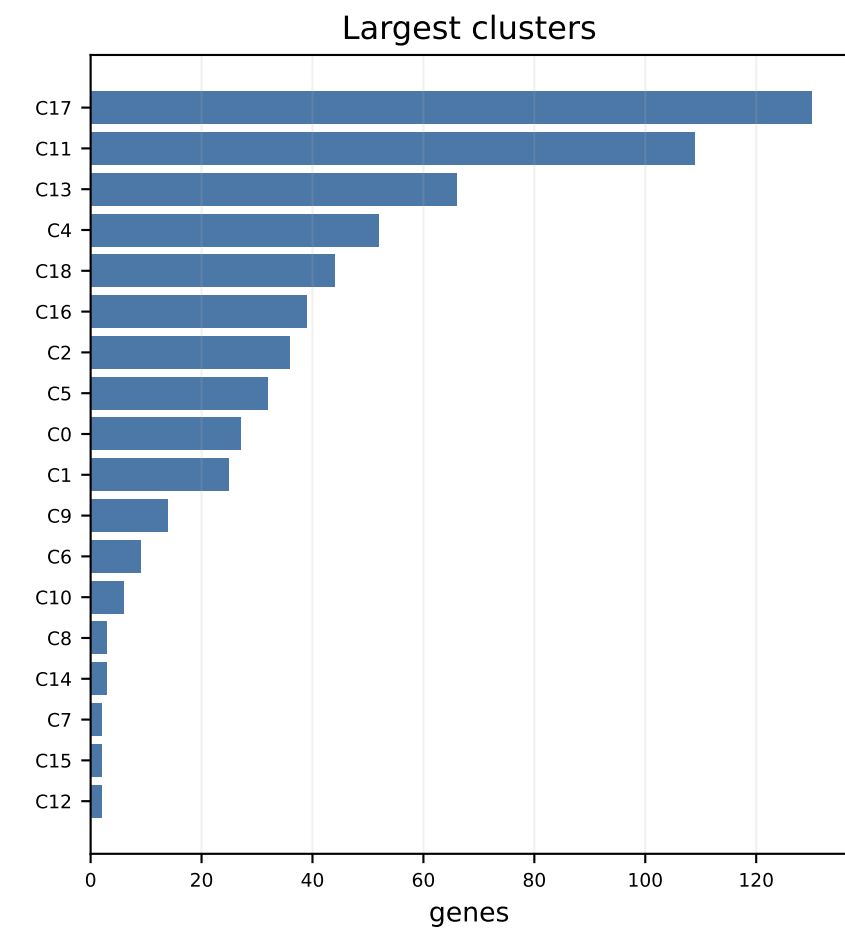
display rank: 2
raw G0-IC rank: 17
quality tier: quality_plausible

method family: legacy_c2ef
weighting: tfidf
block: components_02_05
modes: 2-5

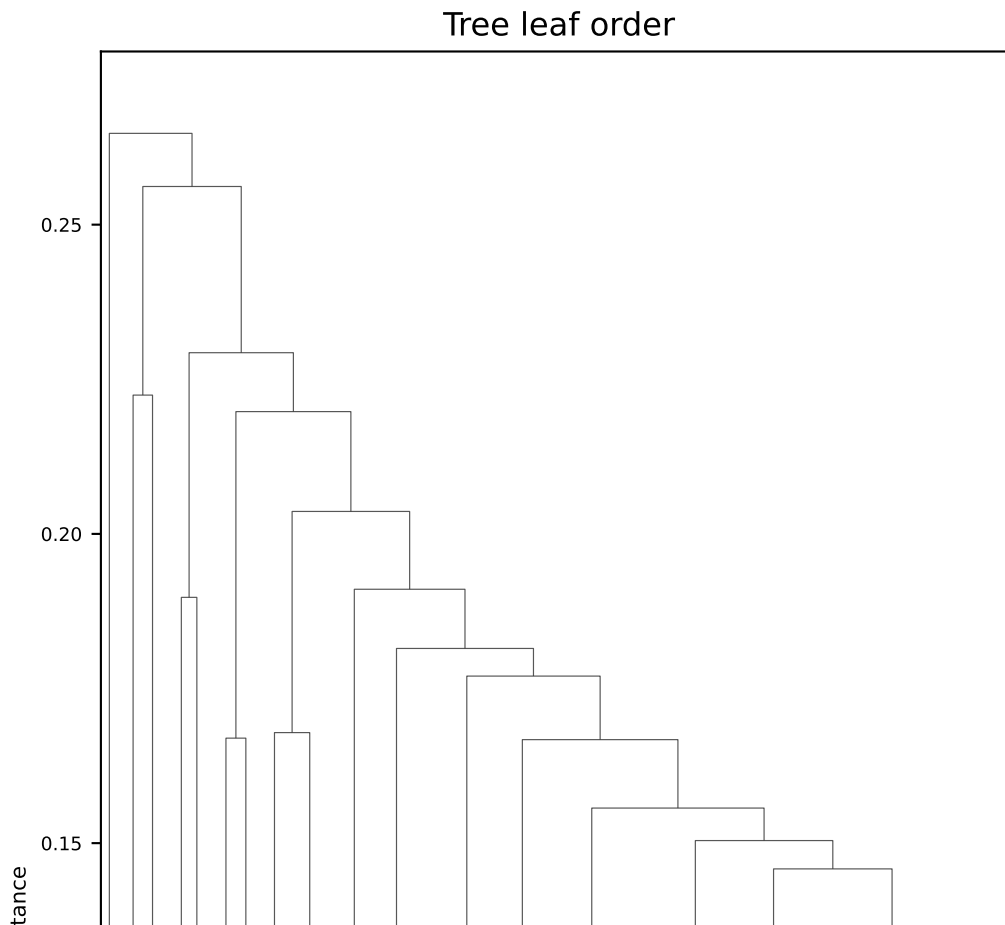
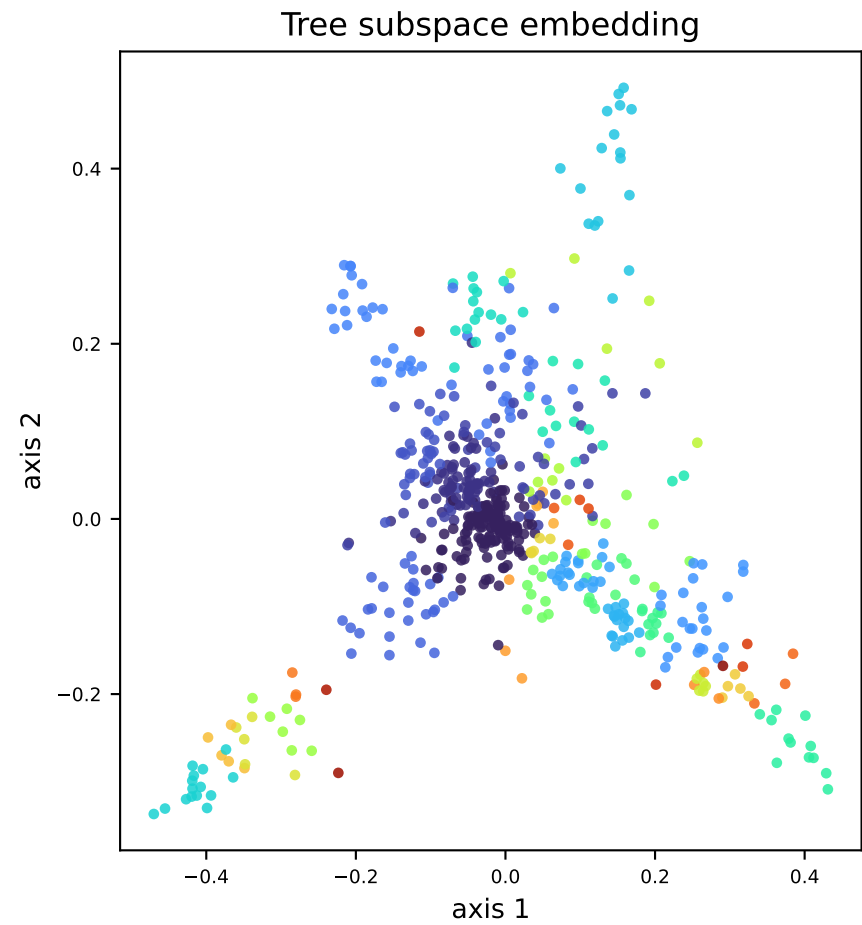
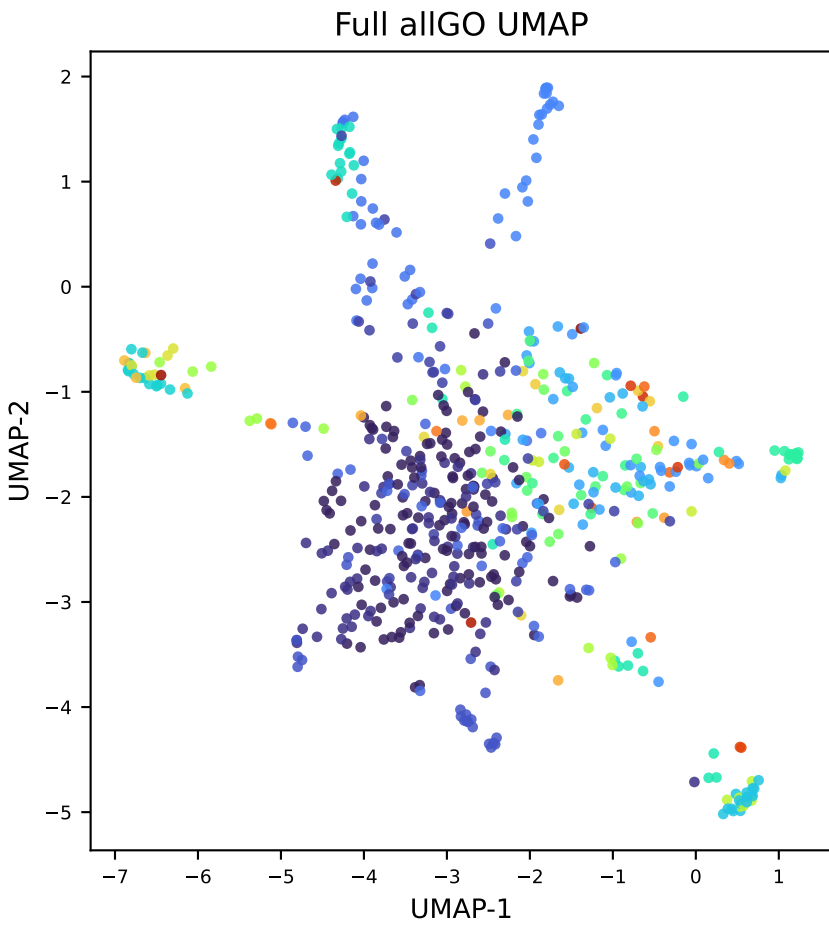
G0-BIC active: 905930.46
G0-BIC active/gene: 1504.87
G0 active params: 25770
G0 logLik: -257637.68

clusters: 19
largest cluster: 130 (0.216)
singletons: 1 (0.053 of clusters)
median cluster size: 25.0

coherent clusters: 11 (0.579)
median significant G0 terms: 49.0
min cluster q: 6.71e-54
median within TF-IDF cosine: 0.1192
weighted within TF-IDF cosine: 0.1342



Rank 3: adaptive diffusion cosine subspace, tfidf, modes 02 05



adaptive_diffusion_cosine_subspace__tfidf__adaptive_modes_02_05

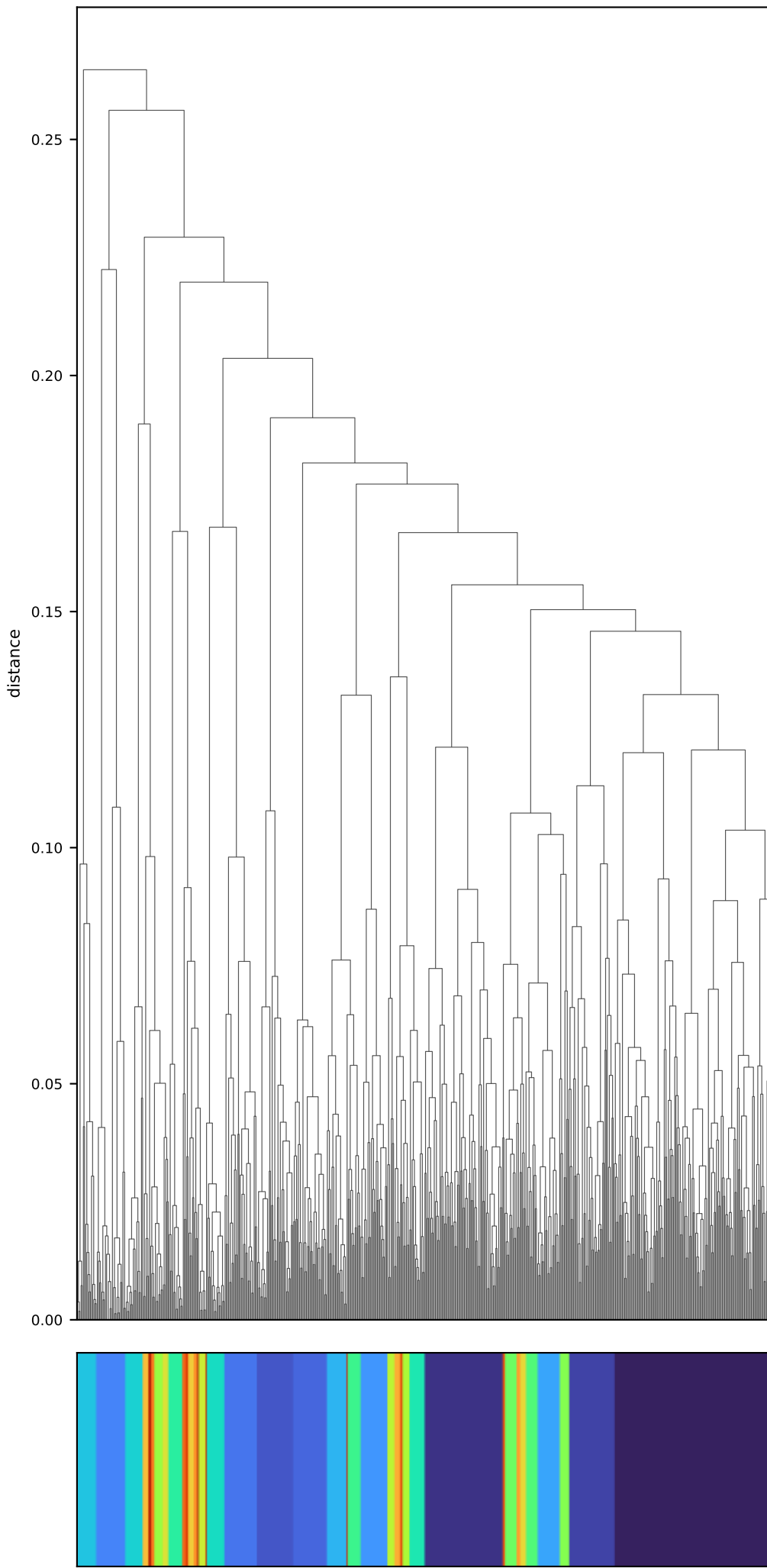
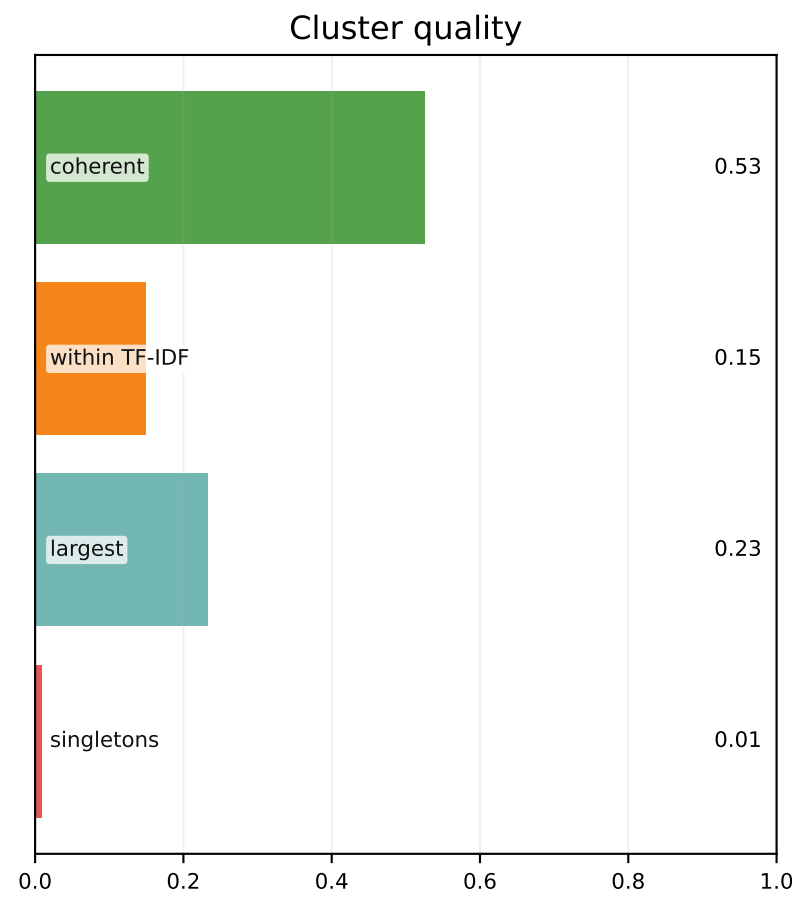
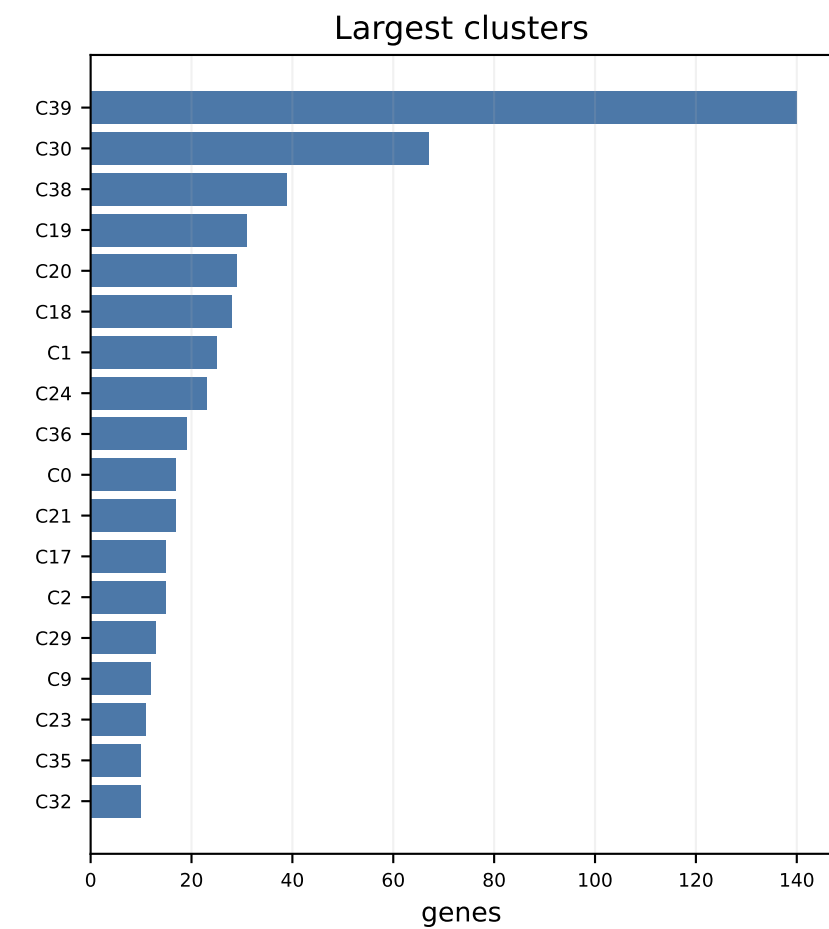
display rank: 3
raw G0-IC rank: 18
quality tier: quality_plausible

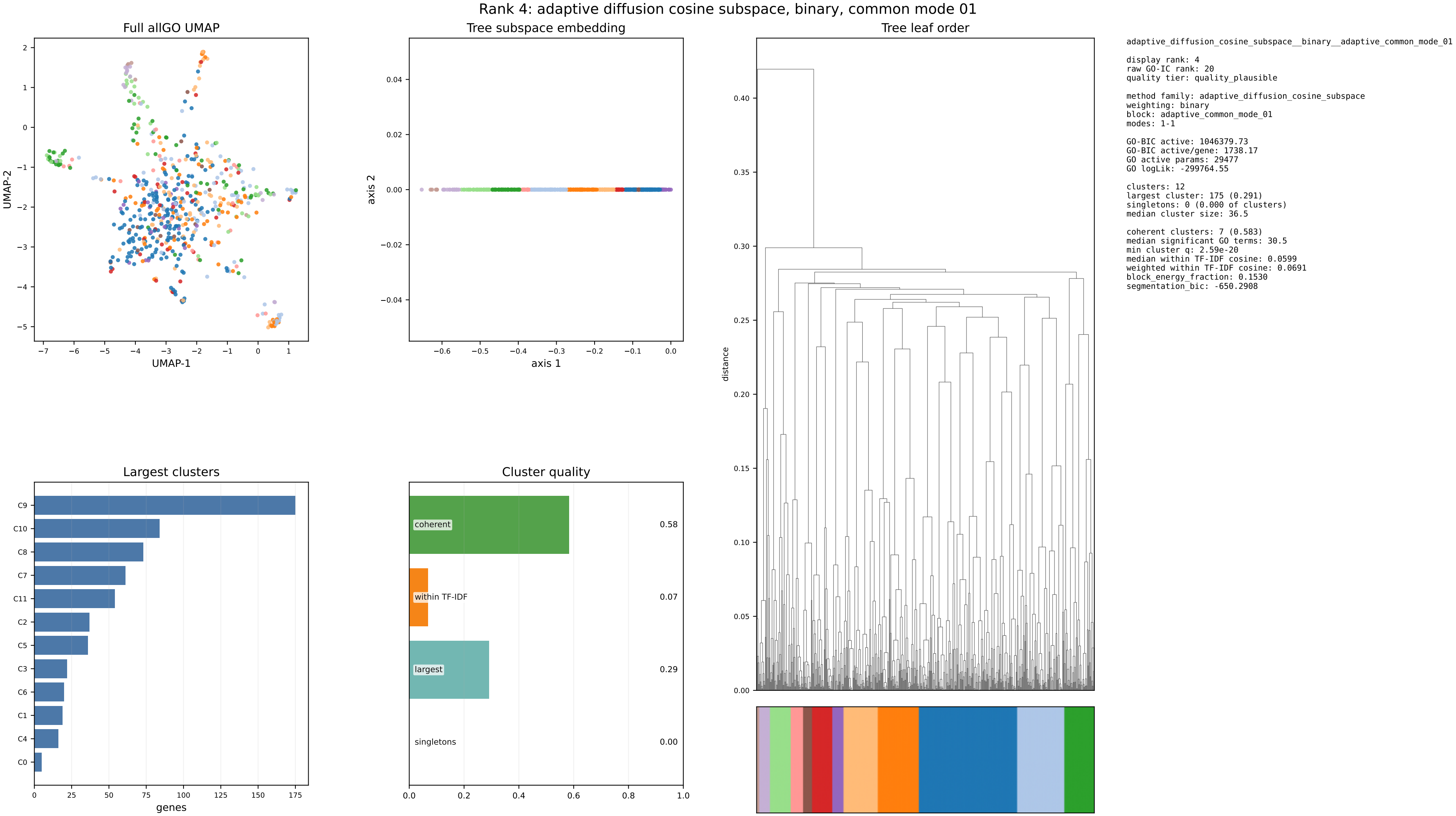
method family: adaptive_diffusion_cosine_subspace
weighting: tfidf
block: adaptive_modes_02_05
modes: 2-5

G0-BIC active: 1016156.04
G0-BIC active/gene: 1687.97
G0 active params: 37065
G0 logLik: -227138.33

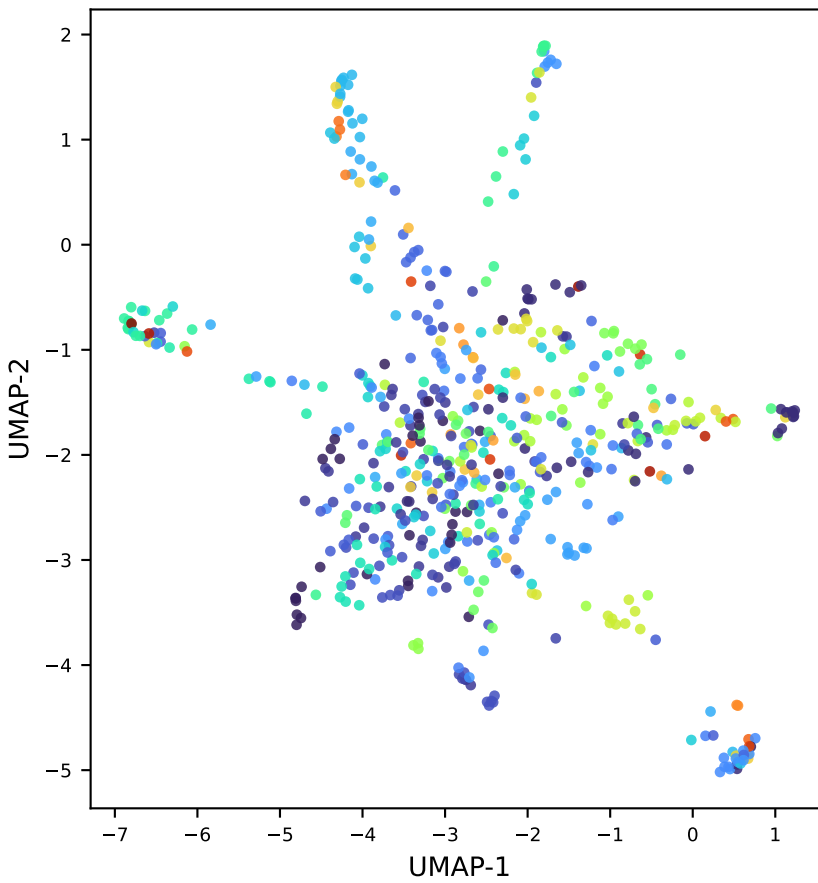
clusters: 40
largest cluster: 140 (0.233)
singletons: 5 (0.125 of clusters)
median cluster size: 6.5

coherent clusters: 21 (0.525)
median significant G0 terms: 21.5
min cluster q: 4.91e-32
median within TF-IDF cosine: 0.1873
weighted within TF-IDF cosine: 0.1497
block_energy_fraction: 0.1819
segmentation_bic: -657.8000

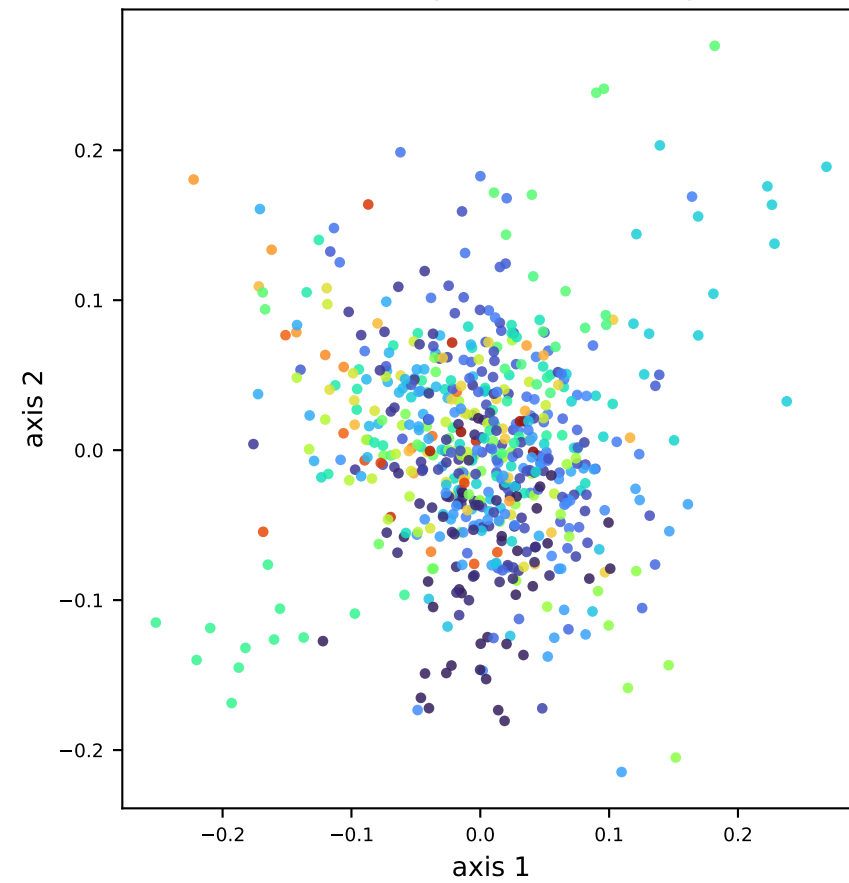




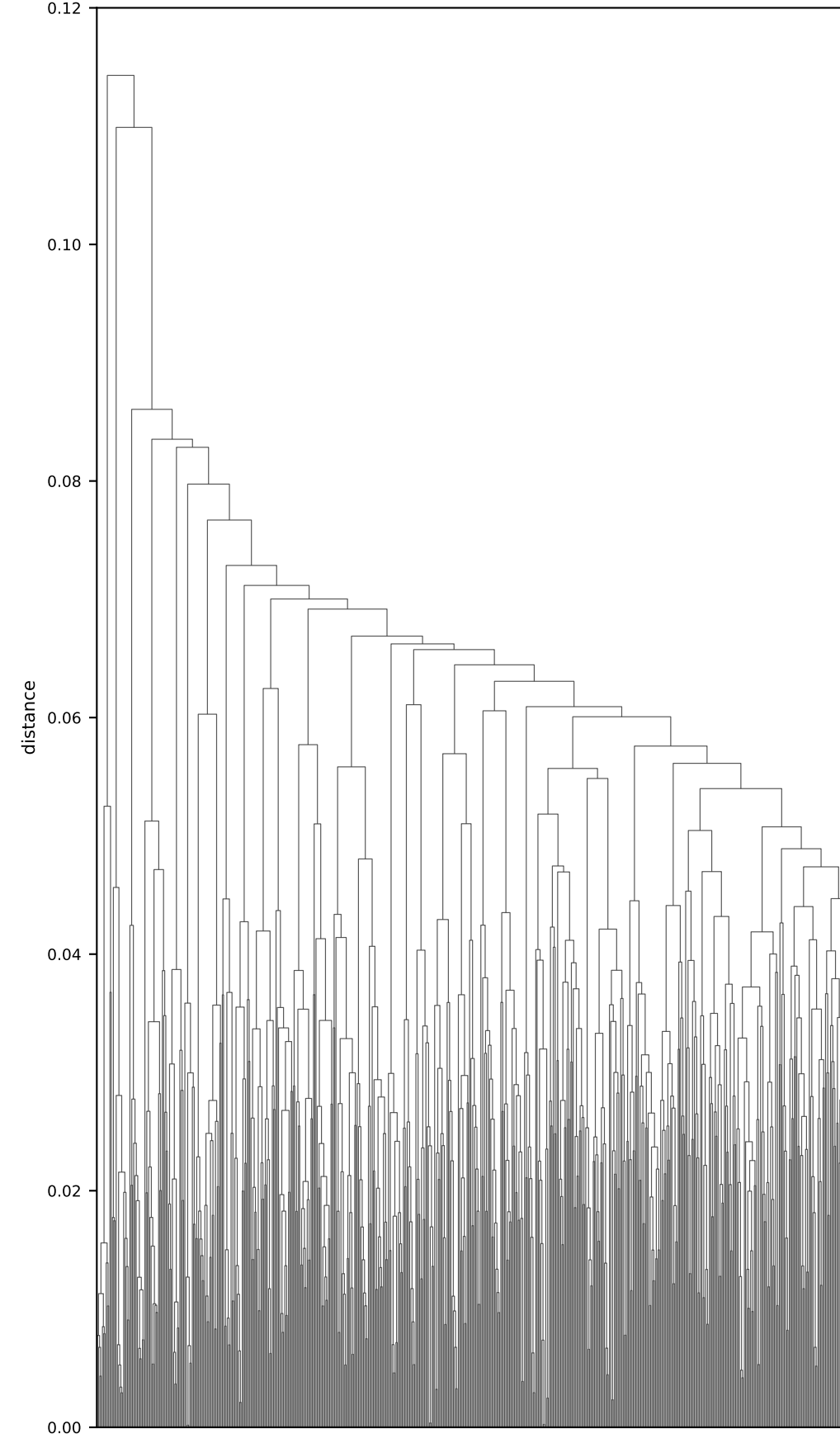
Full allGO UMAP



Tree subspace embedding



Tree leaf order



adaptive_diffusion_cosine_subspace__binary__adaptive_modes_22_40

```
display rank: 5
raw G0-IC rank: 23
quality tier: quality_plausible
```

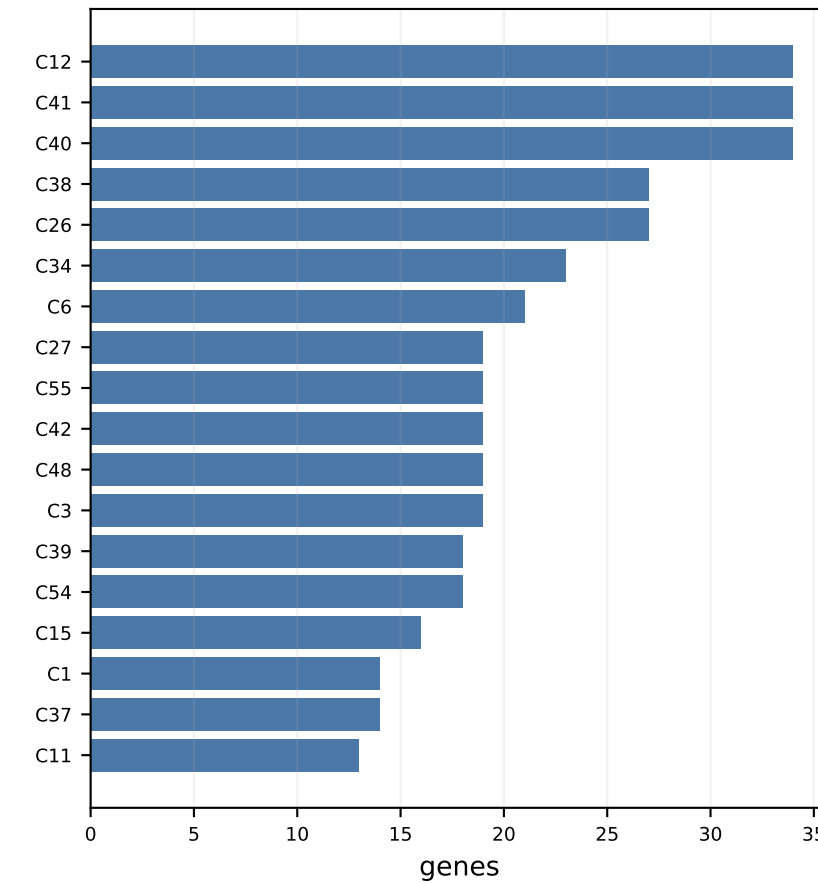
```
method family: adaptive_diffusion_cosine_subspace
weighting: binary
block: adaptive_modes_22_40
modes: 22-40
```

G0-BIC active: 1086541.58
G0-BIC active/gene: 1804.89
G0 active params: 42940
G0 logLik: -217800.66

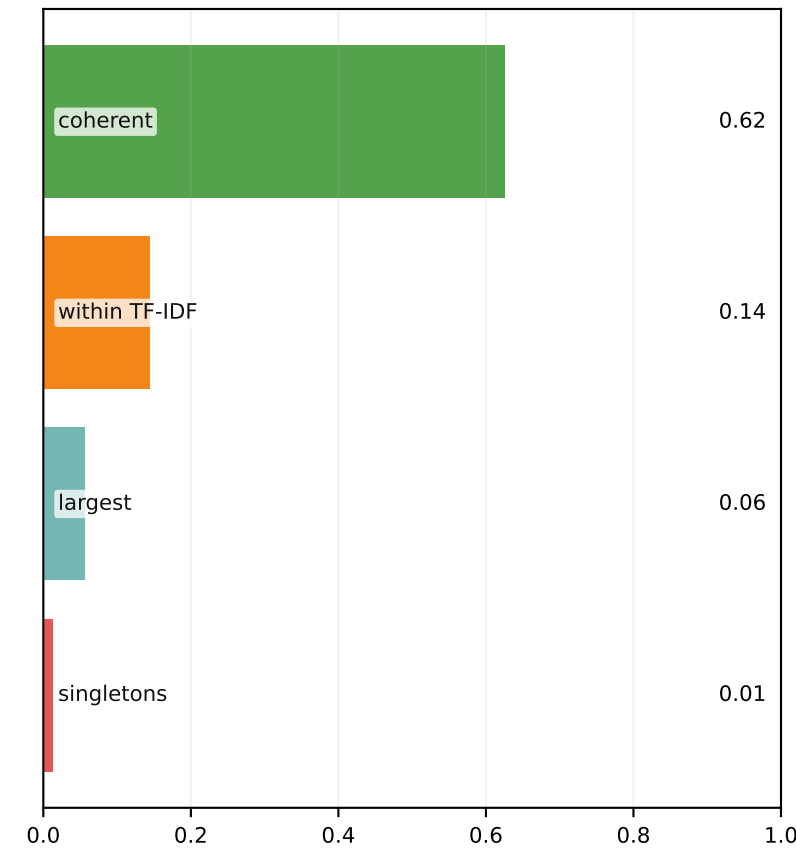
```
clusters: 56
largest cluster: 34 (0.056)
singletons: 8 (0.143 of clusters)
median cluster size: 9.0
```

```
coherent clusters: 35 (0.625)
median significant G0 terms: 21.5
min cluster q: 1.72e-16
median within TF-IDF cosine: 0.1496
weighted within TF-IDF cosine: 0.1449
block_energy_fraction: 0.1591
segmentation_bic: -650.2908
```

Largest clusters



Cluster quality

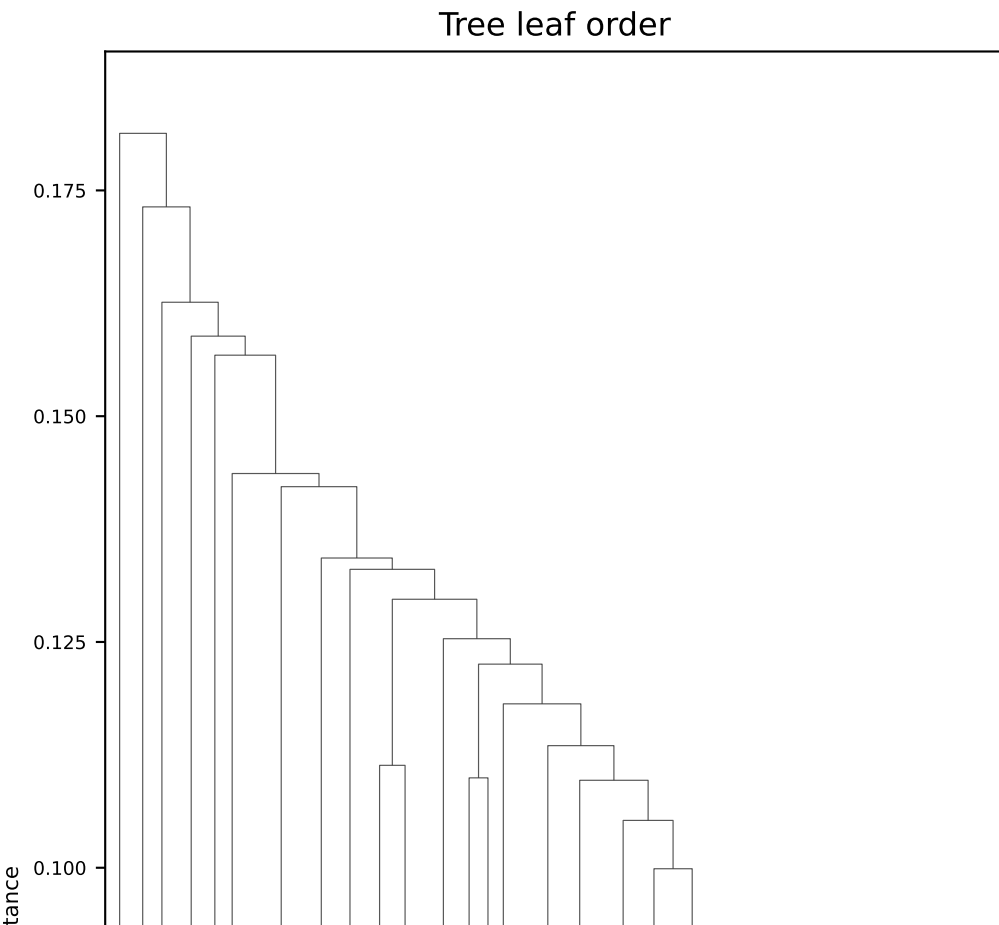
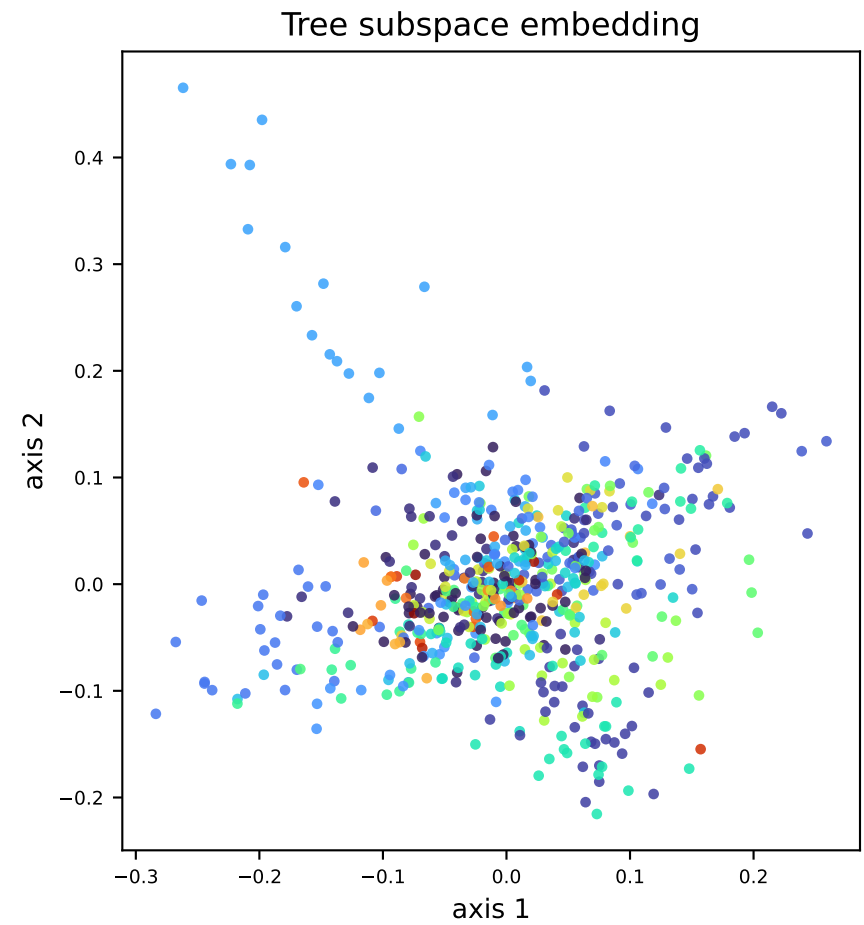
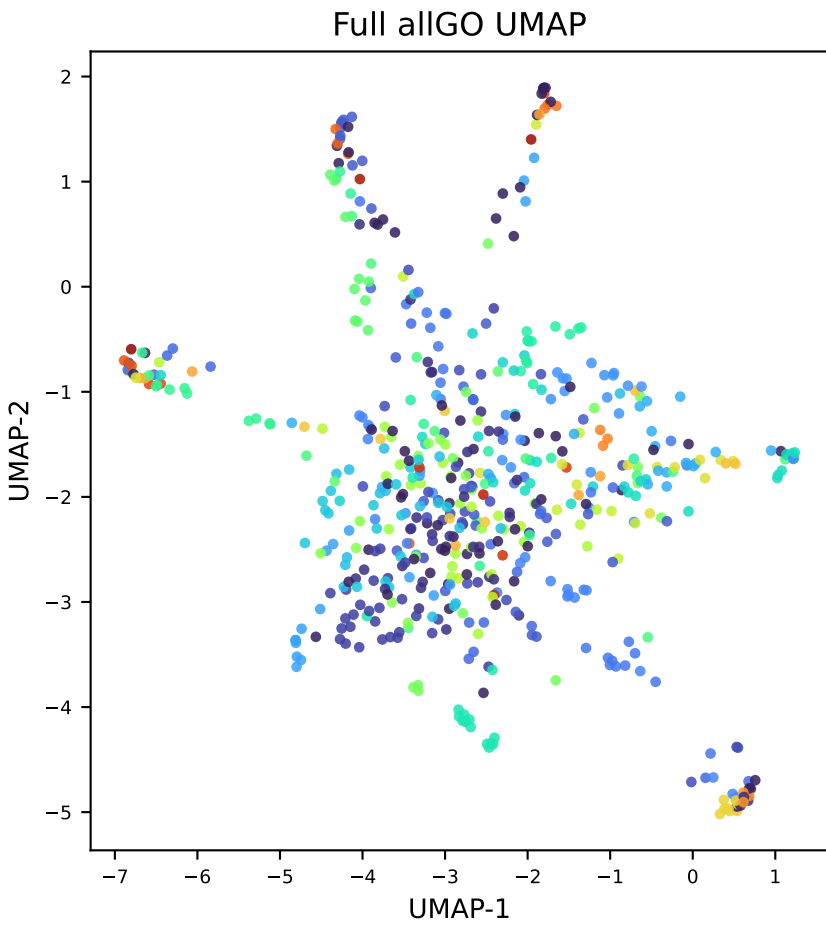


within TF-IDF

largest

singletons

Rank 6: adaptive diffusion cosine subspace, binary, modes 12 21



adaptive_diffusion_cosine_subspace__binary__adaptive_modes_12_21

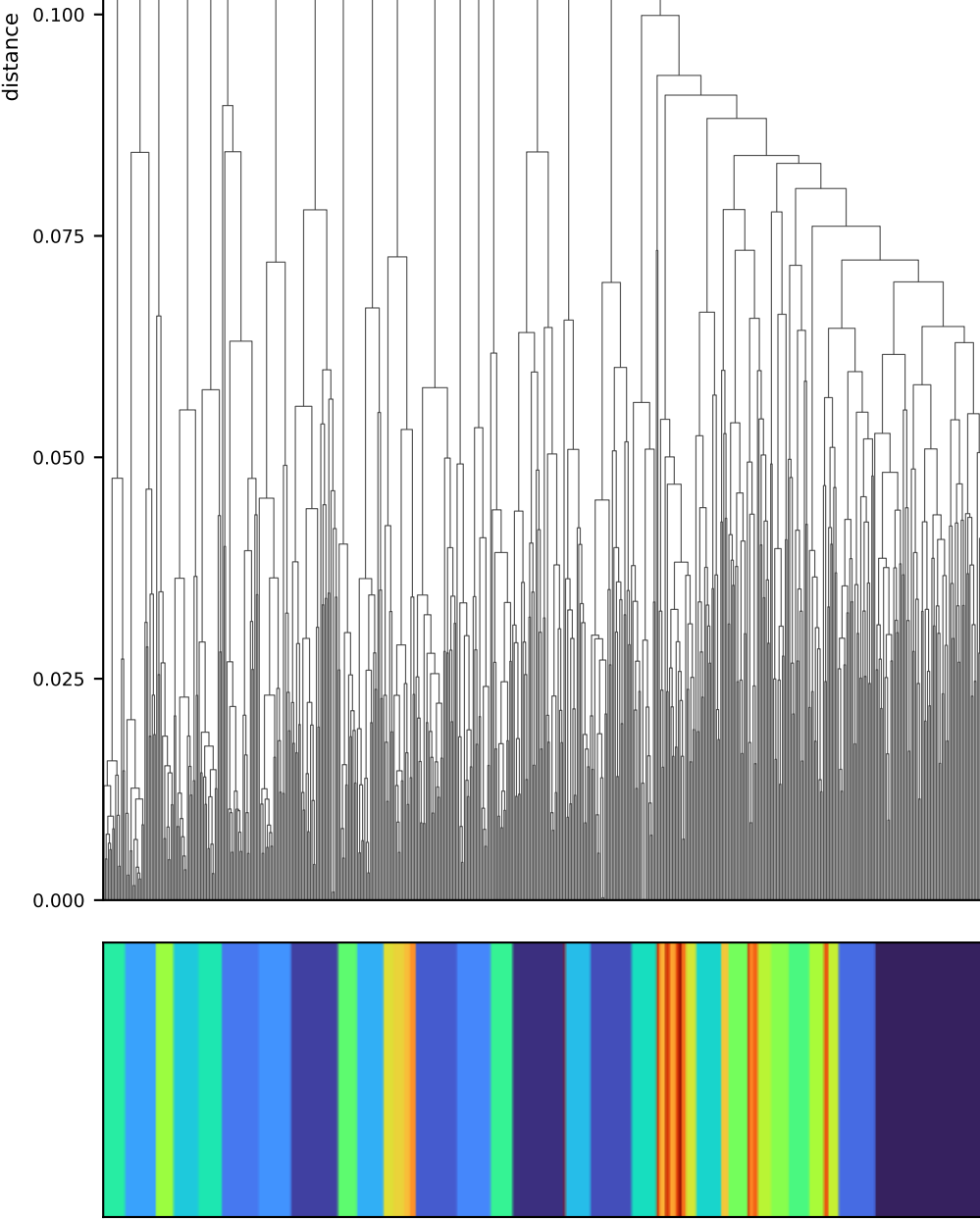
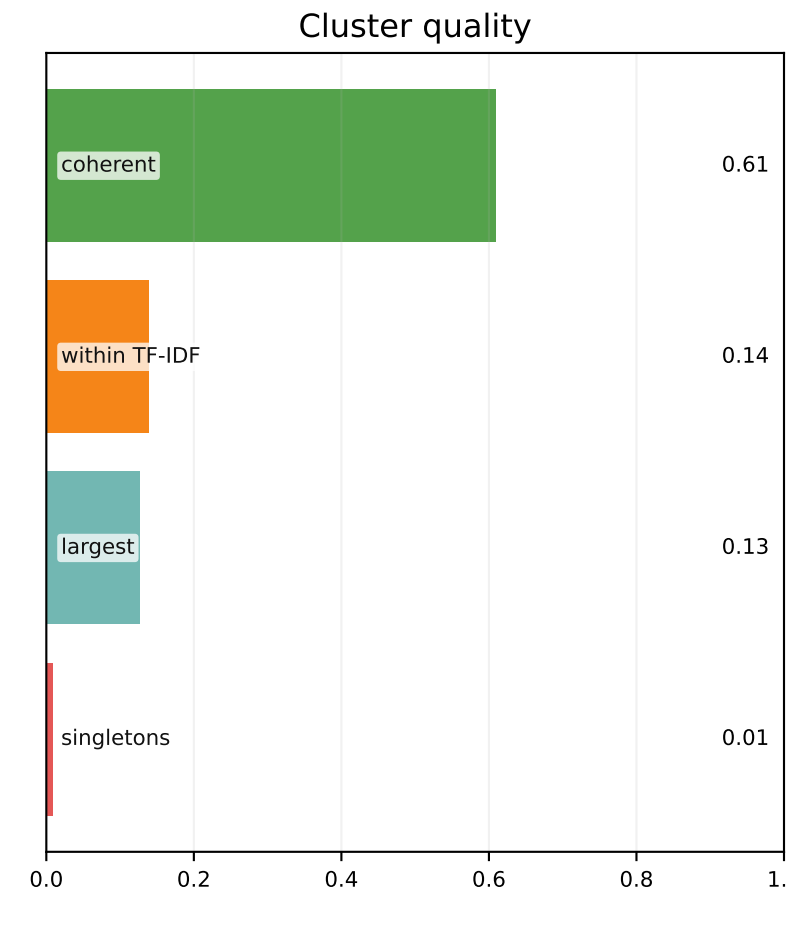
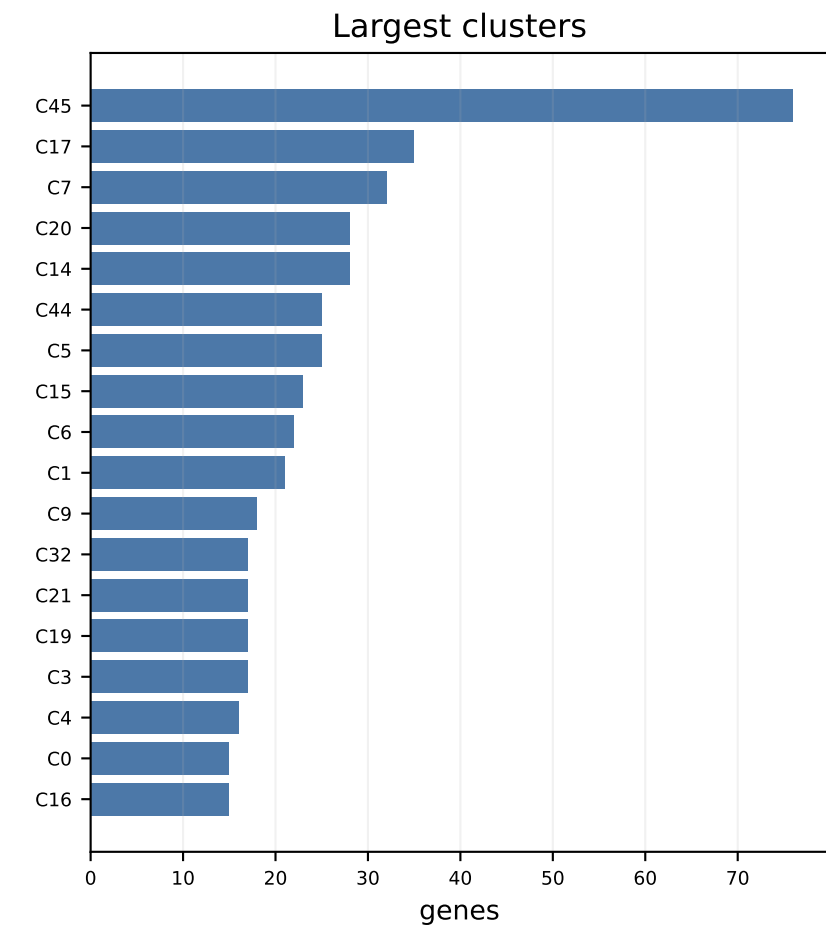
display rank: 6
raw GO-IC rank: 24
quality tier: quality_plausible

method family: adaptive_diffusion_cosine_subspace
weighting: binary
block: adaptive_modes_12_21
modes: 12-21

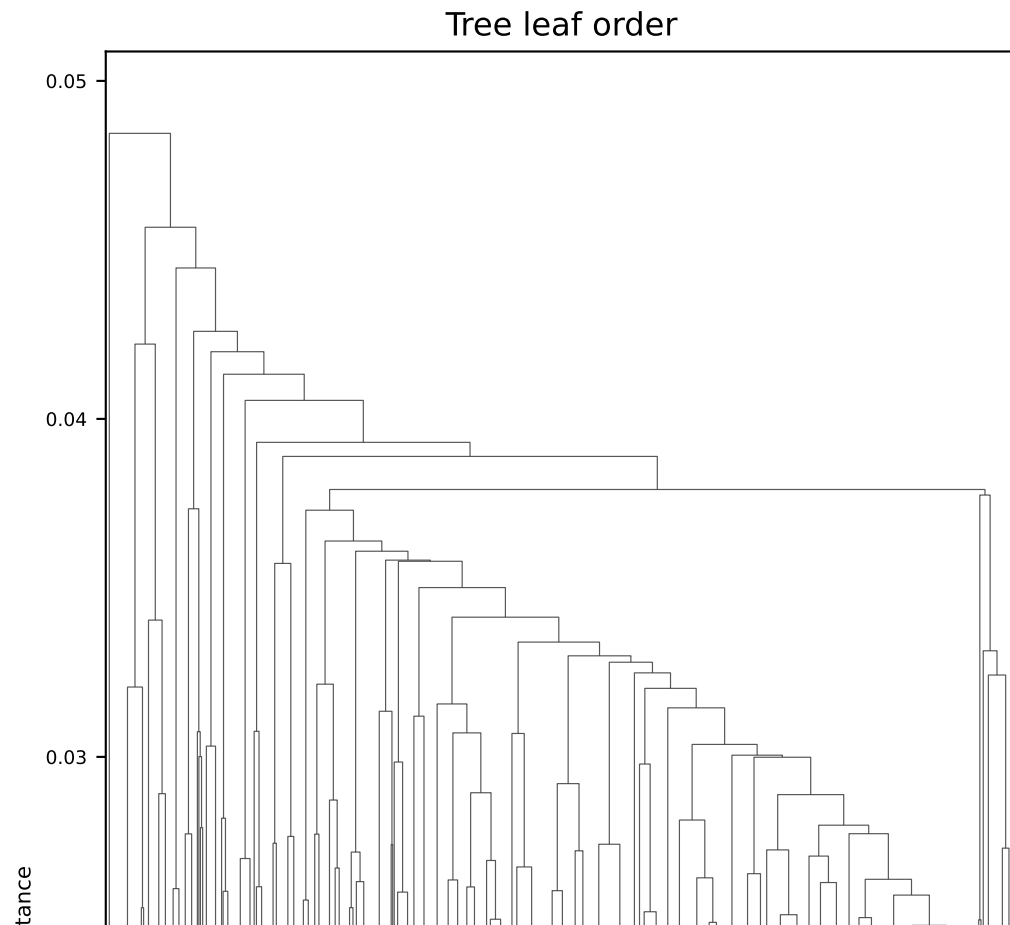
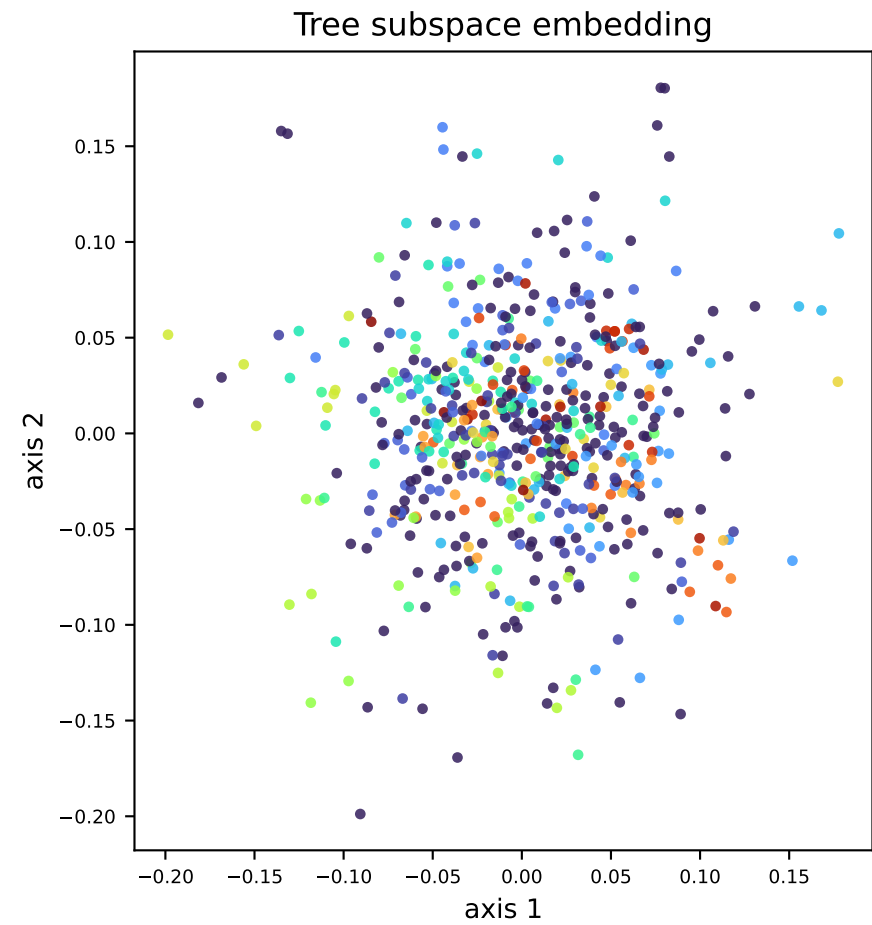
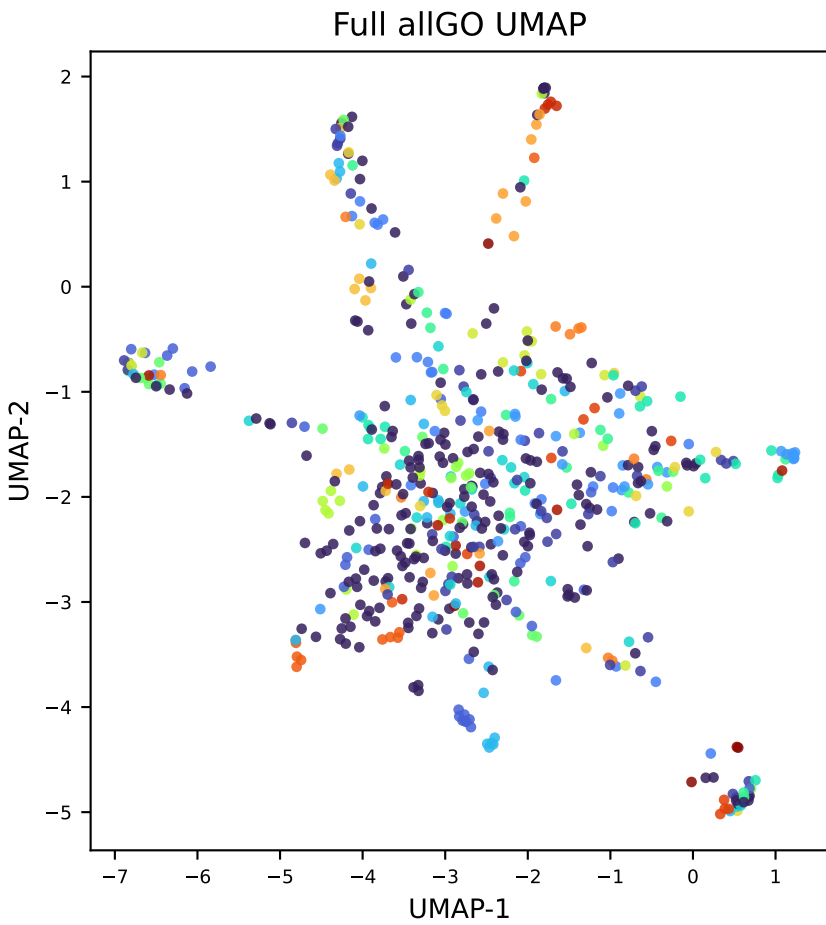
GO-BIC active: 1127563.65
GO-BIC active/gene: 1873.03
GO active params: 41867
GO logLik: -246444.66

clusters: 46
largest cluster: 76 (0.126)
singletons: 5 (0.109 of clusters)
median cluster size: 11.0

coherent clusters: 28 (0.609)
median significant GO terms: 33.5
min cluster q: 1.68e-21
median within TF-IDF cosine: 0.1547
weighted within TF-IDF cosine: 0.1388
block_energy_fraction: 0.1234
segmentation_bic: -650.2908



Rank 7: adaptive diffusion cosine subspace, tfidf, modes 53 80



adaptive_diffusion_cosine_subspace__tfidf__adaptive_modes_53_80

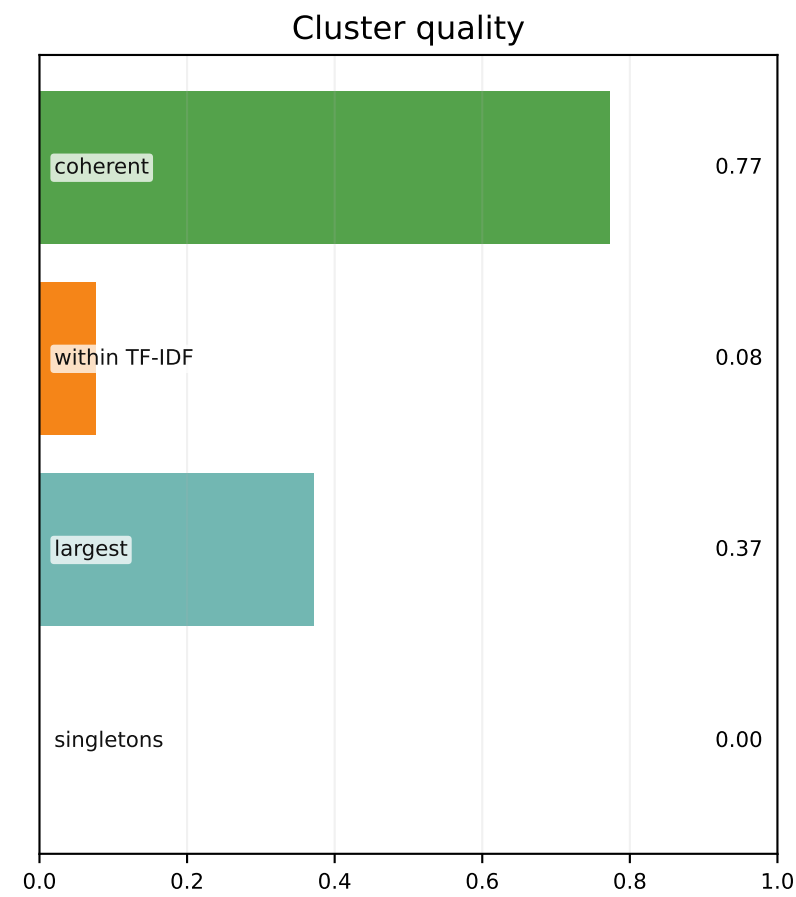
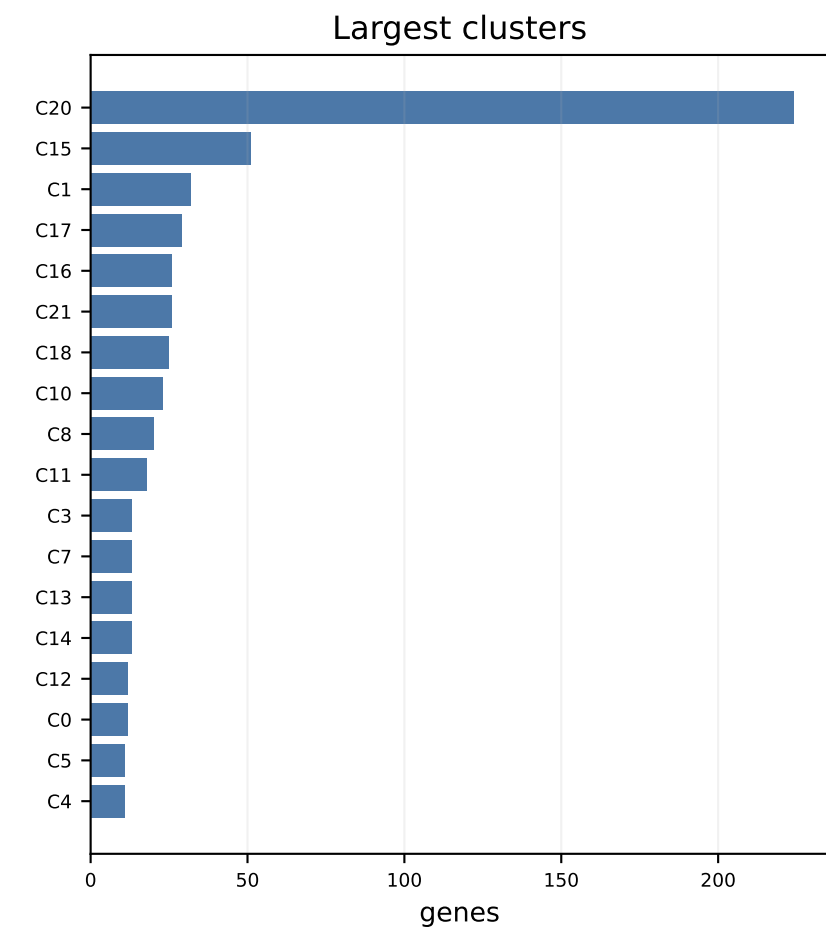
display rank: 7
raw G0-IC rank: 25
quality tier: quality_plausible

method family: adaptive_diffusion_cosine_subspace
weighting: tfidf
block: adaptive_modes_53_80
modes: 53-80

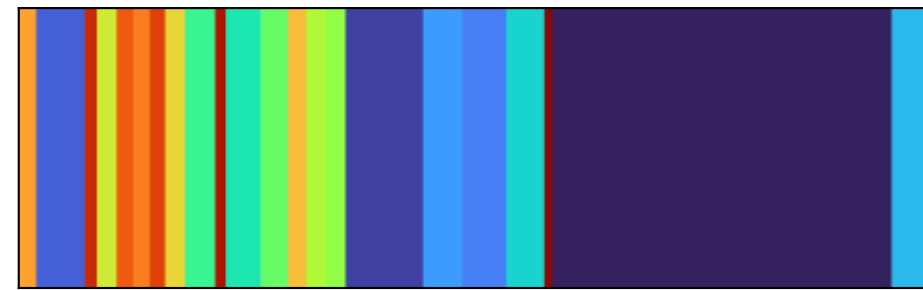
G0-BIC active: 1149957.71
G0-BIC active/gene: 1910.23
G0 active params: 35191
G0 logLik: -308243.42

clusters: 22
largest cluster: 224 (0.372)
singletons: 0 (0.000 of clusters)
median cluster size: 13.0

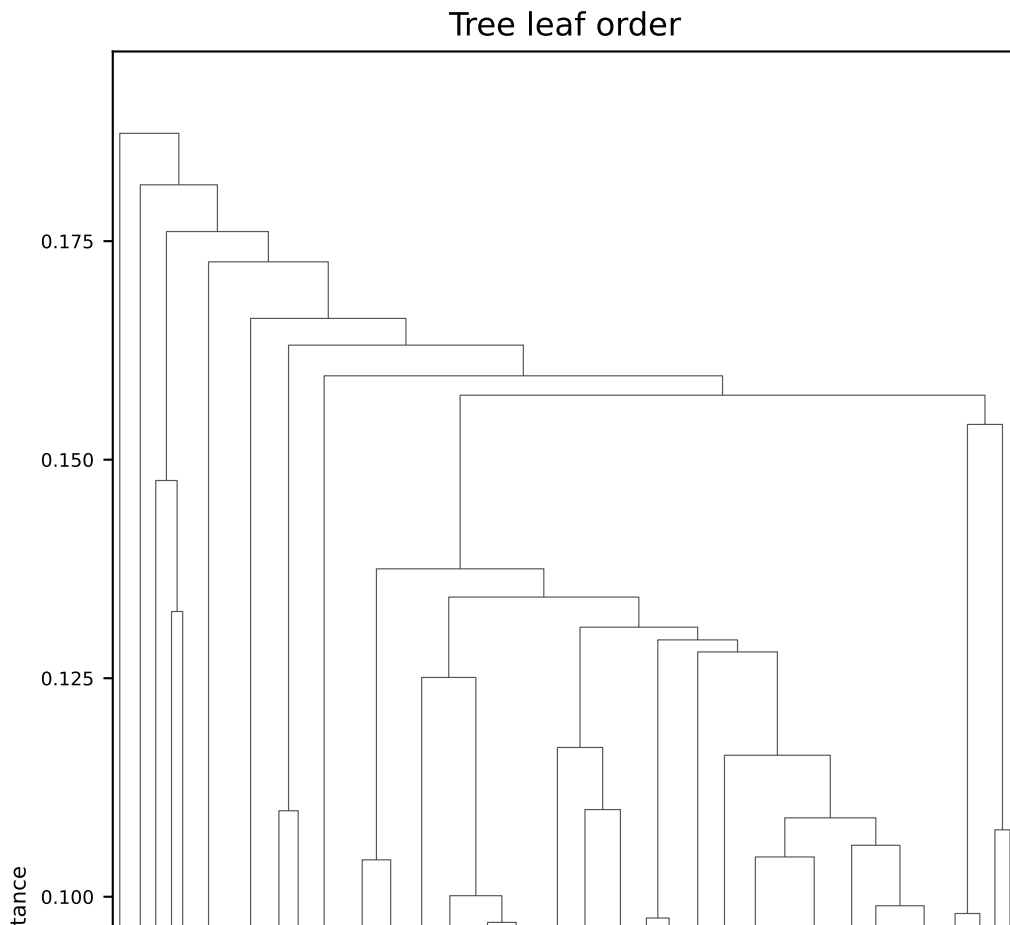
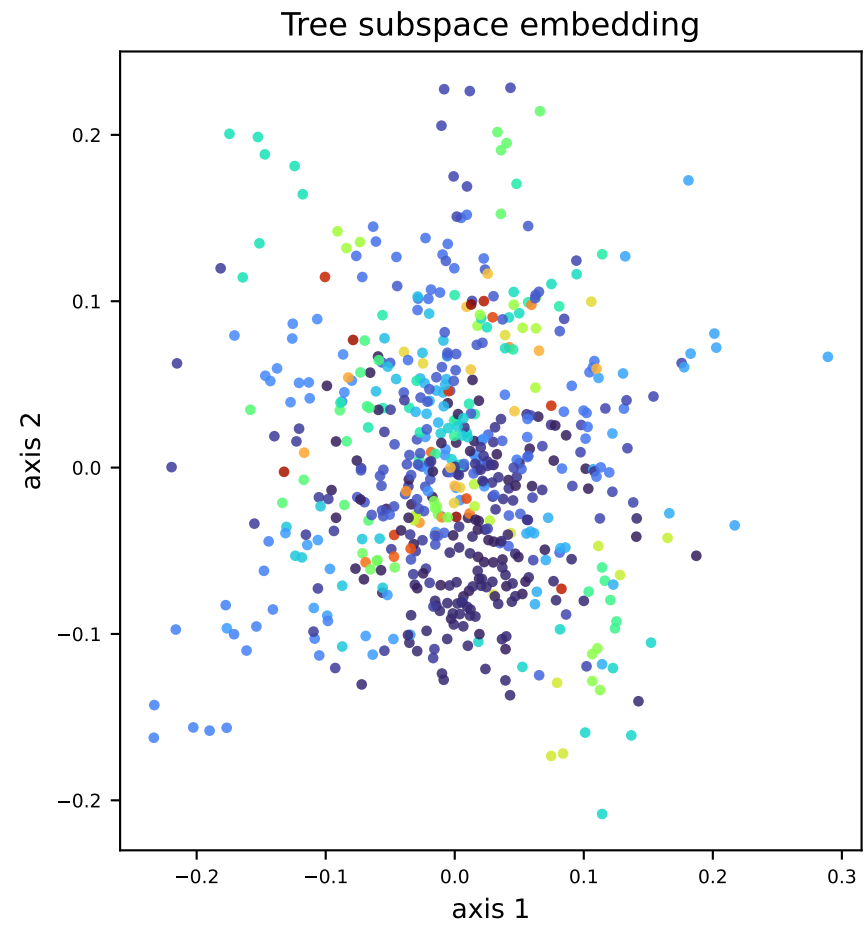
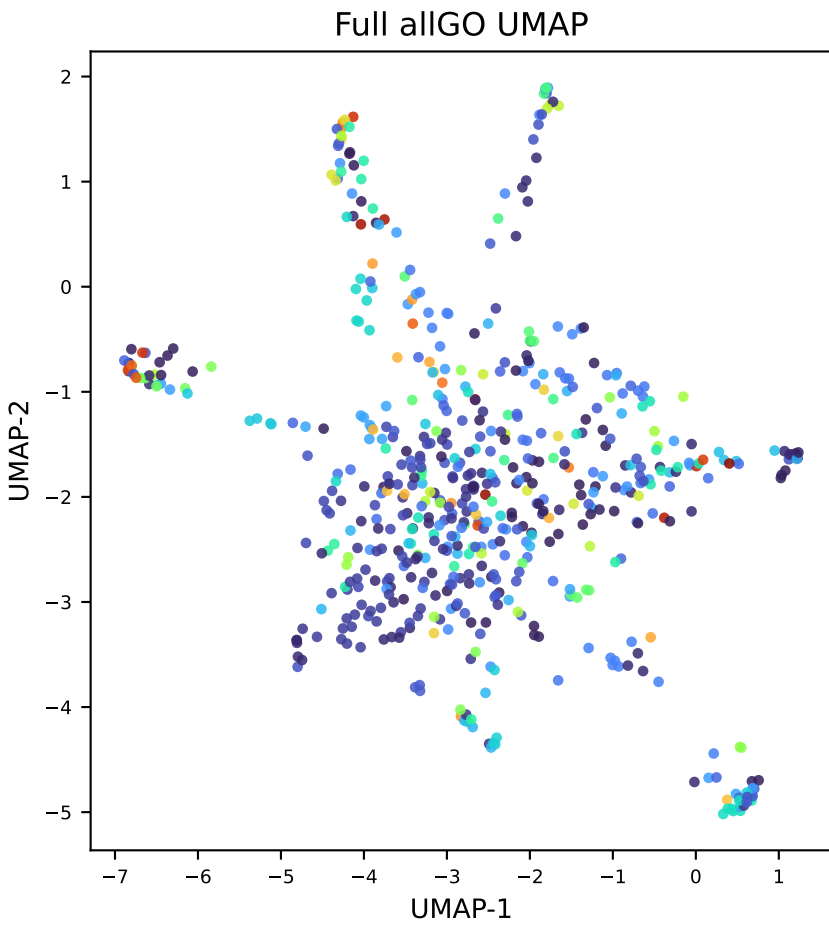
coherent clusters: 17 (0.773)
median significant G0 terms: 64.0
min cluster q: 1.18e-10
median within TF-IDF cosine: 0.1010
weighted within TF-IDF cosine: 0.0772
block_energy_fraction: 0.1753
segmentation_bic: -657.8000



distance



Rank 8: adaptive diffusion cosine subspace, tfidf, modes 16 19



adaptive_diffusion_cosine_subspace__tfidf__adaptive_modes_16_19

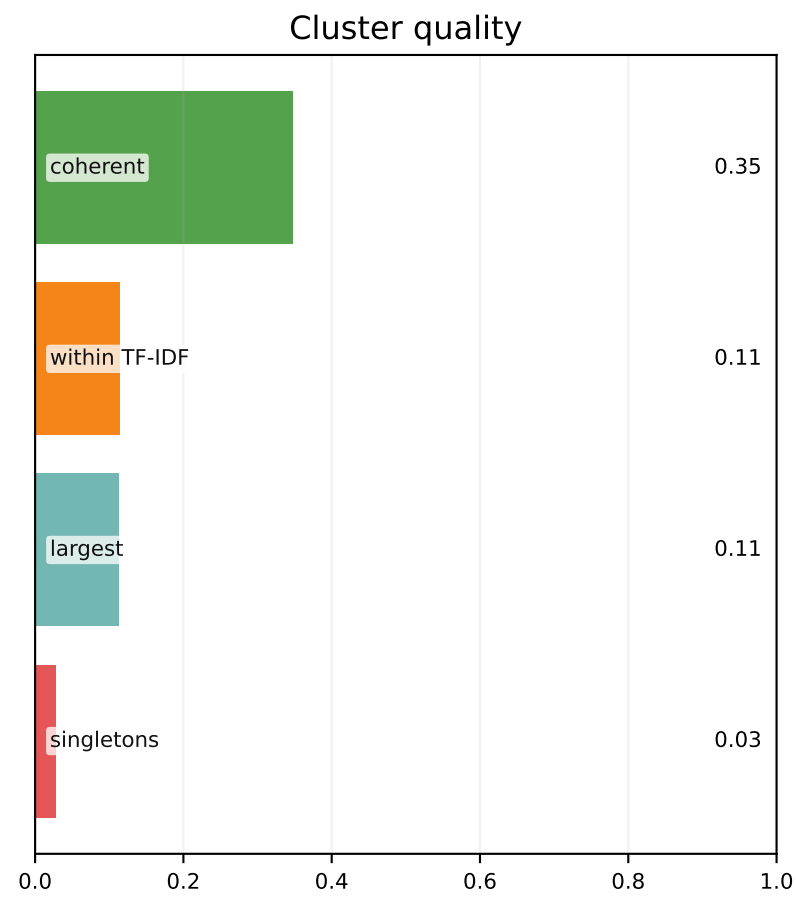
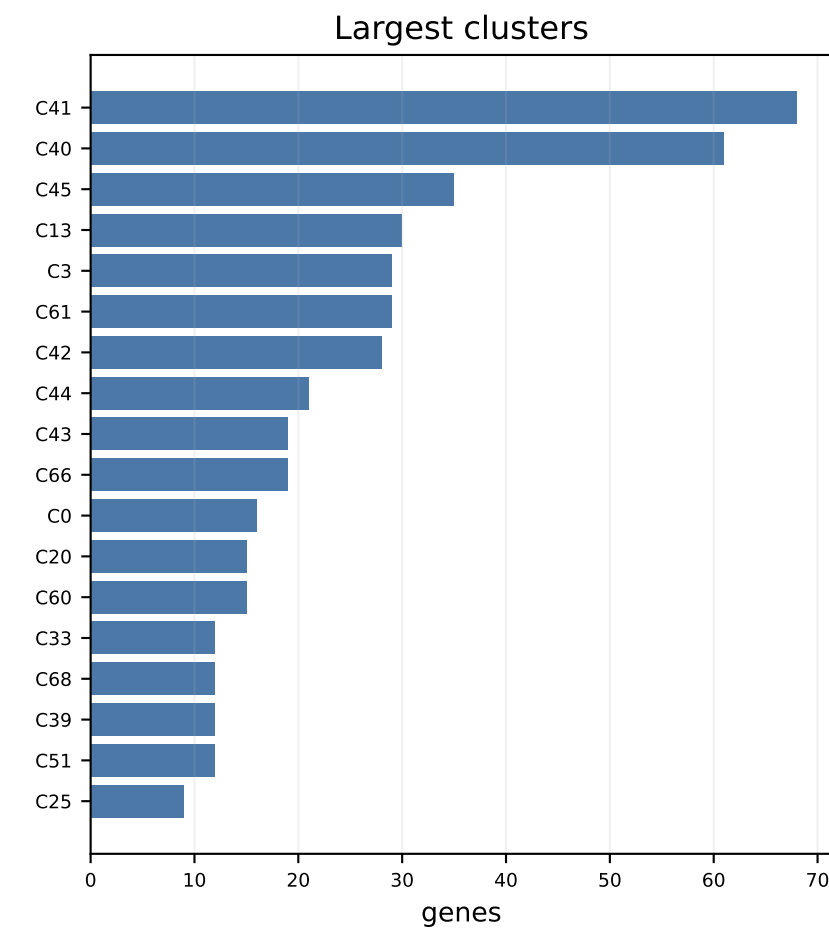
display rank: 8
raw G0-IC rank: 26
quality tier: quality_plausible

method family: adaptive_diffusion_cosine_subspace
weighting: tfidf
block: adaptive_modes_16_19
modes: 16-19

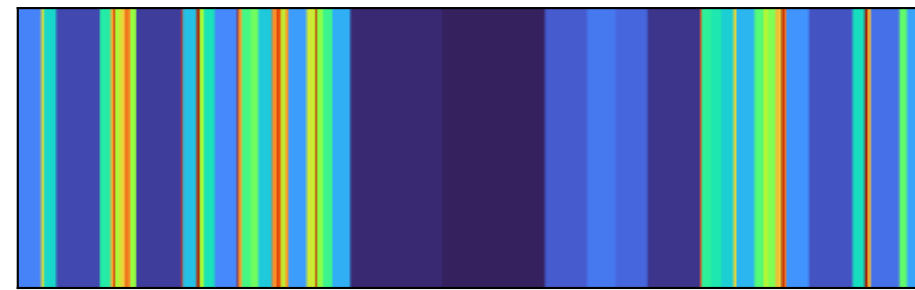
G0-BIC active: 1150604.11
G0-BIC active/gene: 1911.30
G0 active params: 44408
G0 logLik: -238705.00

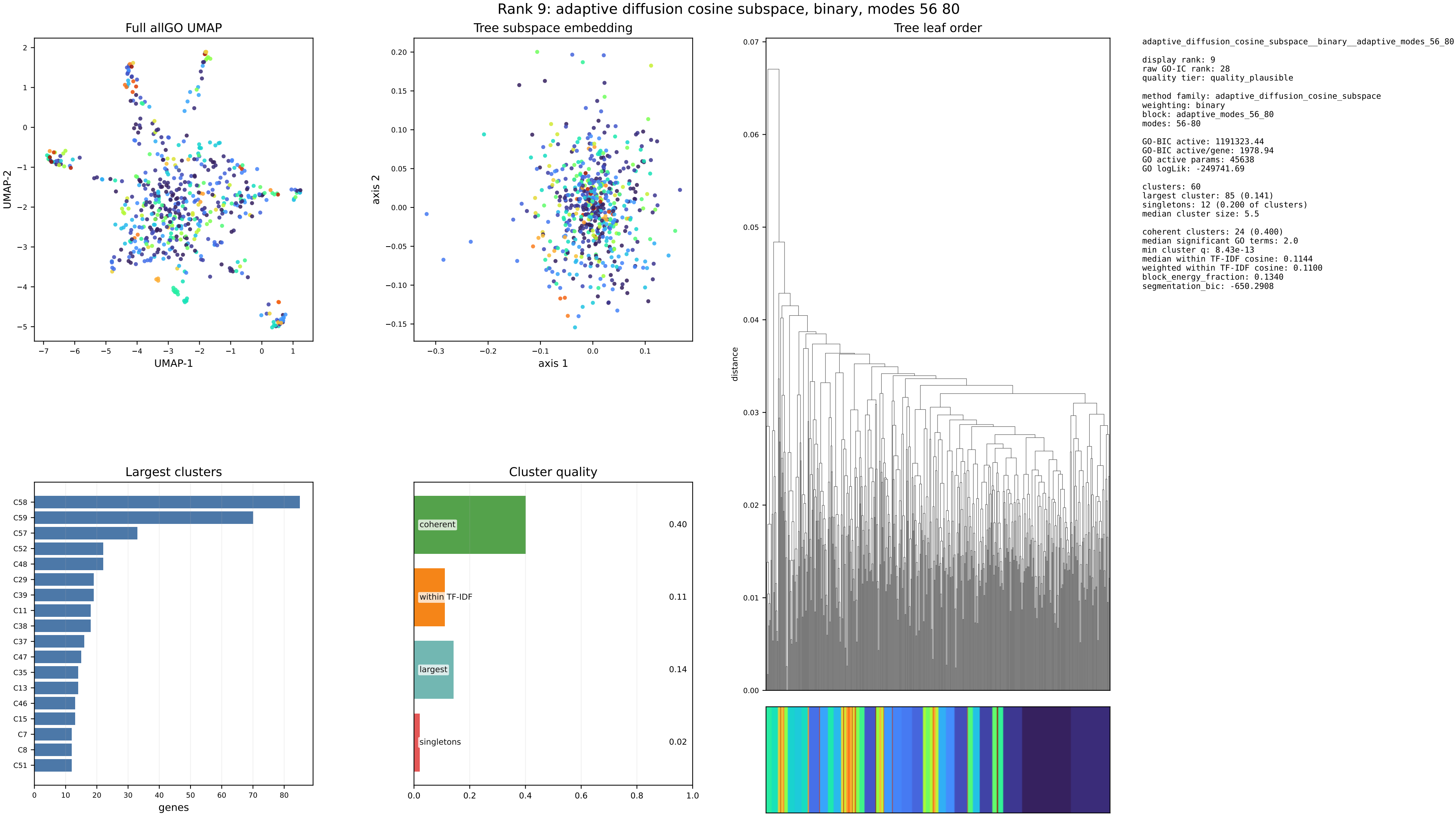
clusters: 69
largest cluster: 68 (0.113)
singletons: 17 (0.246 of clusters)
median cluster size: 3.0

coherent clusters: 24 (0.348)
median significant G0 terms: 0.0
min cluster q: 5.67e-16
median within TF-IDF cosine: 0.1164
weighted within TF-IDF cosine: 0.1138
block_energy_fraction: 0.0499
segmentation_bic: -657.8000

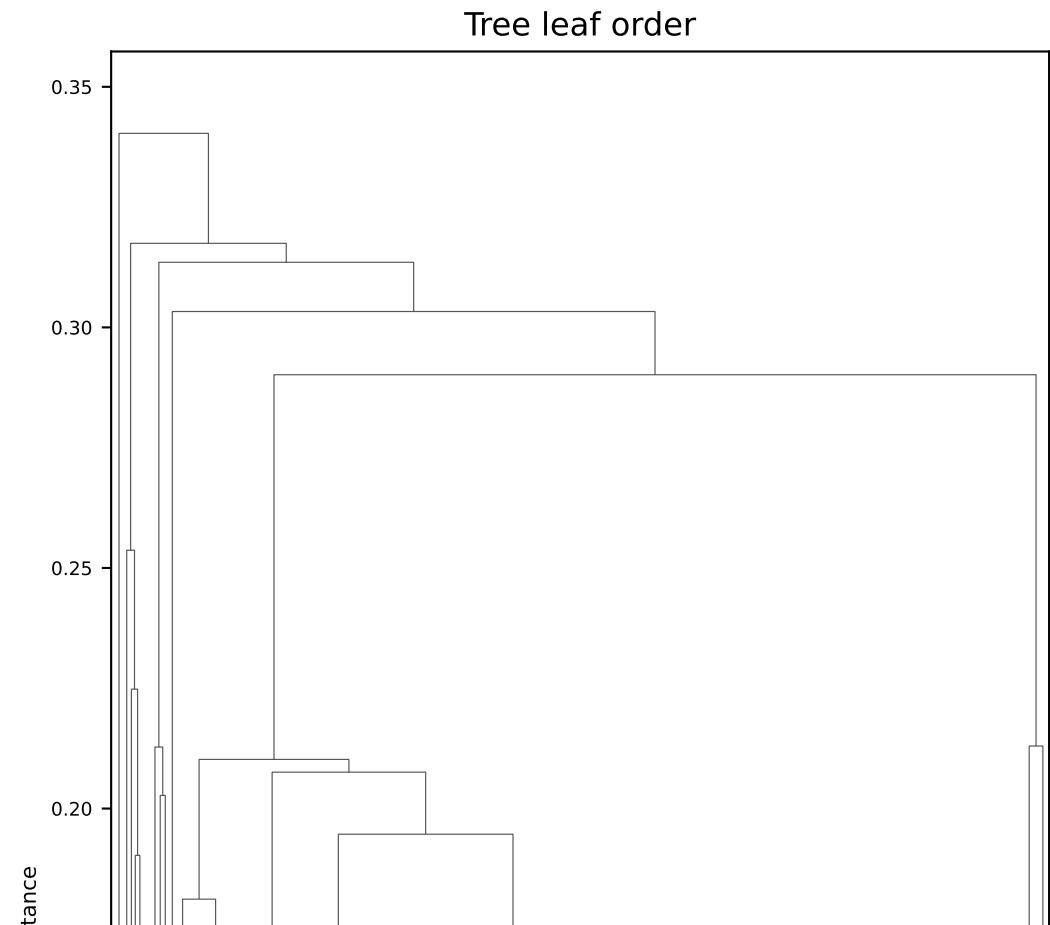
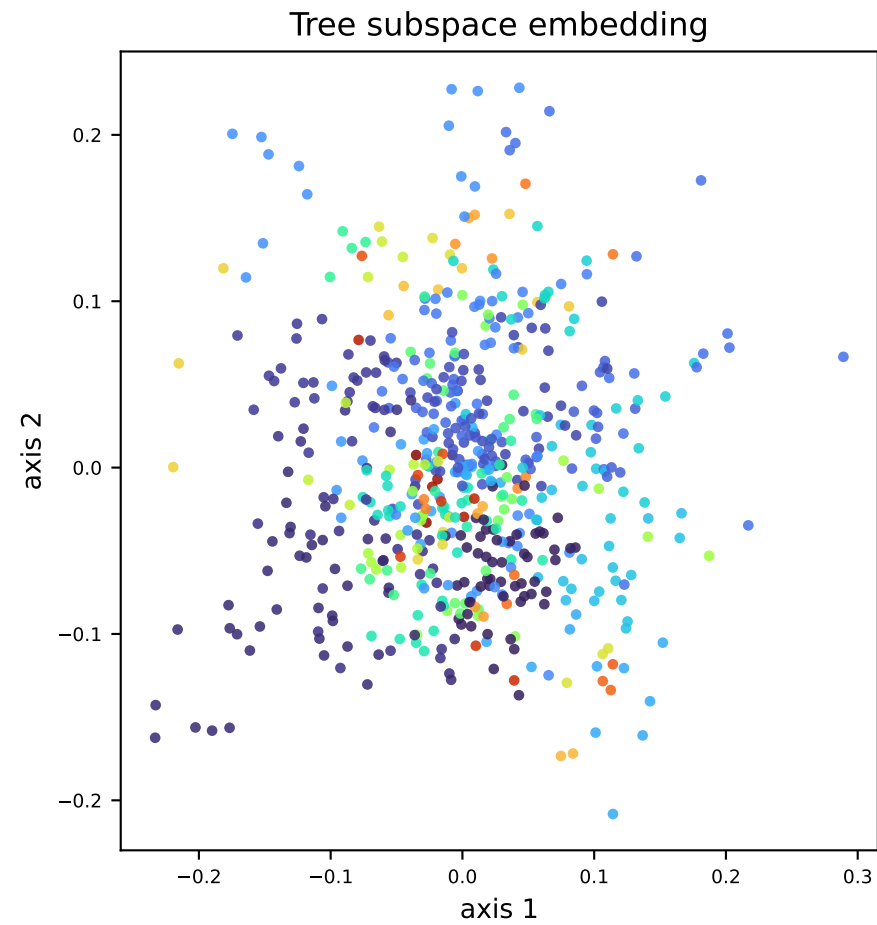
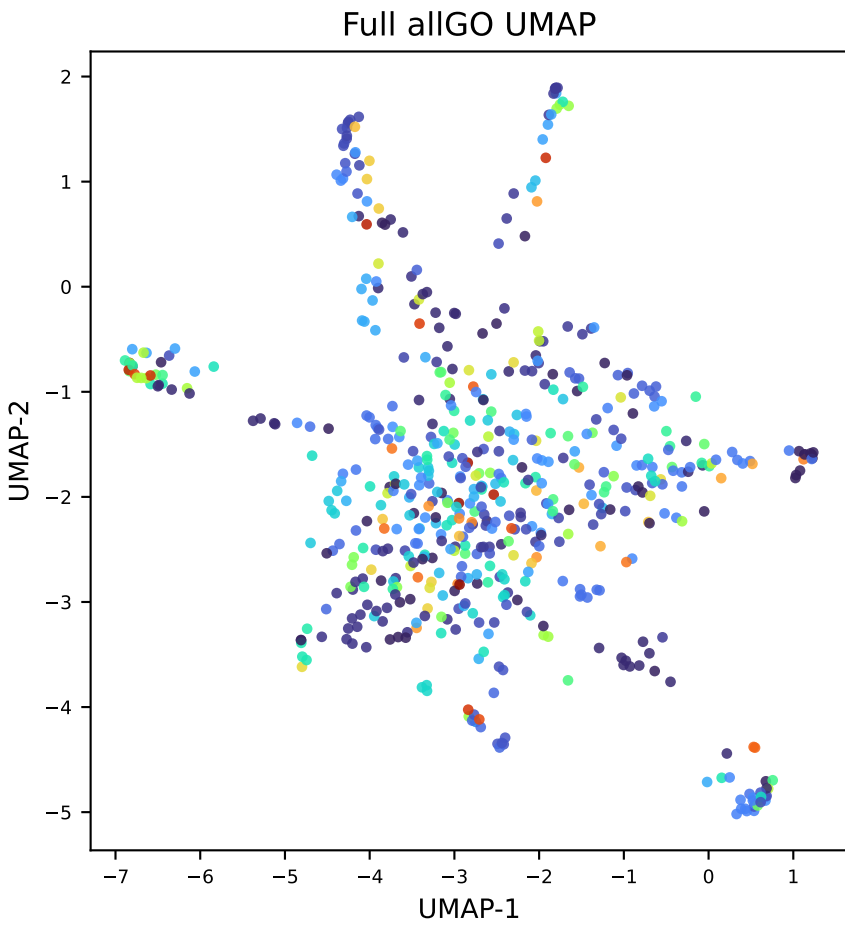


distance





Rank 10: raw cosine subspace, tfidf, modes 16 19



raw_cosine_subspace__tfidf__adaptive_modes_16_19

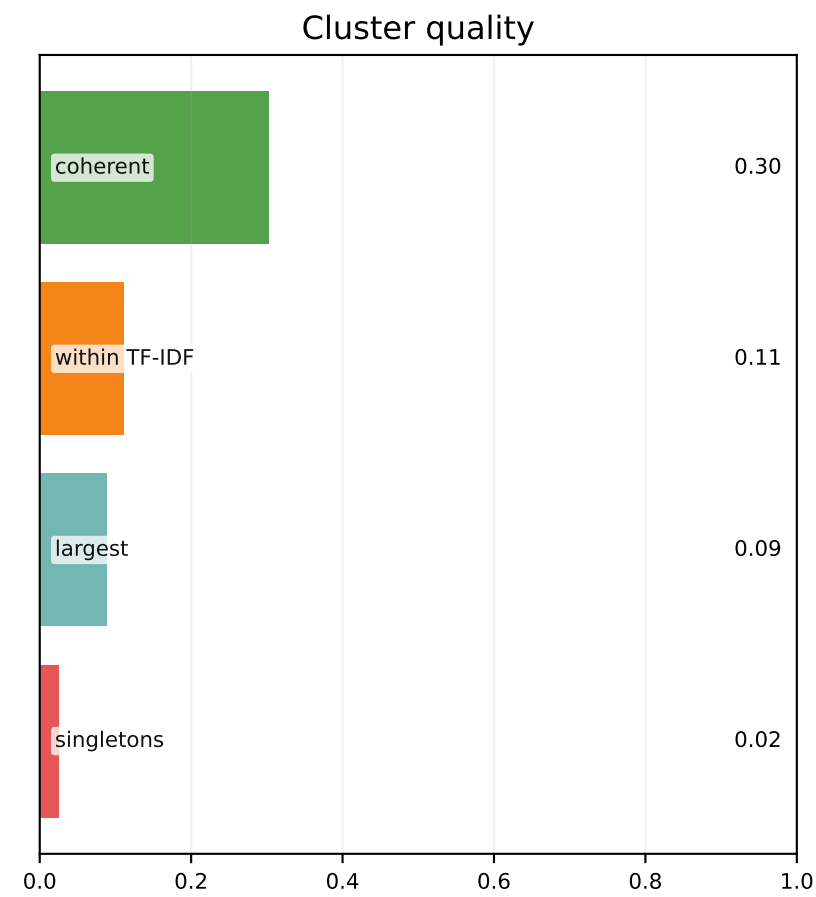
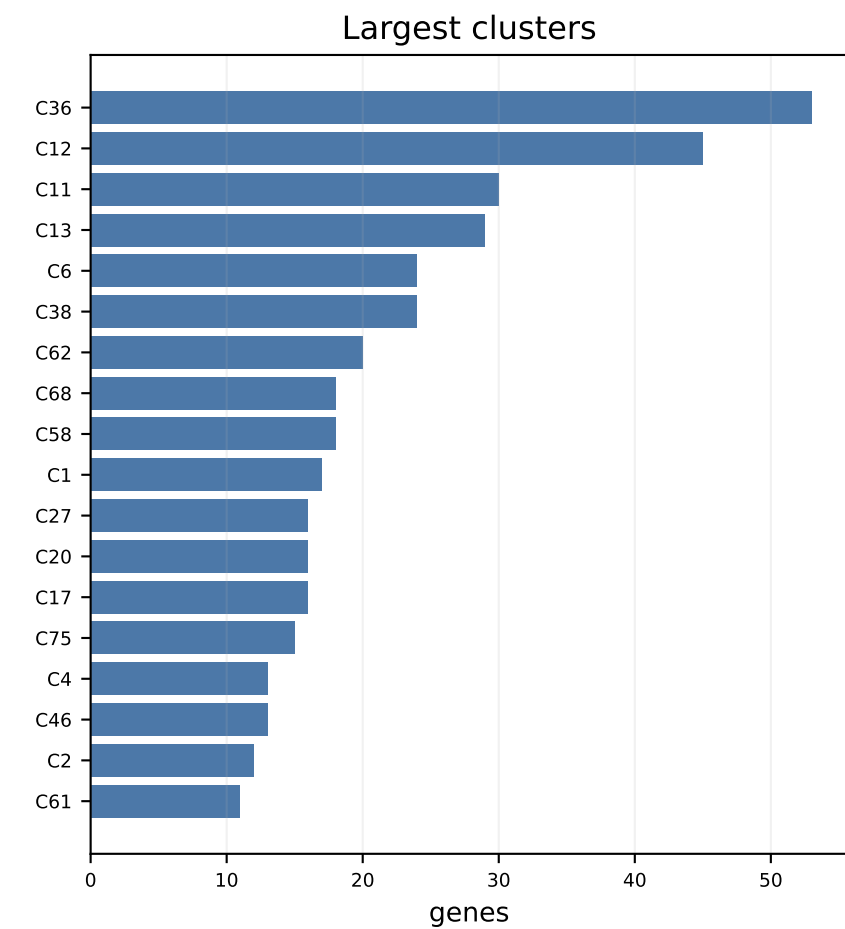
display rank: 10
raw G0-IC rank: 29
quality tier: quality_plausible

method family: raw_cosine_subspace
weighting: tfidf
block: adaptive_modes_16_19
modes: 16-19

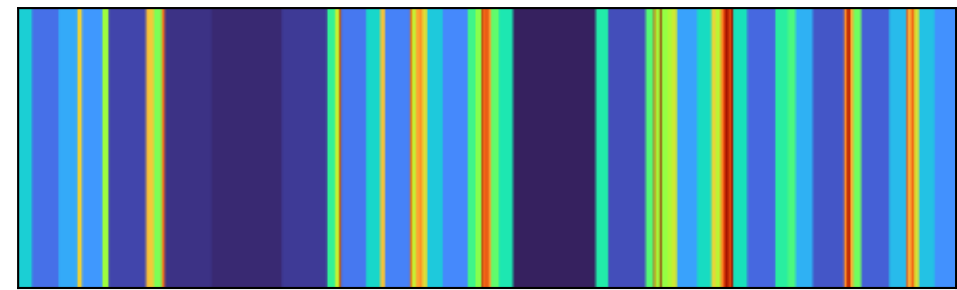
G0-BIC active: 1207532.44
G0-BIC active/gene: 2005.87
G0 active params: 48671
G0 logLik: -234857.12

clusters: 76
largest cluster: 53 (0.088)
singletons: 15 (0.197 of clusters)
median cluster size: 3.5

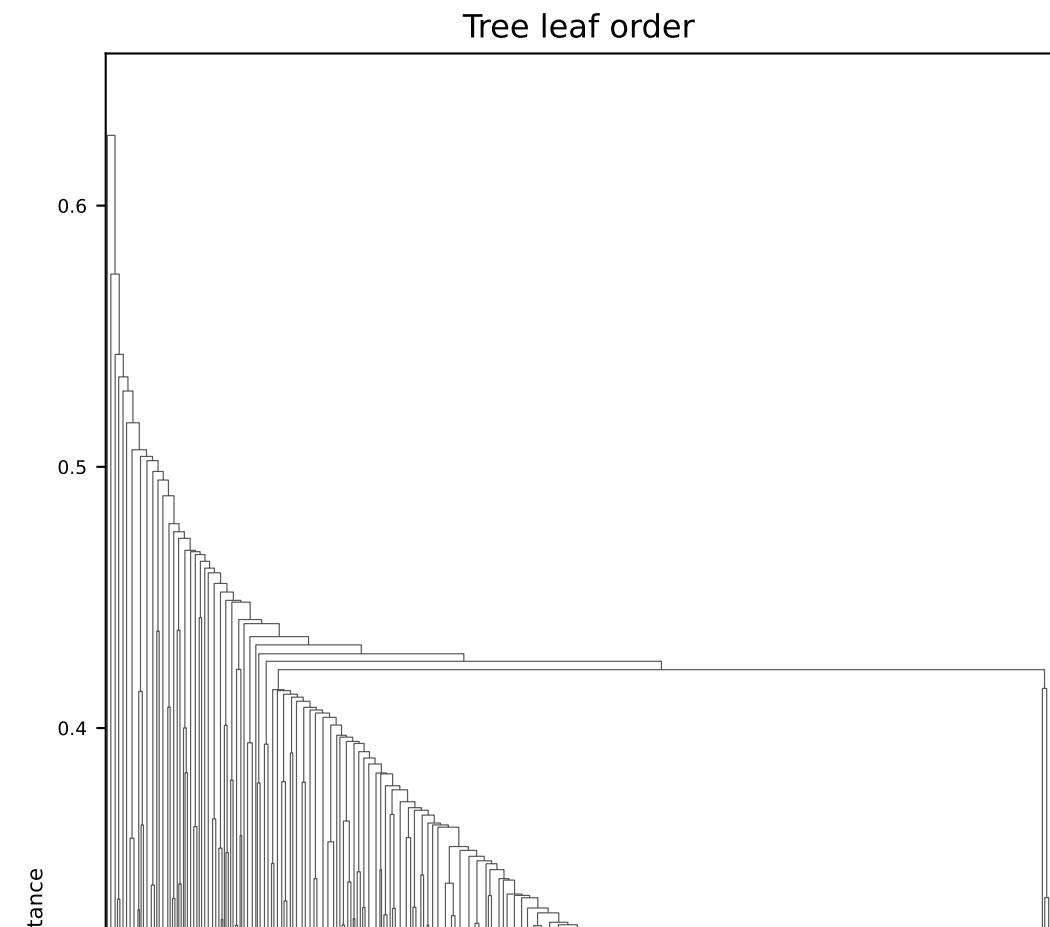
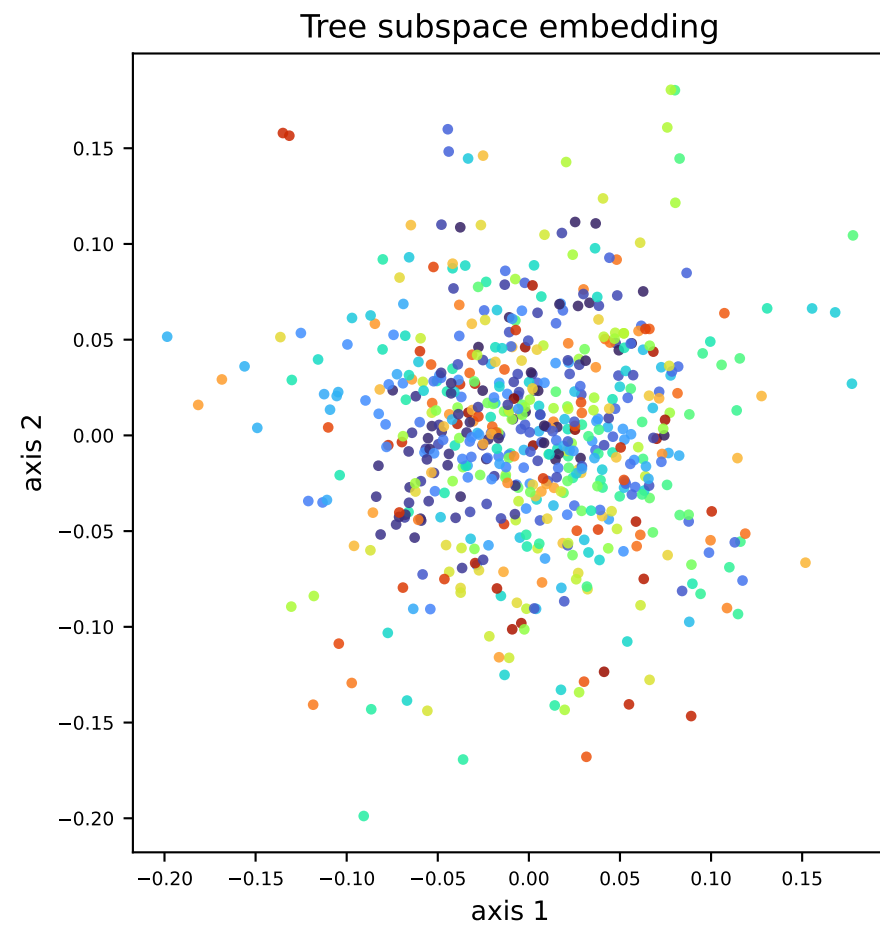
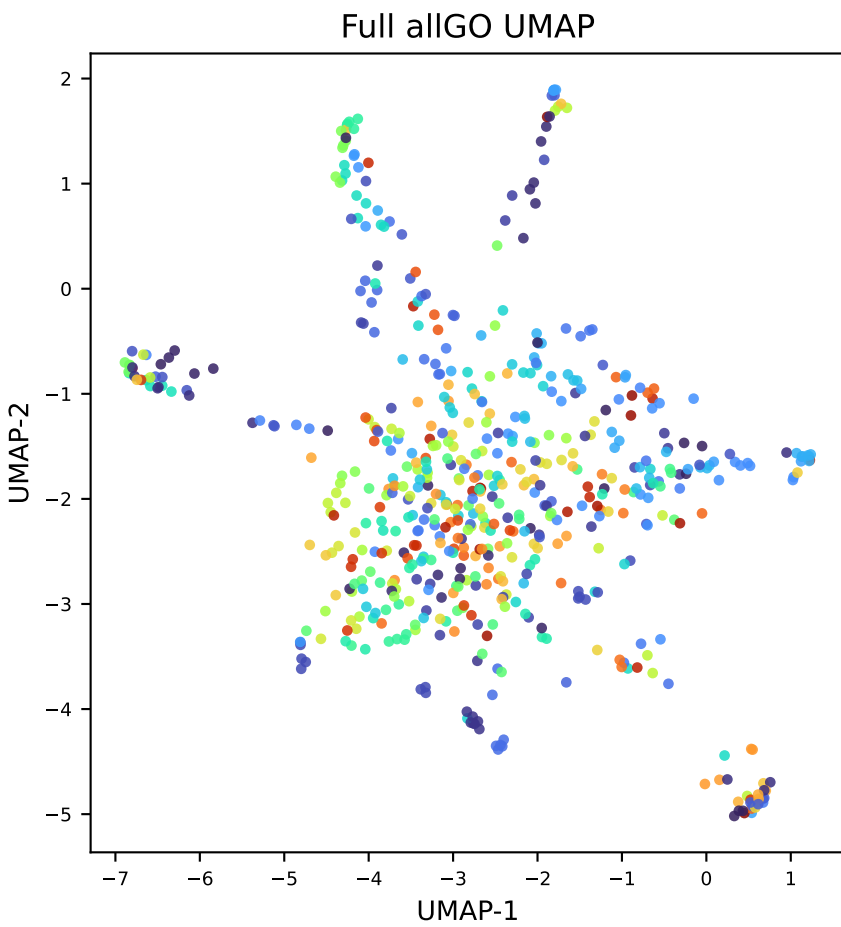
coherent clusters: 23 (0.303)
median significant G0 terms: 0.0
min cluster q: 4.07e-14
median within TF-IDF cosine: 0.0963
weighted within TF-IDF cosine: 0.1112
block_energy_fraction: 0.0499
segmentation_bic: -657.8000



distance



Rank 11: raw cosine subspace, tfidf, modes 53 80



raw_cosine_subspace__tfidf__adaptive_modes_53_80

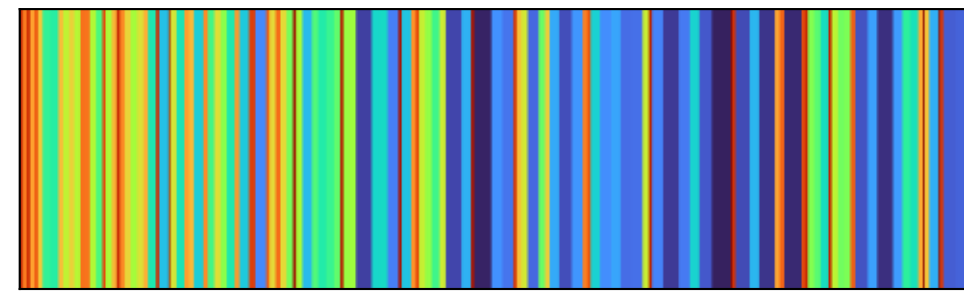
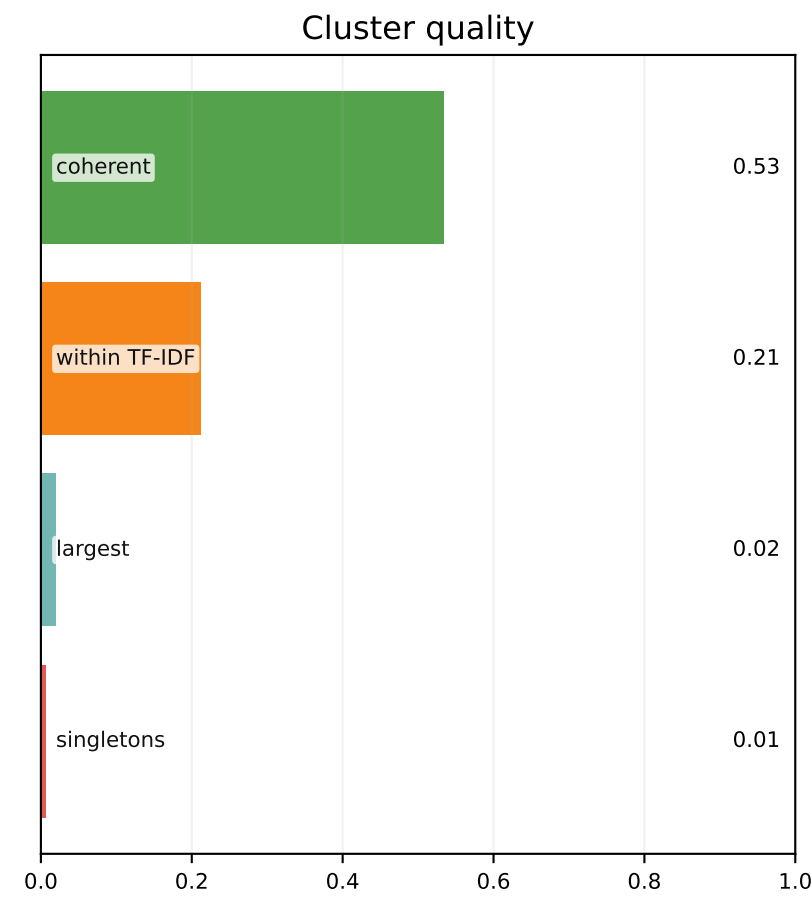
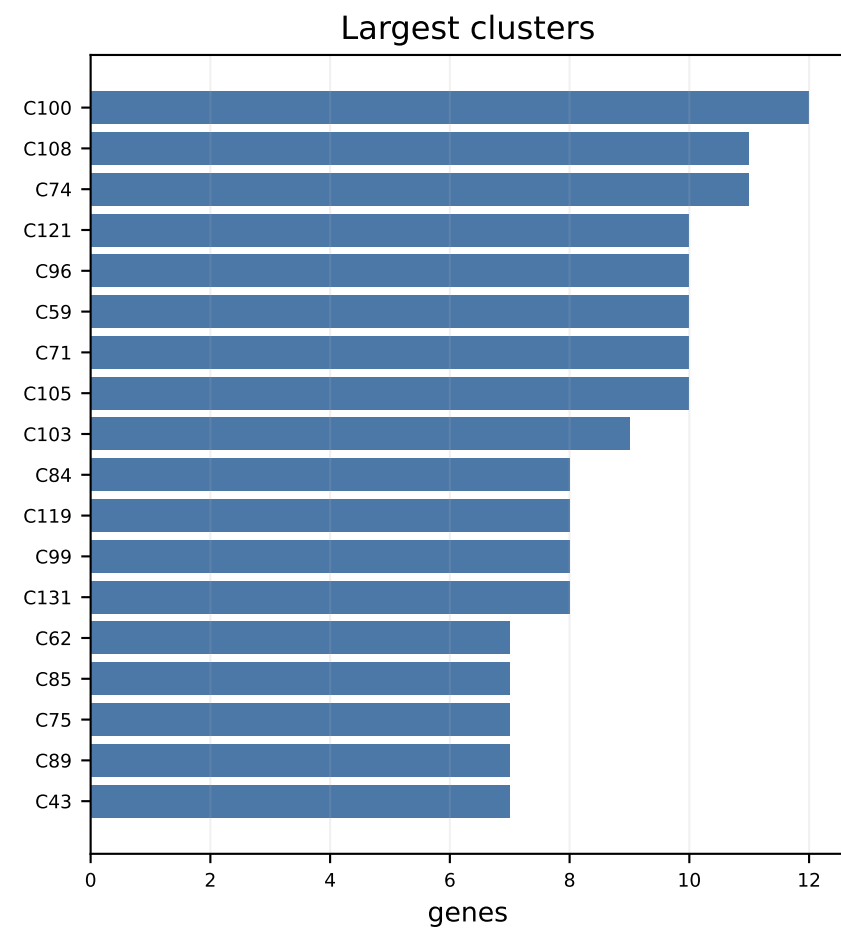
display rank: 11
raw G0-IC rank: 30
quality tier: quality_plausible

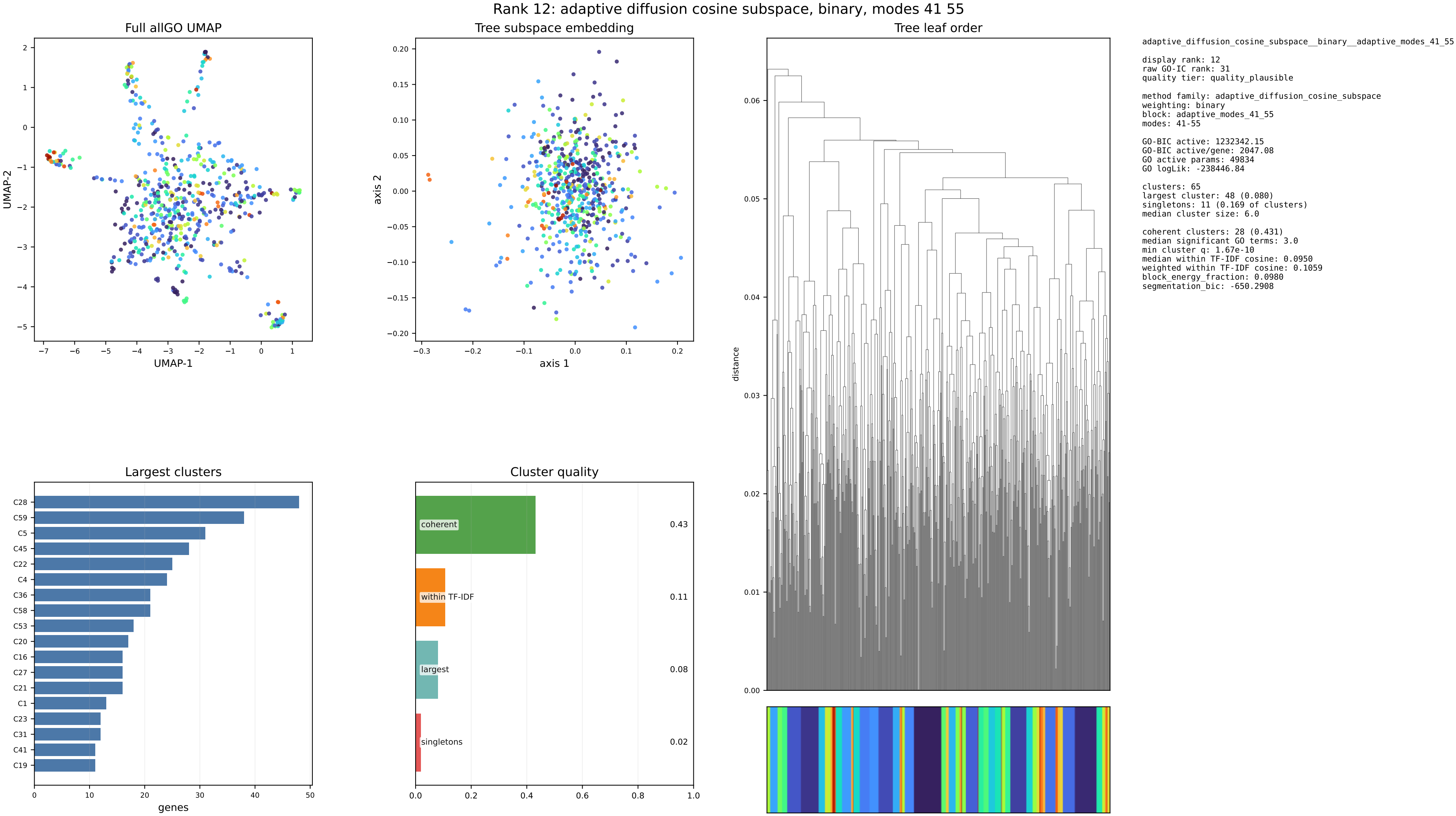
method family: raw_cosine_subspace
weighting: tfidf
block: adaptive_modes_53_80
modes: 53-80

G0-BIC active: 1221629.18
G0-BIC active/gene: 2029.28
G0 active params: 57997
G0 logLik: -171217.68

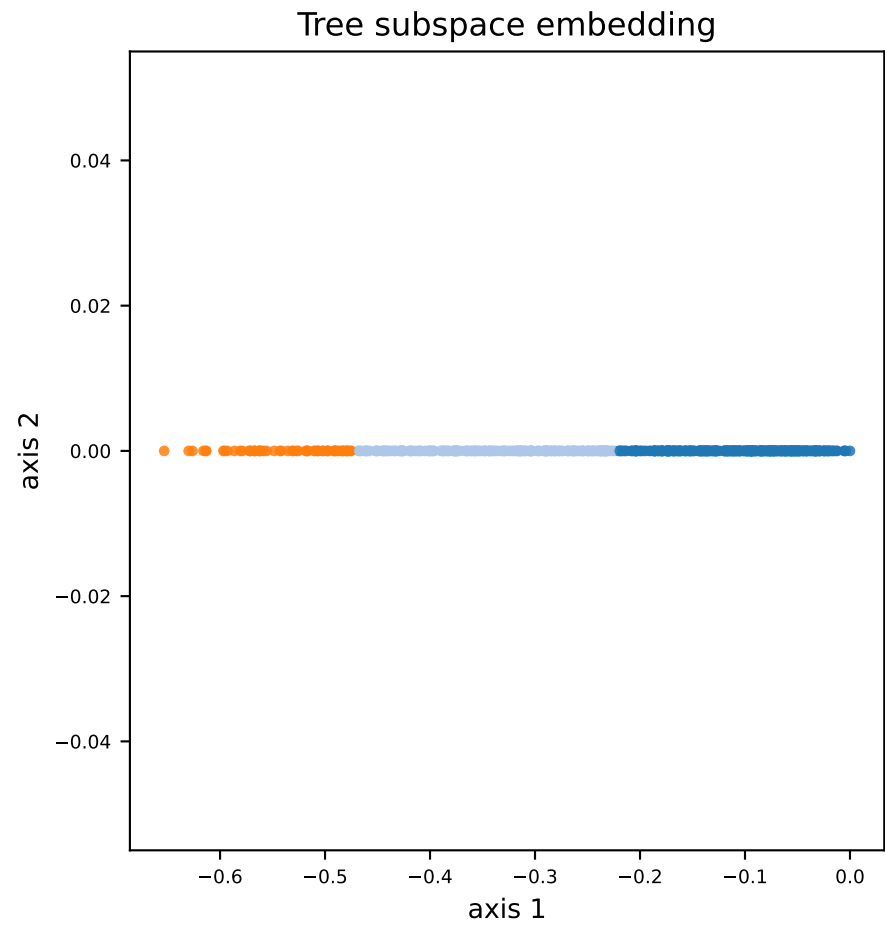
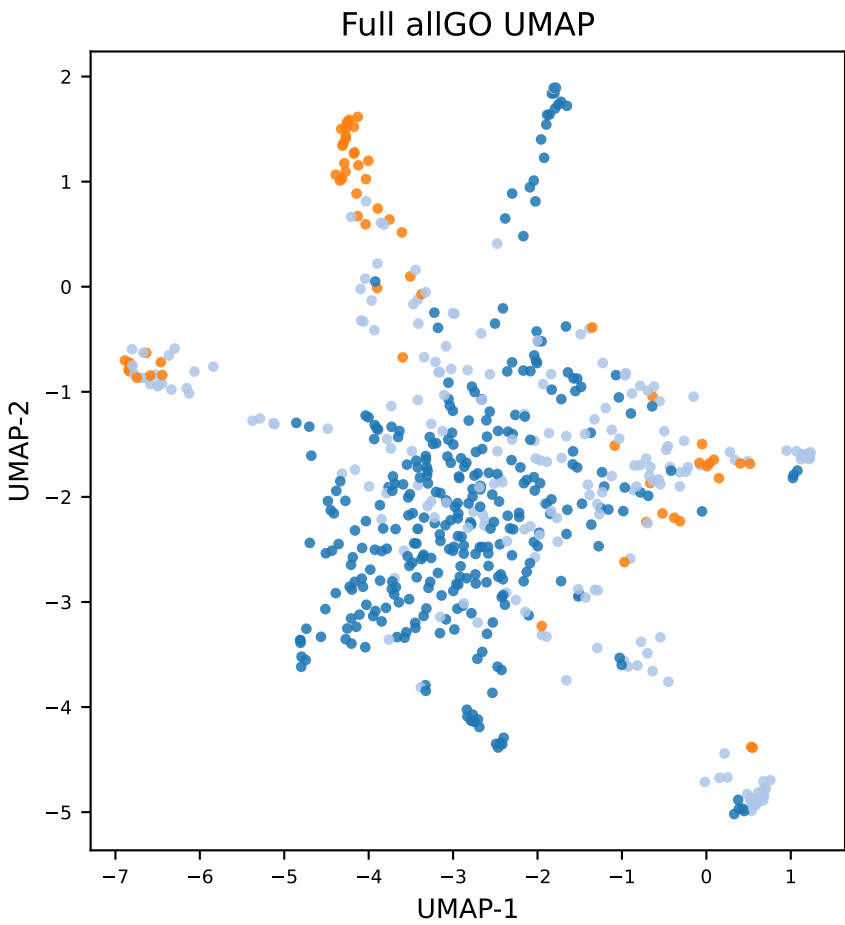
clusters: 133
largest cluster: 12 (0.020)
singletons: 4 (0.030 of clusters)
median cluster size: 4.0

coherent clusters: 71 (0.534)
median significant G0 terms: 4.0
min cluster q: 4.64e-12
median within TF-IDF cosine: 0.1912
weighted within TF-IDF cosine: 0.2122
block_energy_fraction: 0.1753
segmentation_bic: -657.8000





Rank 13: raw cosine subspace, binary, common mode 01



raw_cosine_subspace_binary_adaptive_common_mode_01

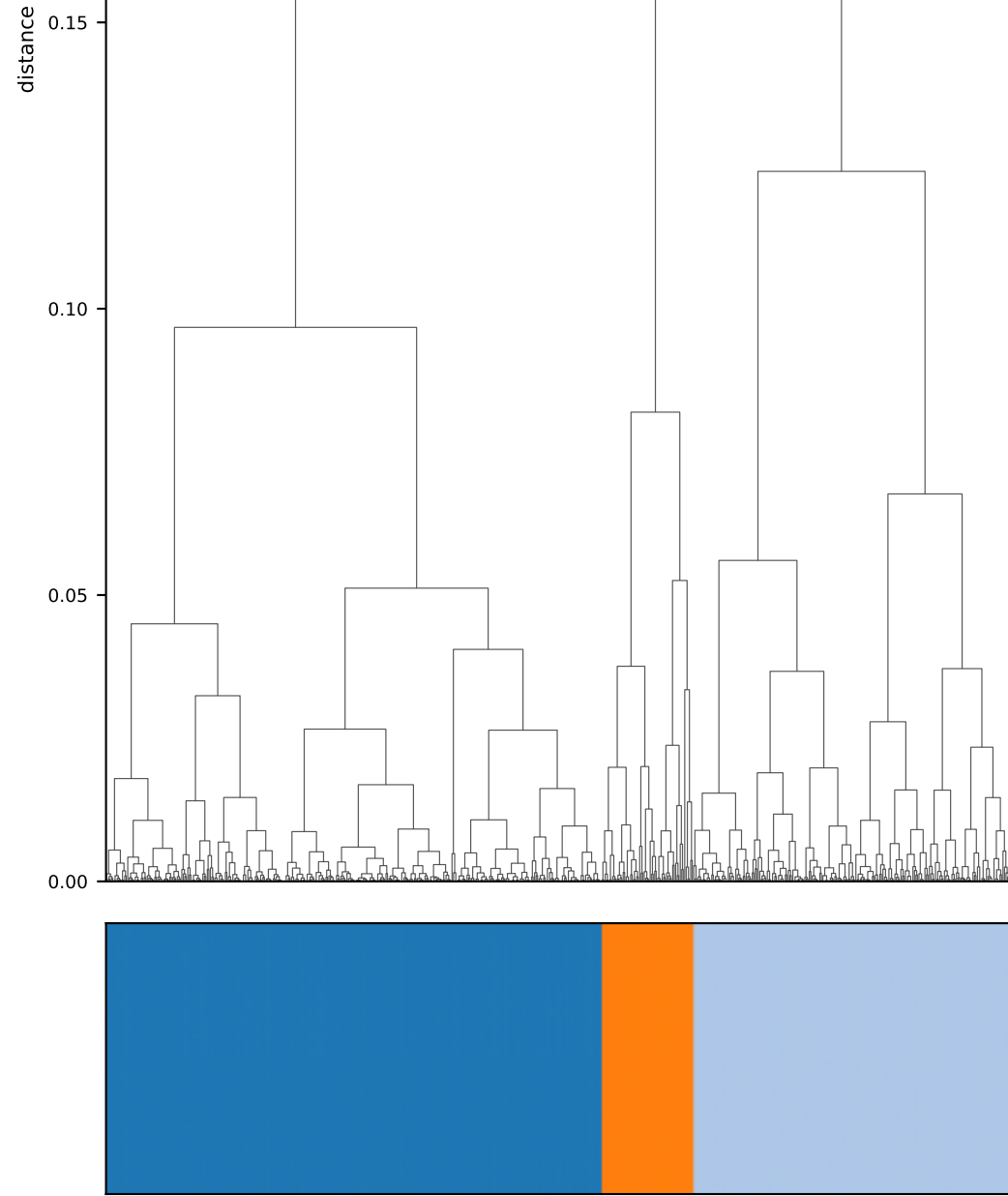
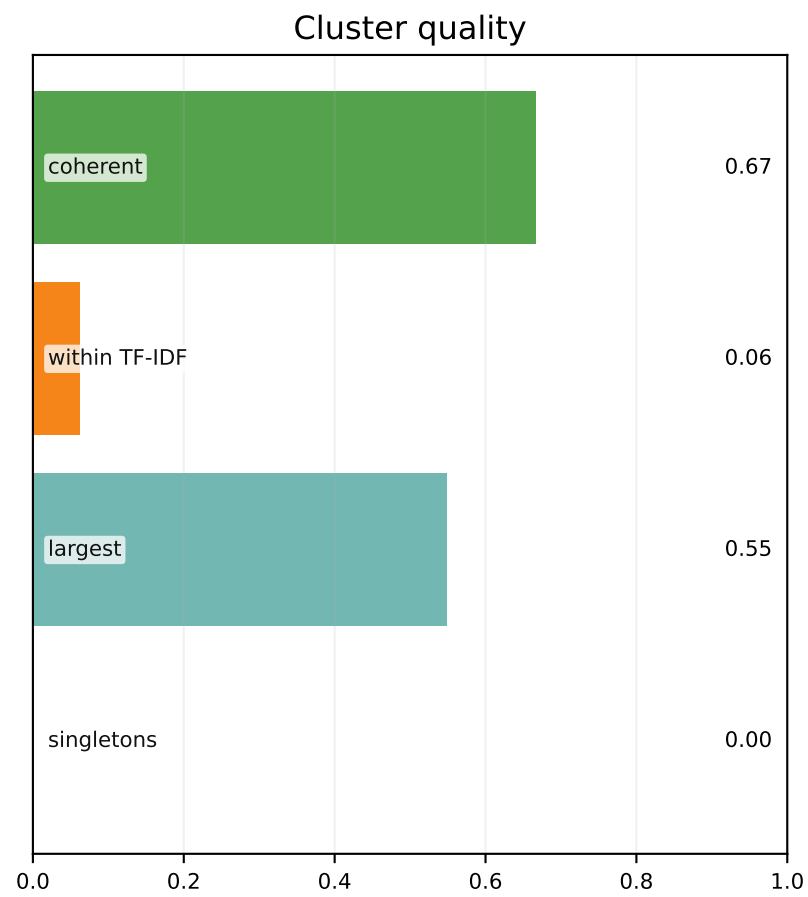
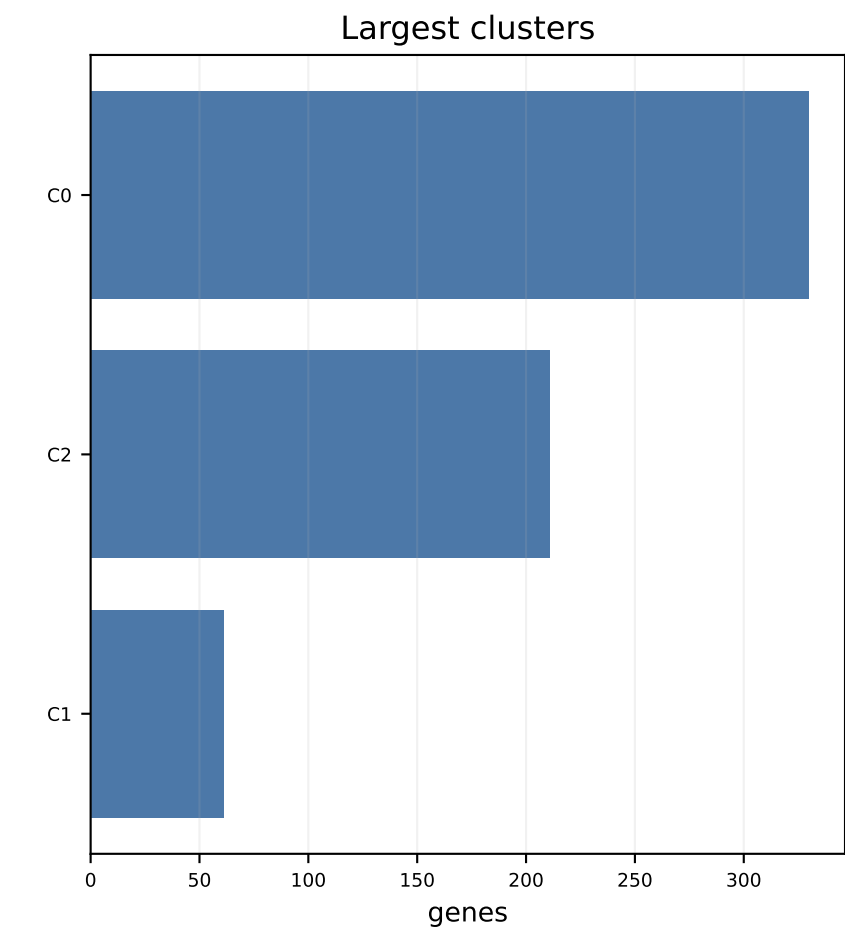
display rank: 13
raw G0-IC rank: 14
quality tier: broad_coherent

method family: raw_cosine_subspace
weighting: binary
block: adaptive_common_mode_01
modes: 1-1

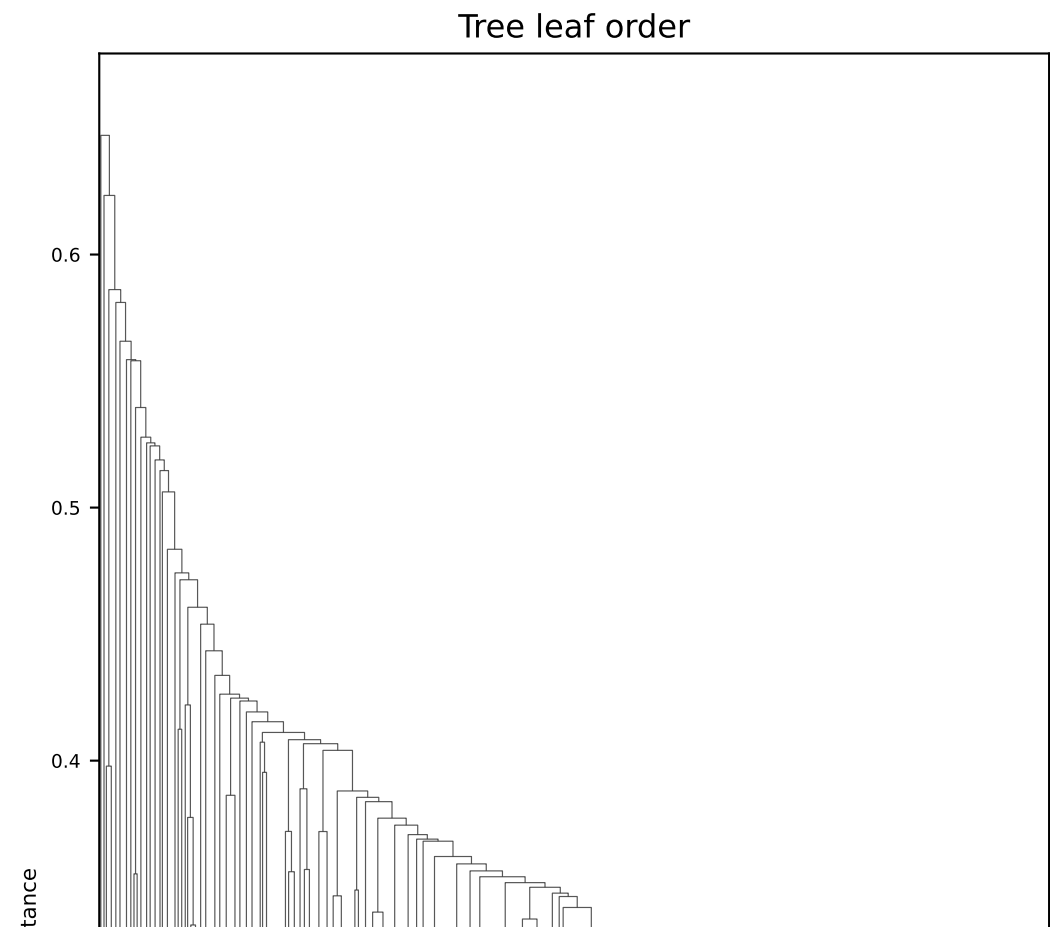
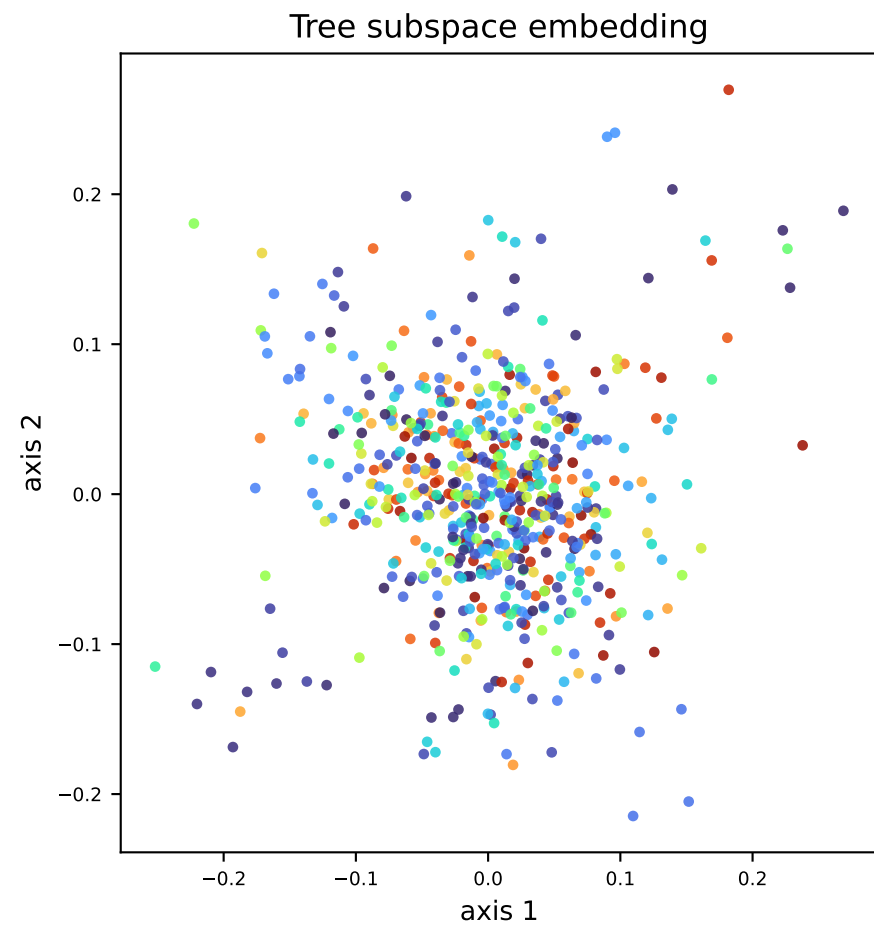
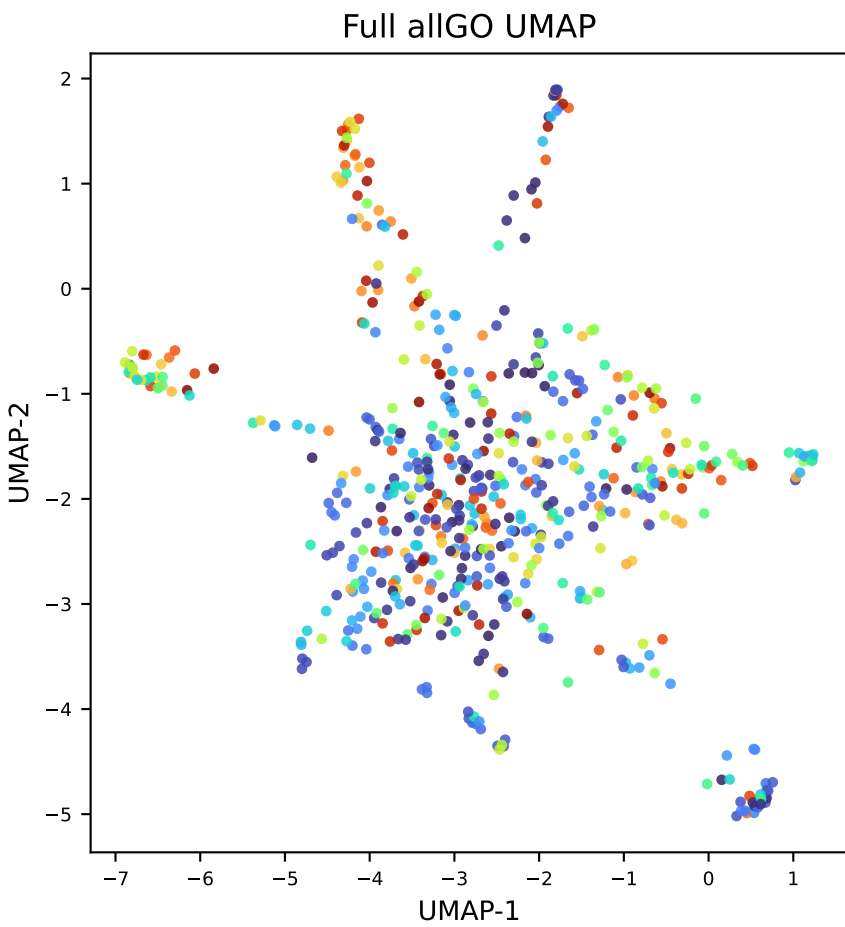
G0-BIC active: 871152.74
G0-BIC active/gene: 1447.10
G0 active params: 13836
G0 logLik: -330704.34

clusters: 3
largest cluster: 330 (0.548)
singletons: 0 (0.000 of clusters)
median cluster size: 211.0

coherent clusters: 2 (0.667)
median significant G0 terms: 654.0
min cluster q: 7.25e-37
median within TF-IDF cosine: 0.0744
weighted within TF-IDF cosine: 0.0617
block_energy_fraction: 0.1530
segmentation_bic: -650.2908



Rank 14: raw cosine subspace, binary, modes 22 40



raw_cosine_subspace__binary__adaptive_modes_22_40

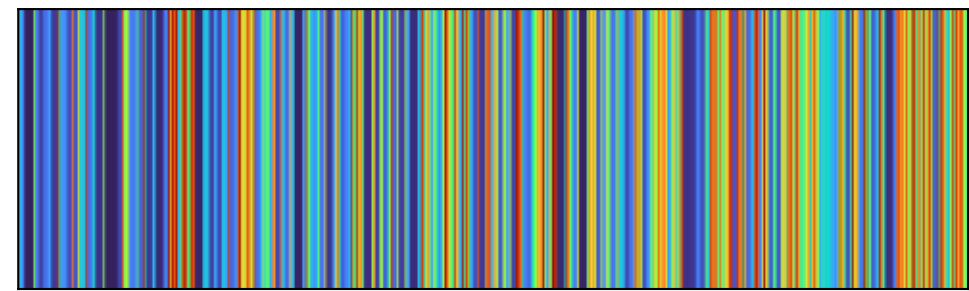
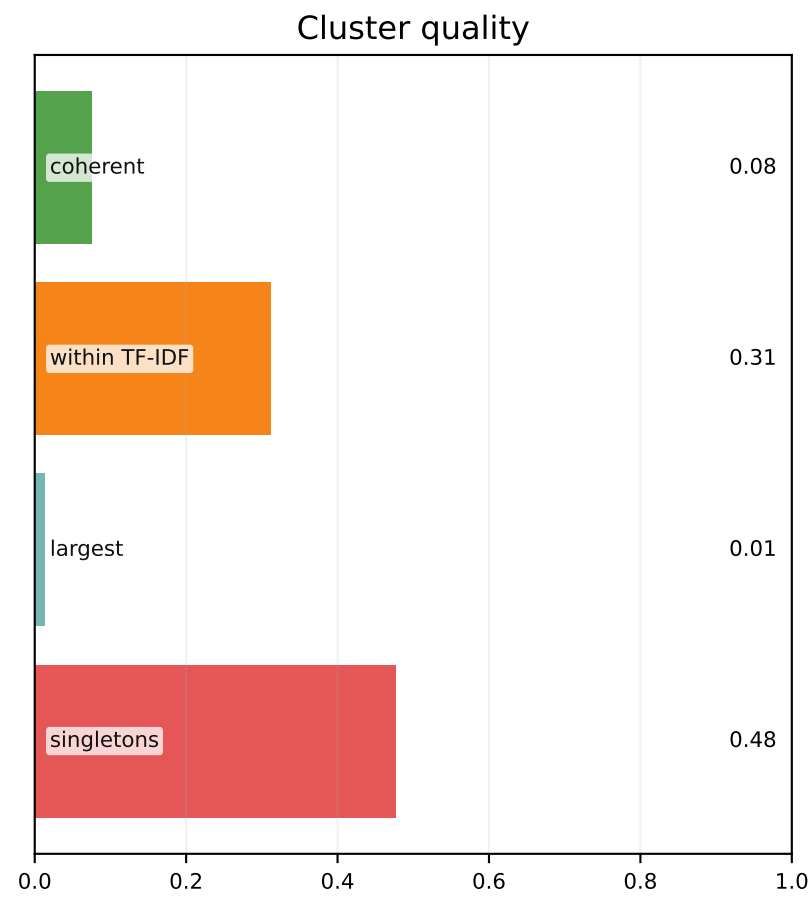
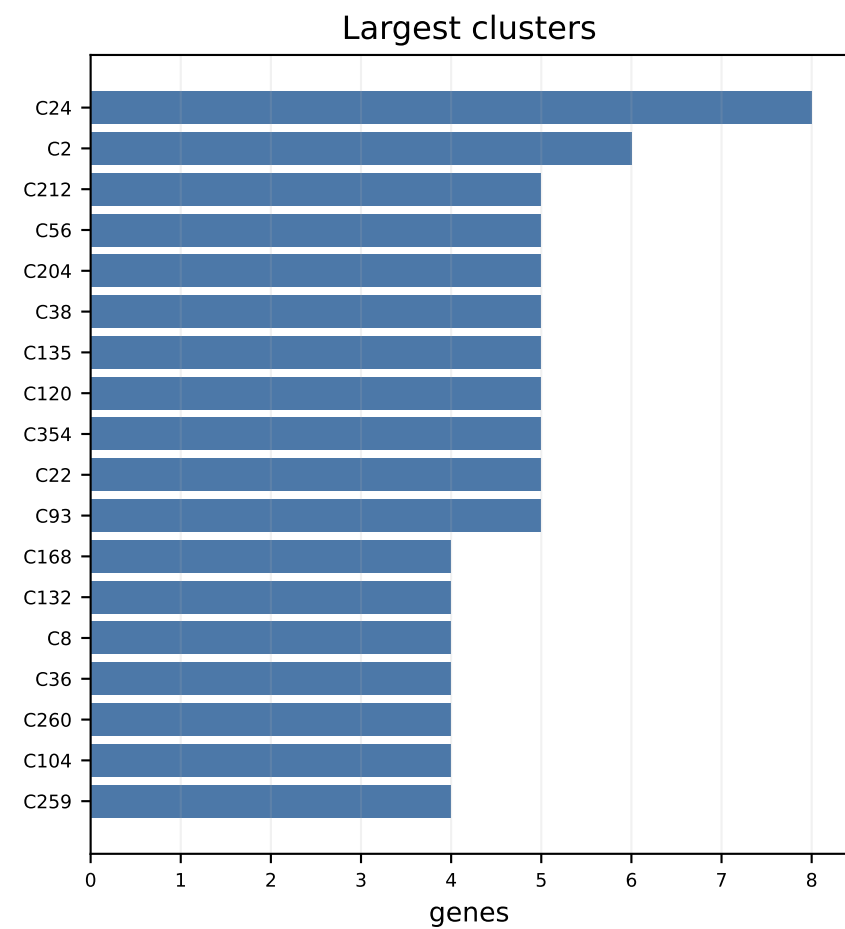
display rank: 14
raw G0-IC rank: 7
quality tier: quality_mixed

method family: raw_cosine_subspace
weighting: binary
block: adaptive_modes_22_40
modes: 22-40

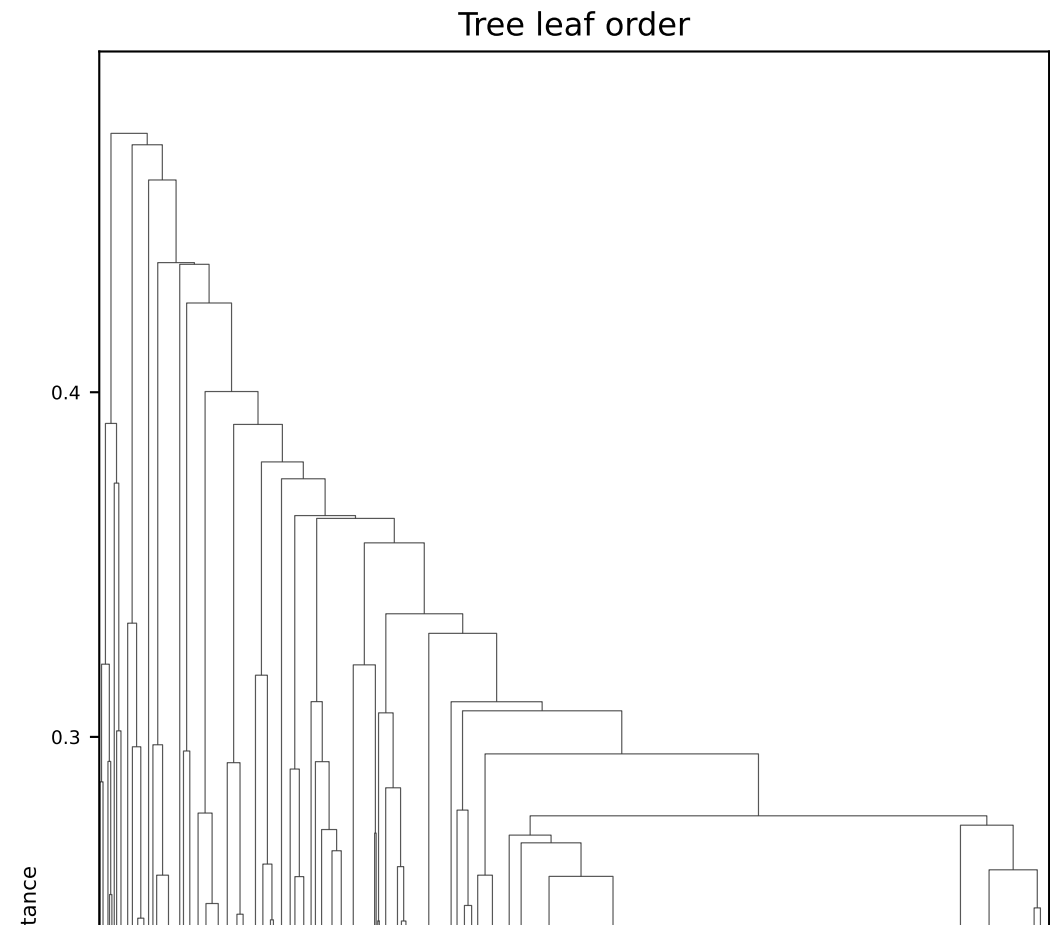
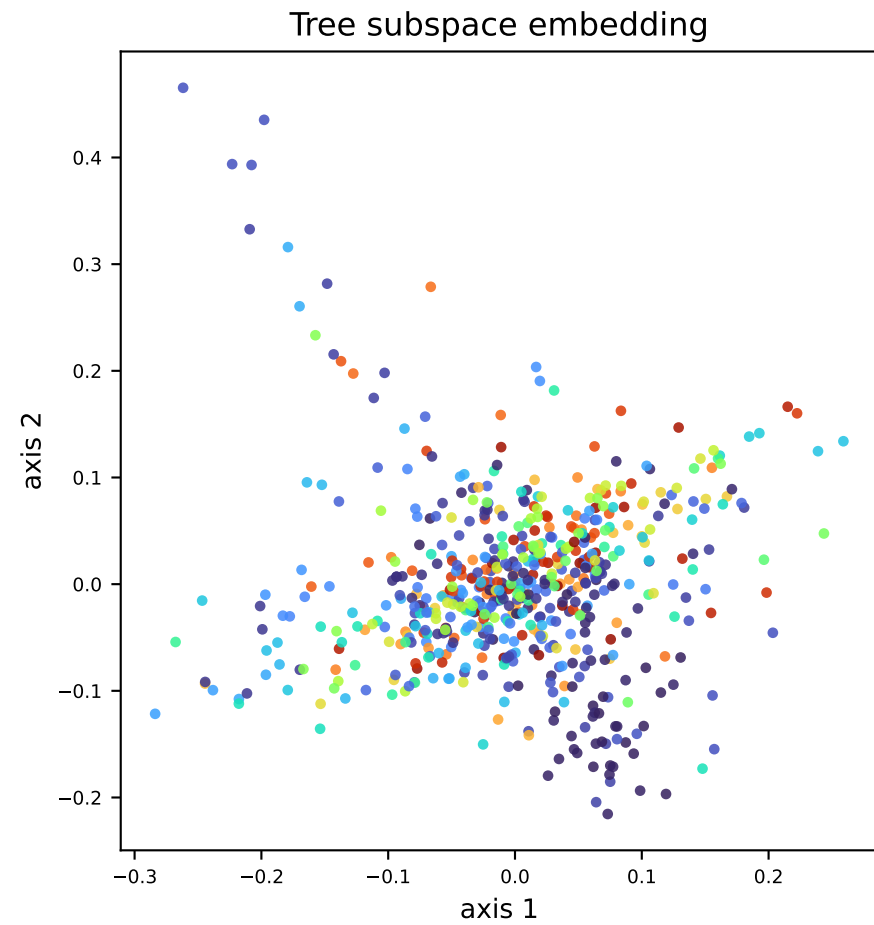
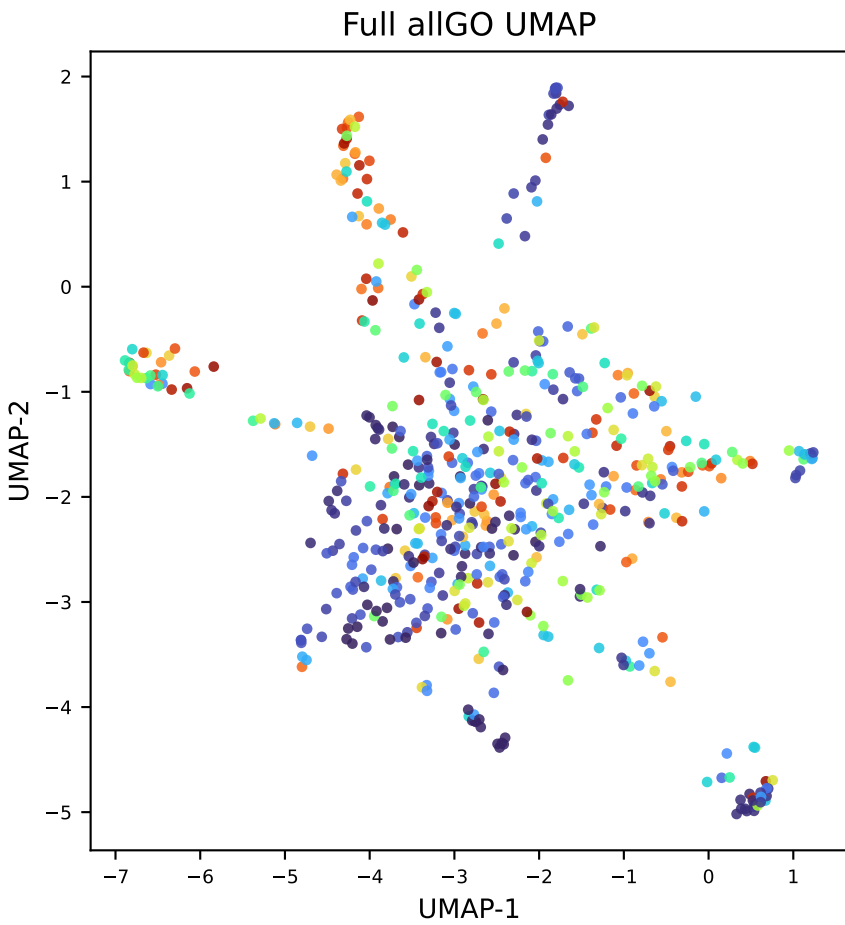
G0-BIC active: 277278.60
G0-BIC active/gene: 460.60
G0 active params: 14626
G0 logLik: -27779.35

clusters: 399
largest cluster: 8 (0.013)
singletons: 287 (0.719 of clusters)
median cluster size: 1.0

coherent clusters: 30 (0.075)
median significant G0 terms: 0.0
min cluster q: 9.83e-09
median within TF-IDF cosine: 0.3204
weighted within TF-IDF cosine: 0.3112
block_energy_fraction: 0.1591
segmentation_bic: -650.2908



Rank 15: raw cosine subspace, binary, modes 12 21



raw_cosine_subspace__binary__adaptive_modes_12_21

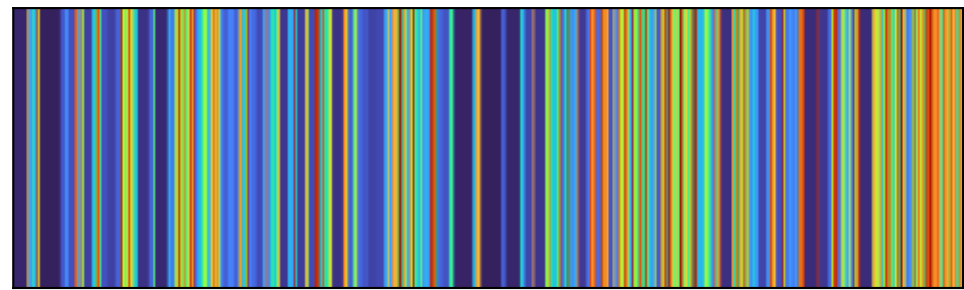
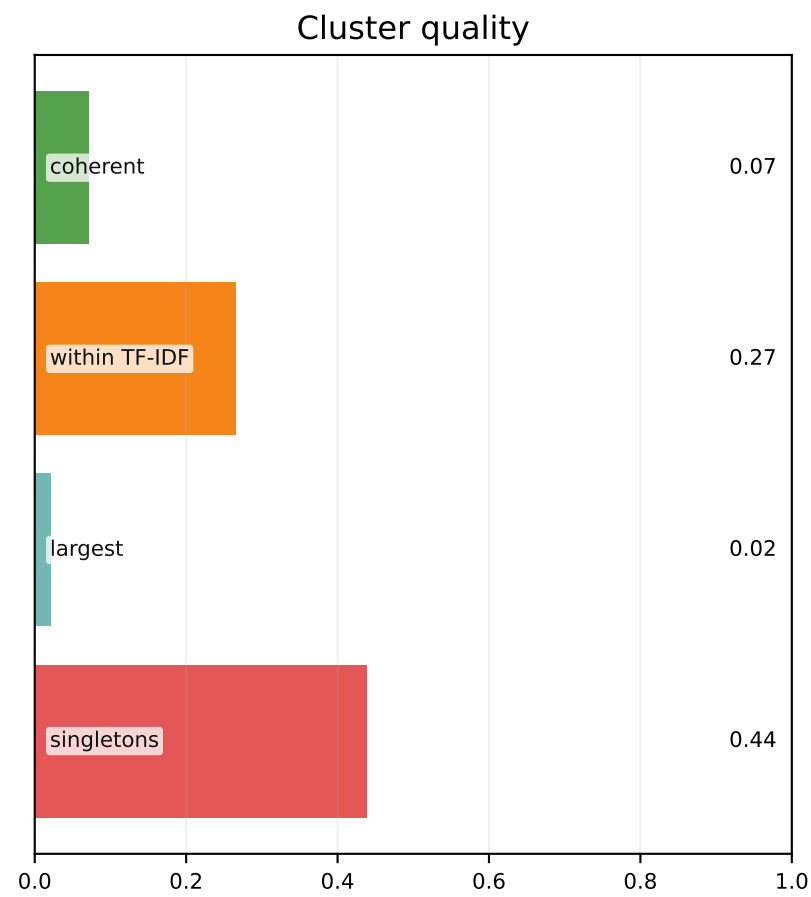
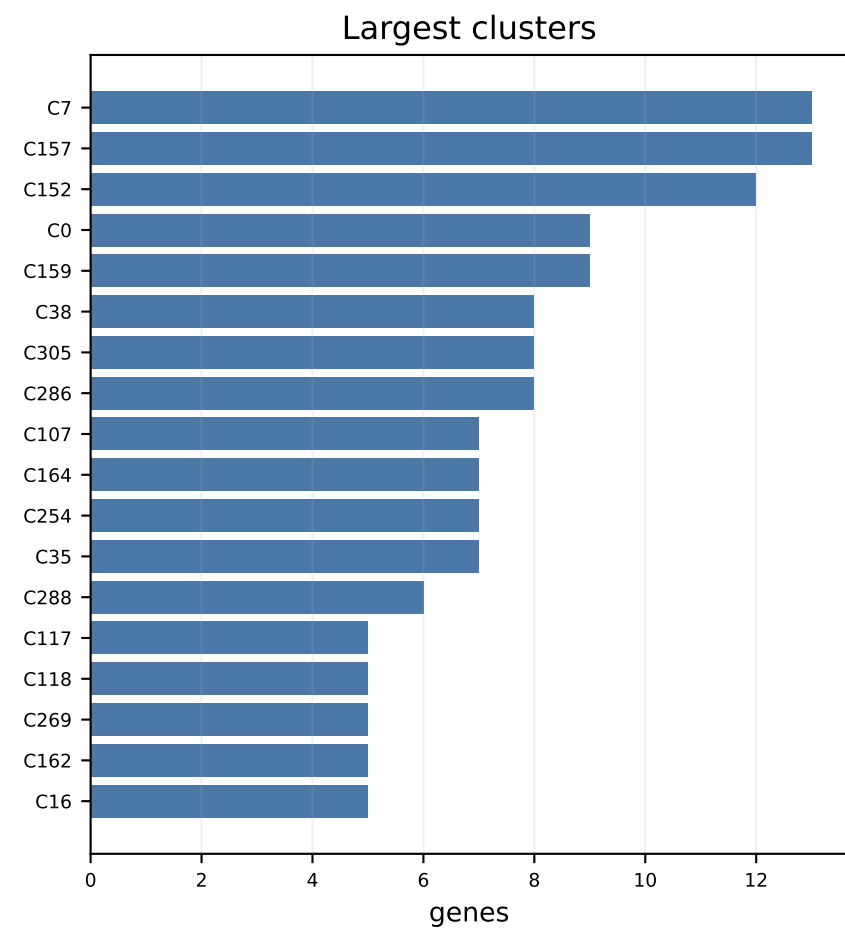
display rank: 15
raw G0-IC rank: 8
quality tier: quality_mixed

method family: raw_cosine_subspace
weighting: binary
block: adaptive_modes_12_21
modes: 12-21

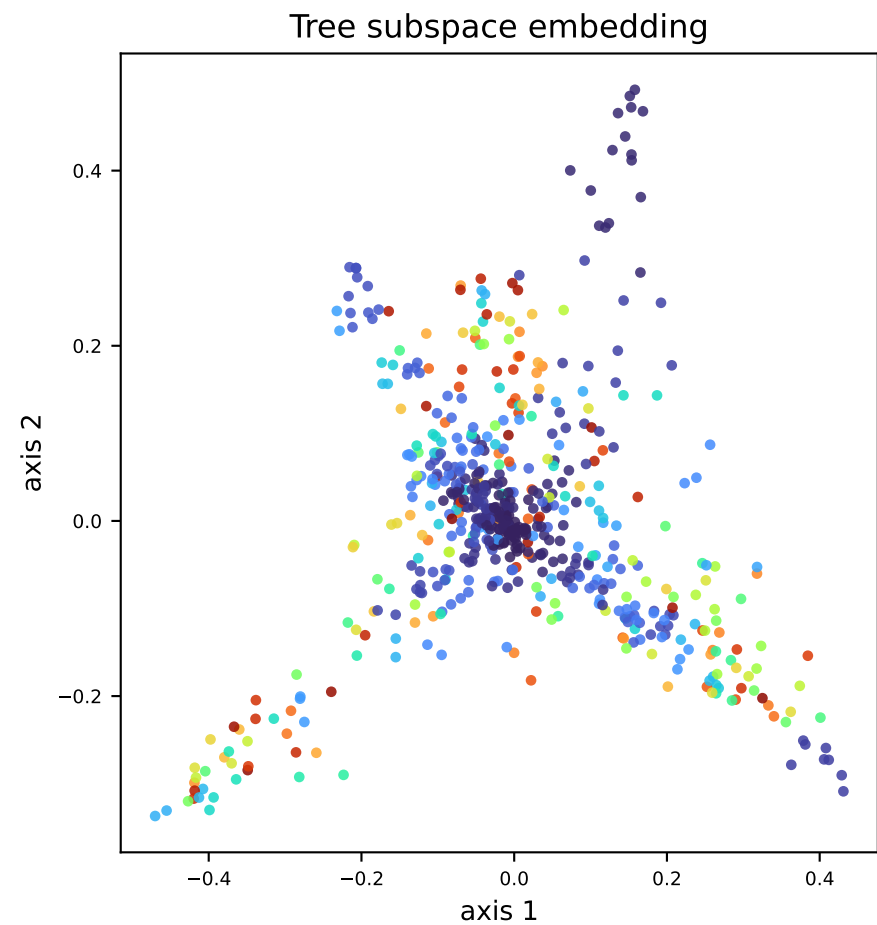
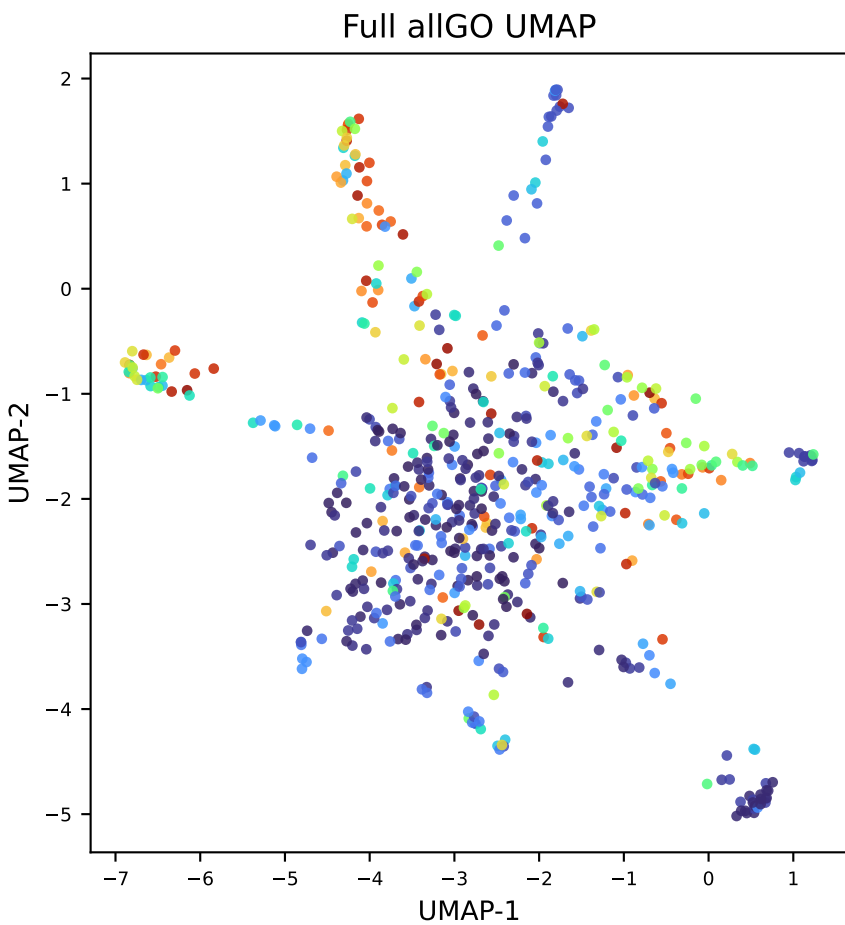
G0-BIC active: 320724.24
G0-BIC active/gene: 532.76
G0 active params: 16229
G0 logLik: -37352.00

clusters: 362
largest cluster: 13 (0.022)
singletons: 264 (0.729 of clusters)
median cluster size: 1.0

coherent clusters: 26 (0.072)
median significant G0 terms: 0.0
min cluster q: 1.28e-15
median within TF-IDF cosine: 0.1903
weighted within TF-IDF cosine: 0.2653
block_energy_fraction: 0.1234
segmentation_bic: -650.2908



Rank 16: raw cosine subspace, tfidf, modes 02 05



raw_cosine_subspace__tfidf__adaptive_modes_02_05

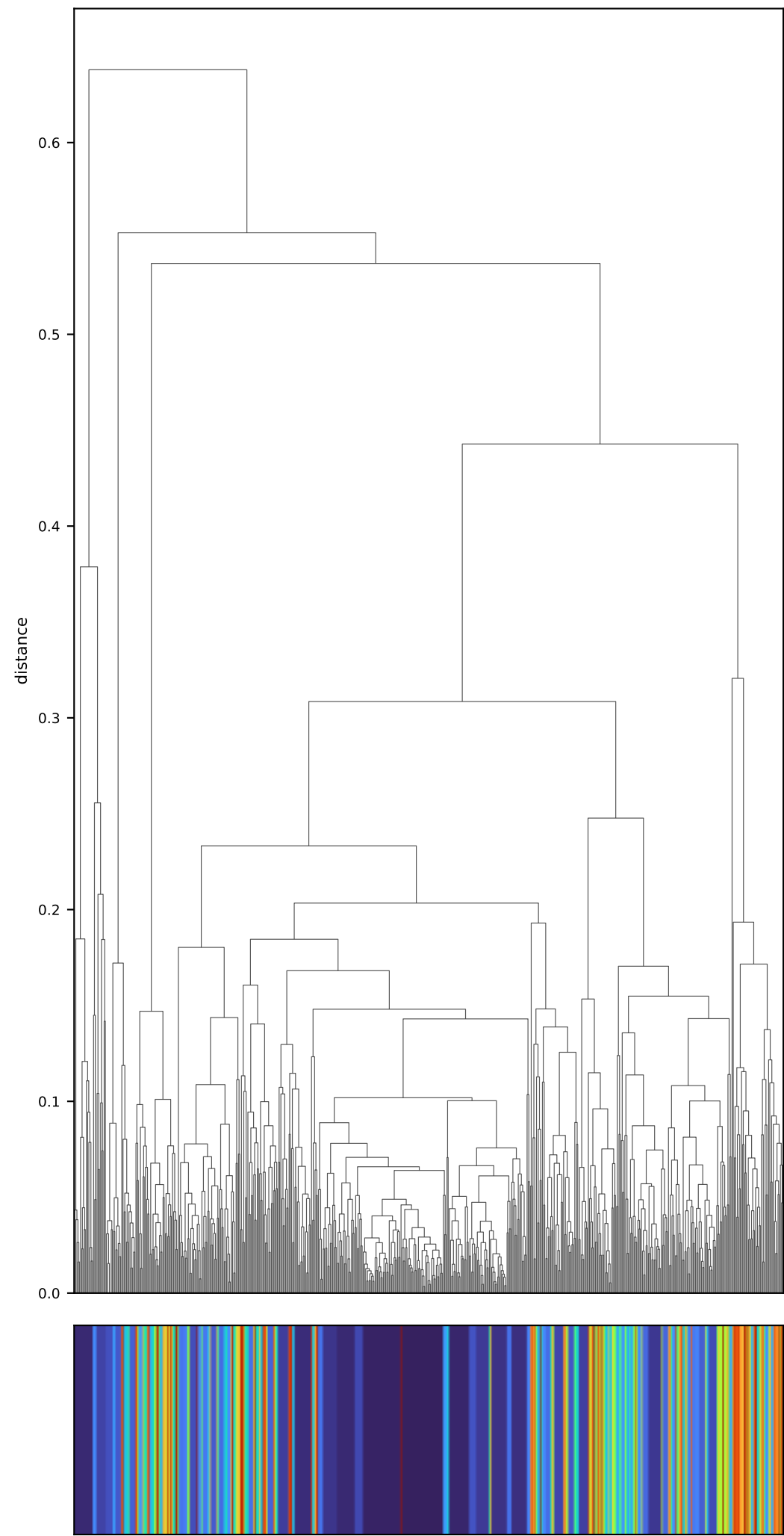
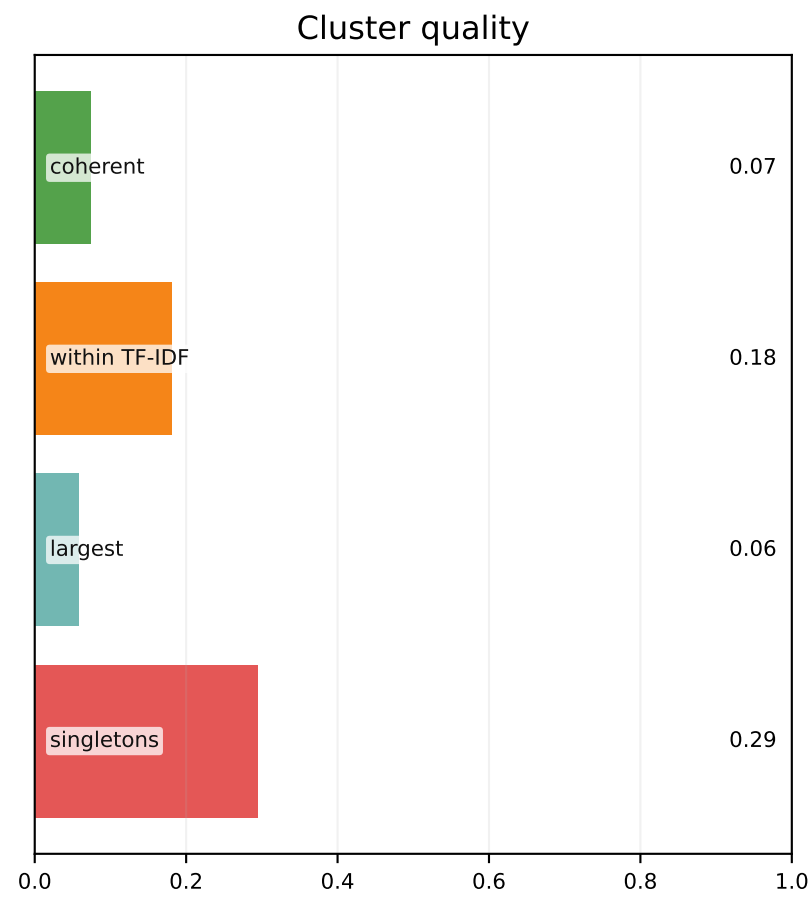
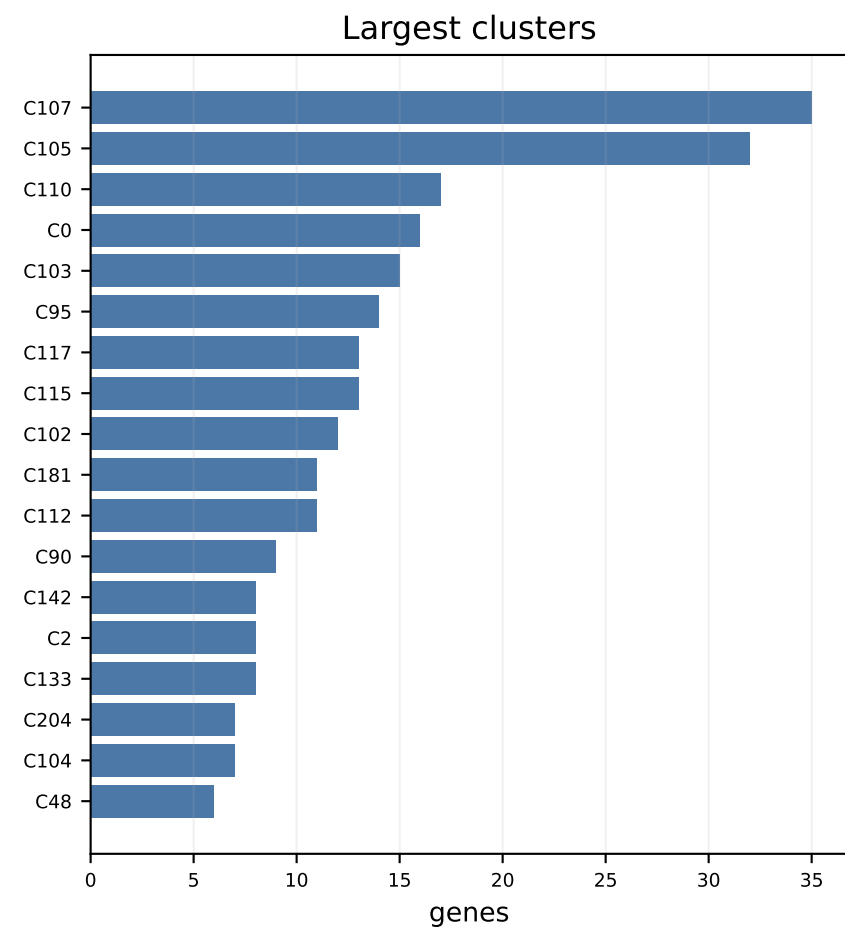
display rank: 16
raw G0-IC rank: 9
quality tier: quality_mixed

method family: raw_cosine_subspace
weighting: tfidf
block: adaptive_modes_02_05
modes: 2-5

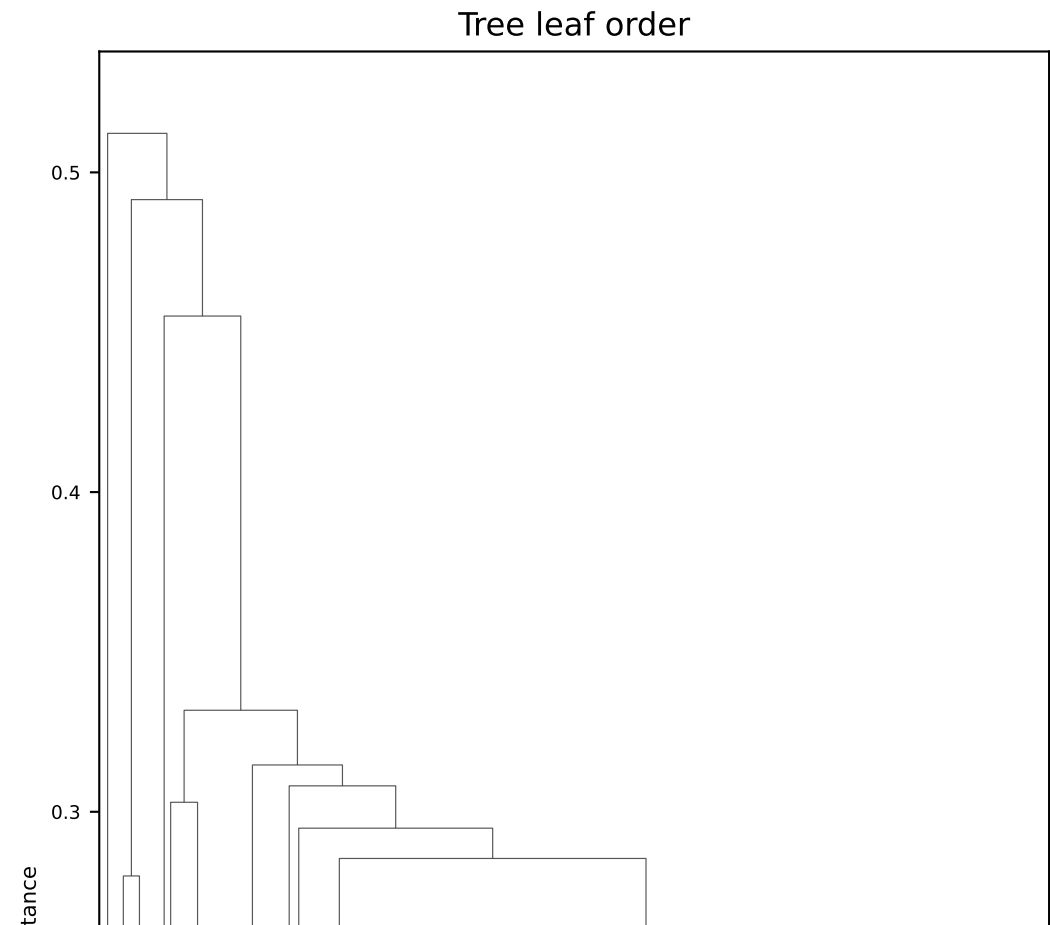
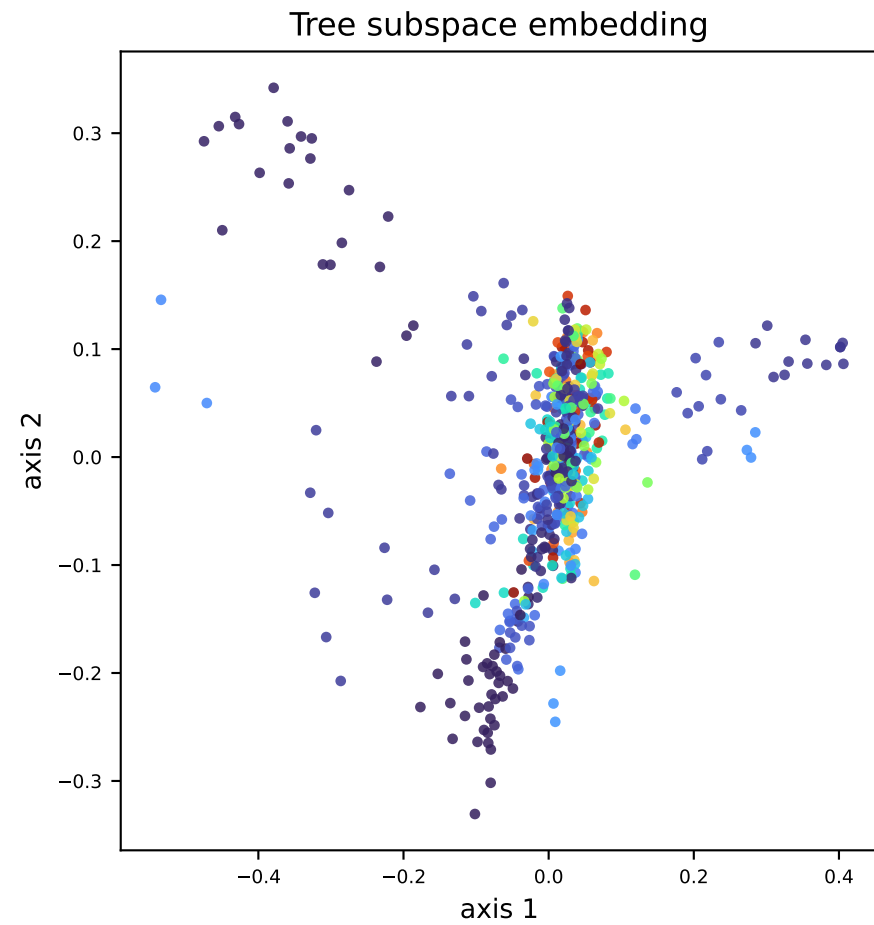
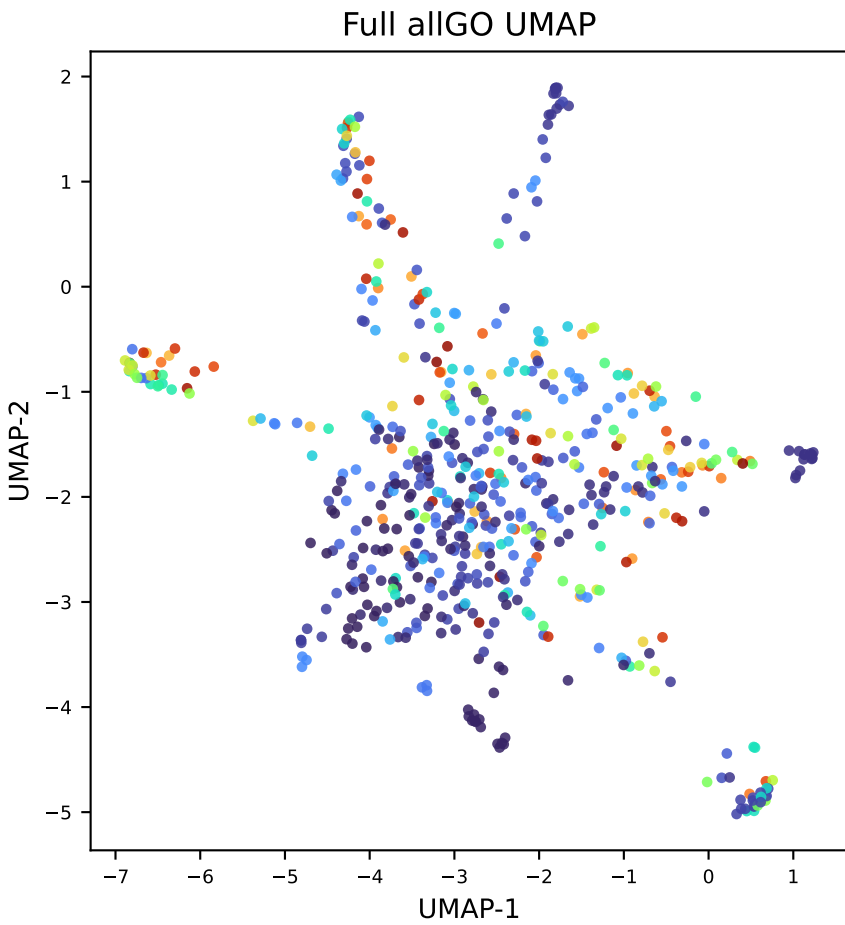
G0-BIC active: 502483.05
G0-BIC active/gene: 834.69
G0 active params: 23925
G0 logLik: -69898.42

clusters: 256
largest cluster: 35 (0.058)
singletons: 177 (0.691 of clusters)
median cluster size: 1.0

coherent clusters: 19 (0.074)
median significant G0 terms: 0.0
min cluster q: 4.03e-22
median within TF-IDF cosine: 0.1707
weighted within TF-IDF cosine: 0.1805
block_energy_fraction: 0.1819
segmentation_bic: -657.8000



Rank 17: raw cosine subspace, tfidf, modes 06 10



raw_cosine_subspace__tfidf__adaptive_modes_06_10

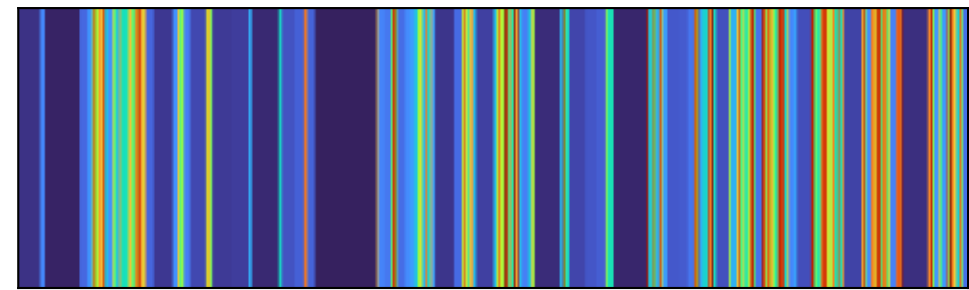
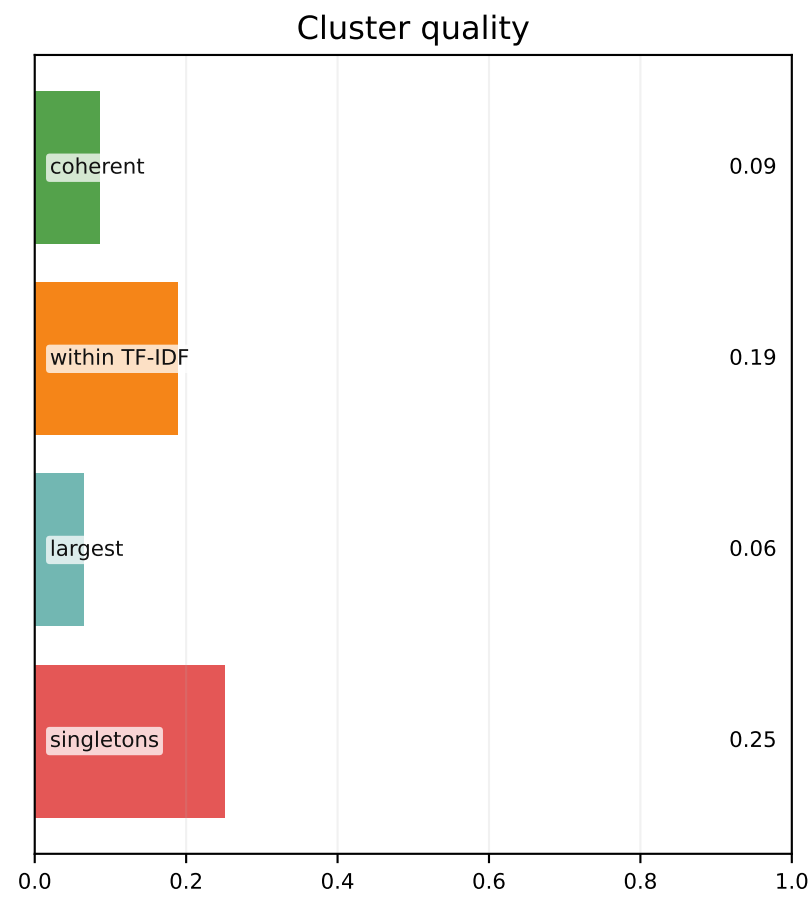
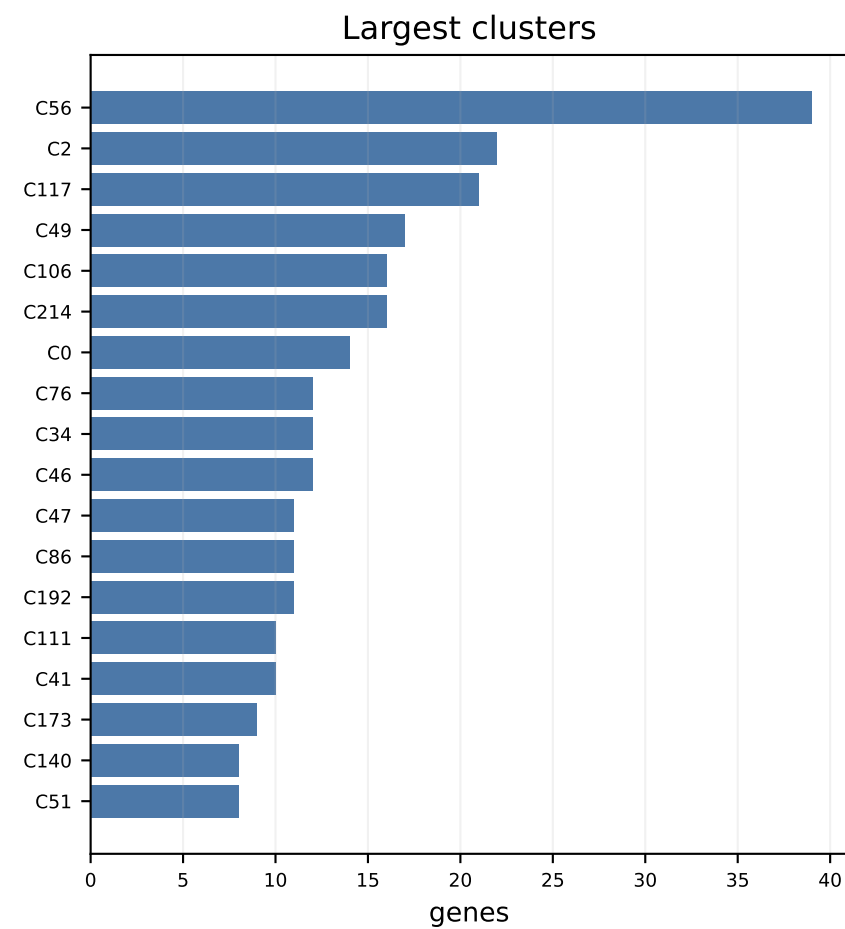
display rank: 17
raw G0-IC rank: 10
quality tier: quality_mixed

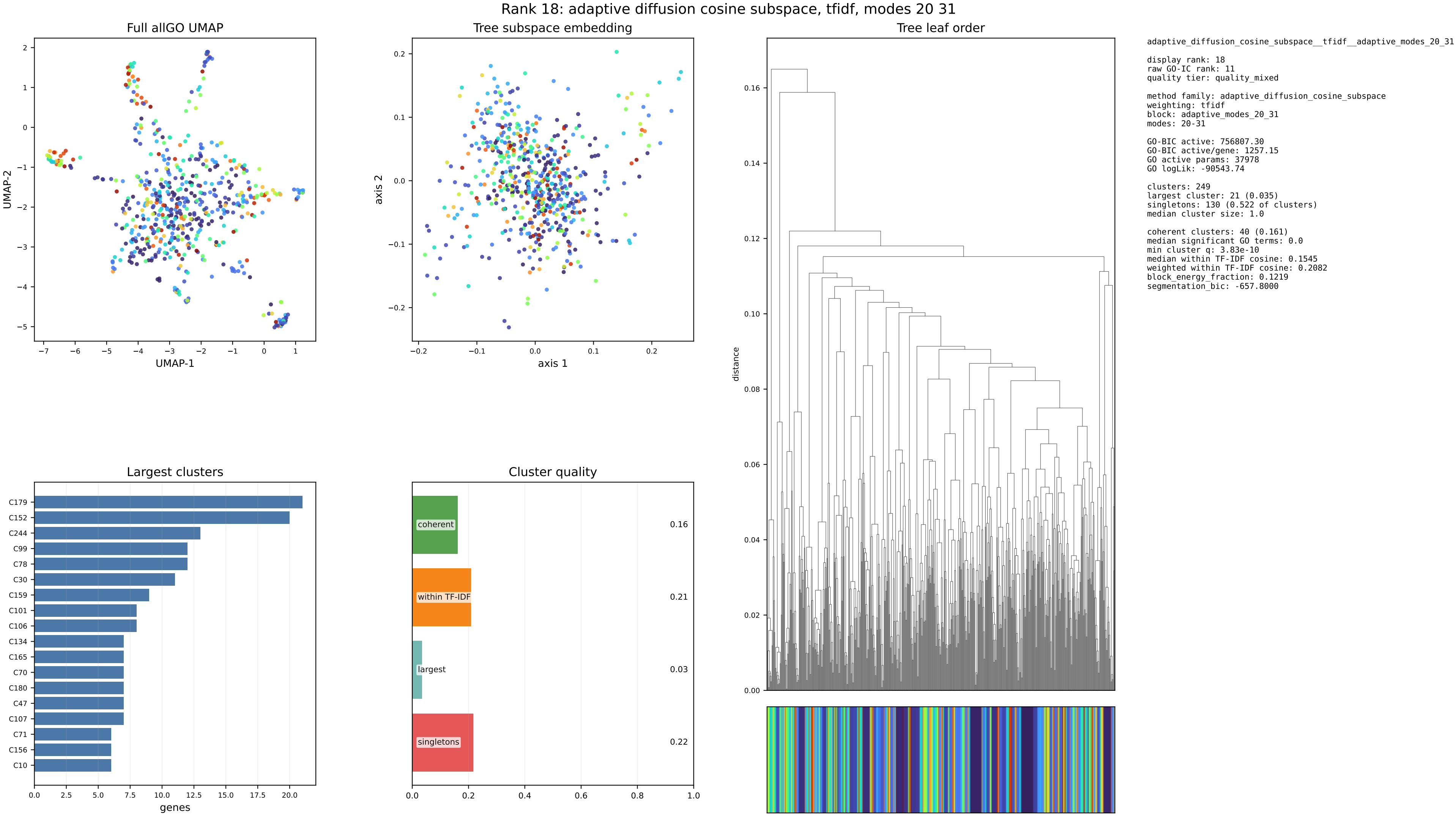
method family: raw_cosine_subspace
weighting: tfidf
block: adaptive_modes_06_10
modes: 6-10

G0-BIC active: 598985.89
G0-BIC active/gene: 994.99
G0 active params: 27543
G0 logLik: -90726.67

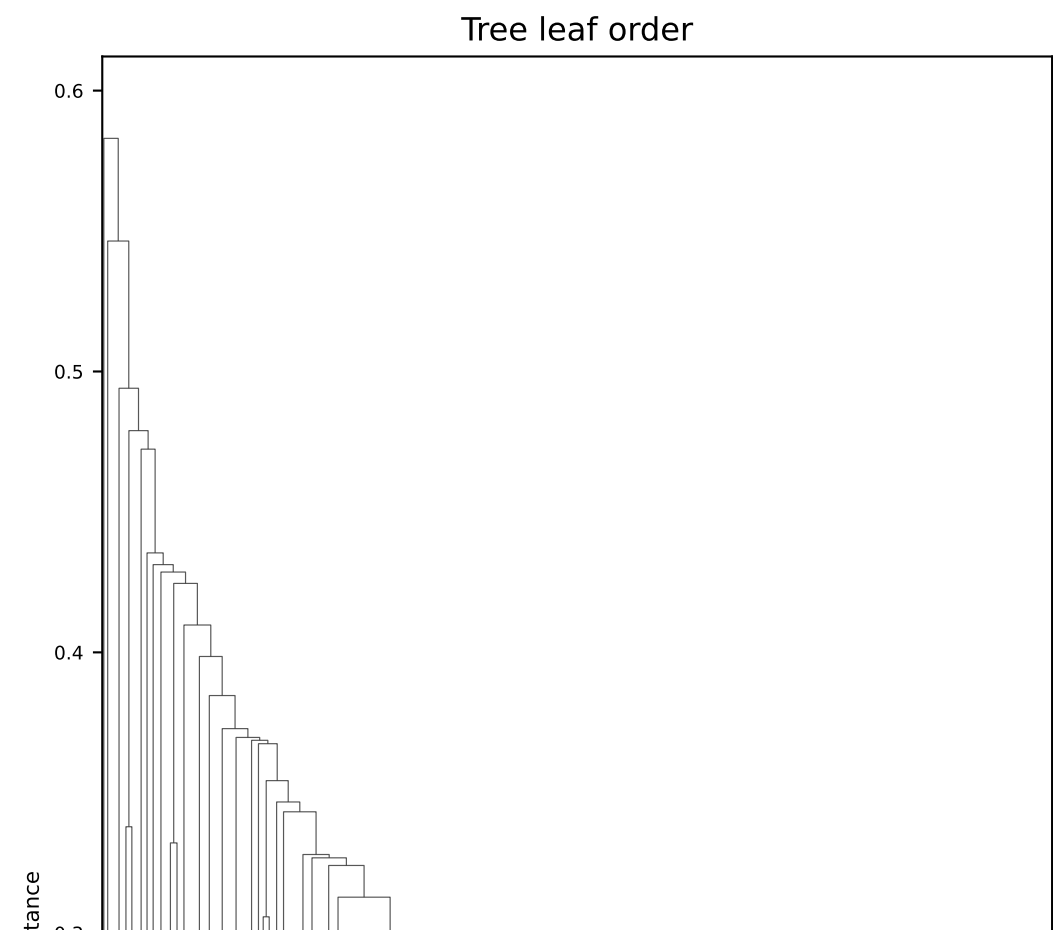
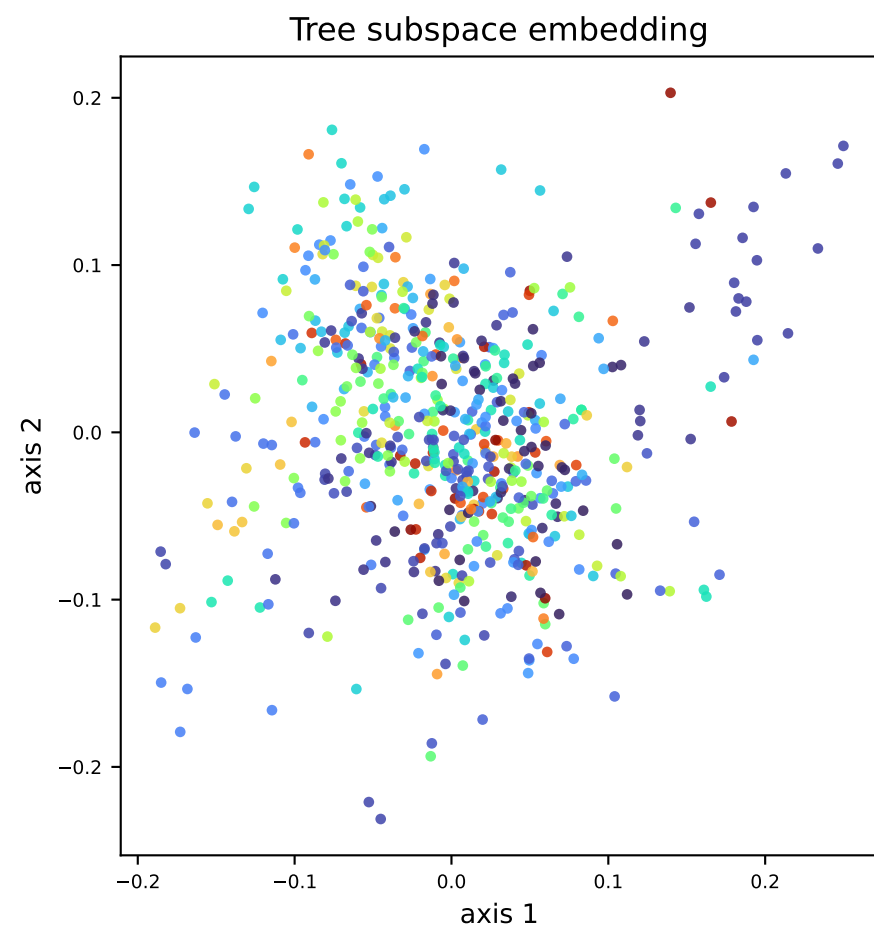
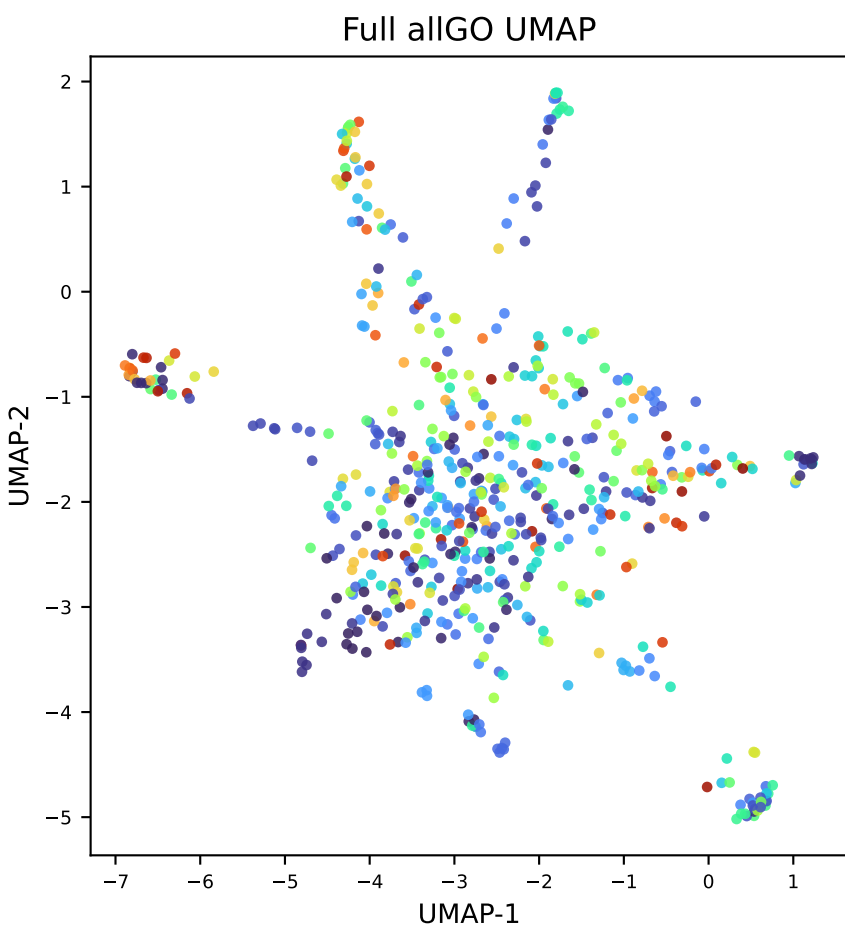
clusters: 233
largest cluster: 39 (0.065)
singletons: 151 (0.648 of clusters)
median cluster size: 1.0

coherent clusters: 20 (0.086)
median significant G0 terms: 0.0
min cluster q: 2.13e-25
median within TF-IDF cosine: 0.1213
weighted within TF-IDF cosine: 0.1888
block_energy_fraction: 0.1128
segmentation_bic: -657.8000





Rank 19: raw cosine subspace, tfidf, modes 20 31



raw_cosine_subspace_tfidf_adaptive_modes_20_31

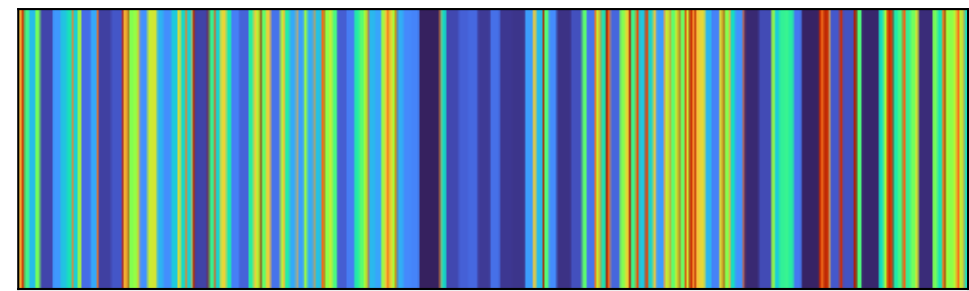
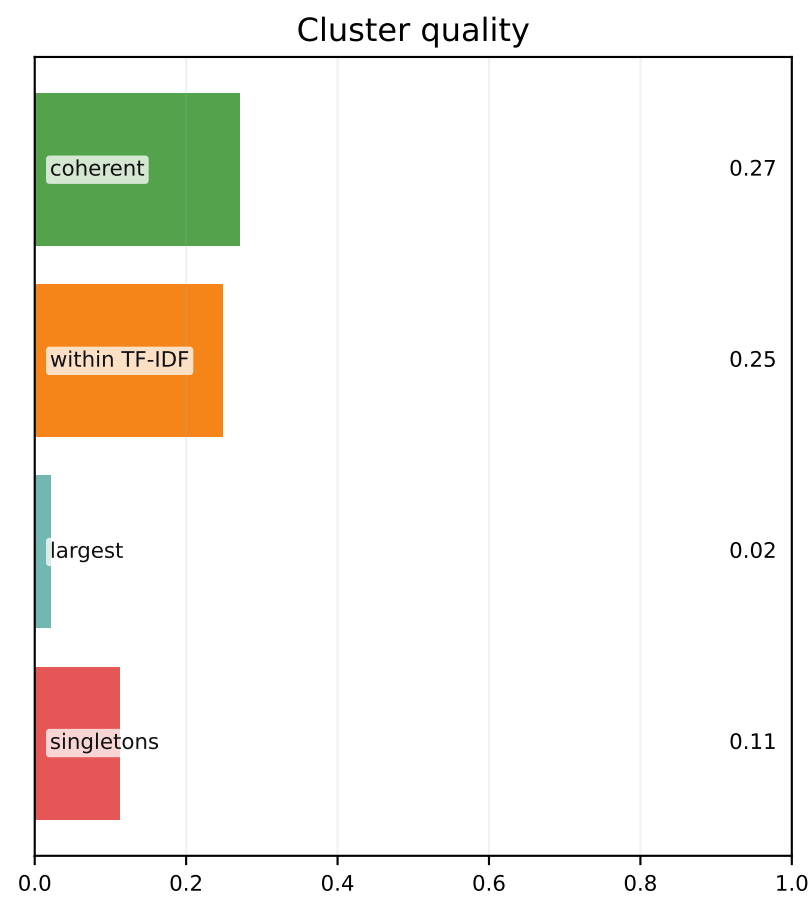
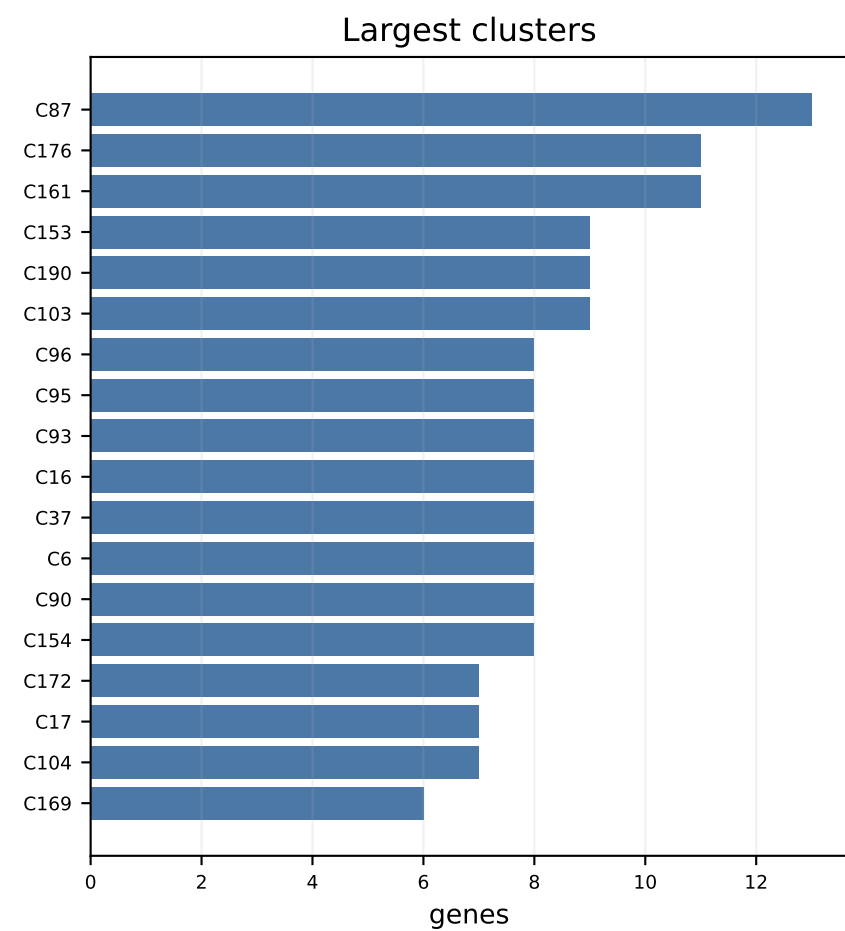
display rank: 19
raw G0-IC rank: 13
quality tier: quality_mixed

method family: raw_cosine_subspace
weighting: tfidf
block: adaptive_modes_20_31
modes: 20-31

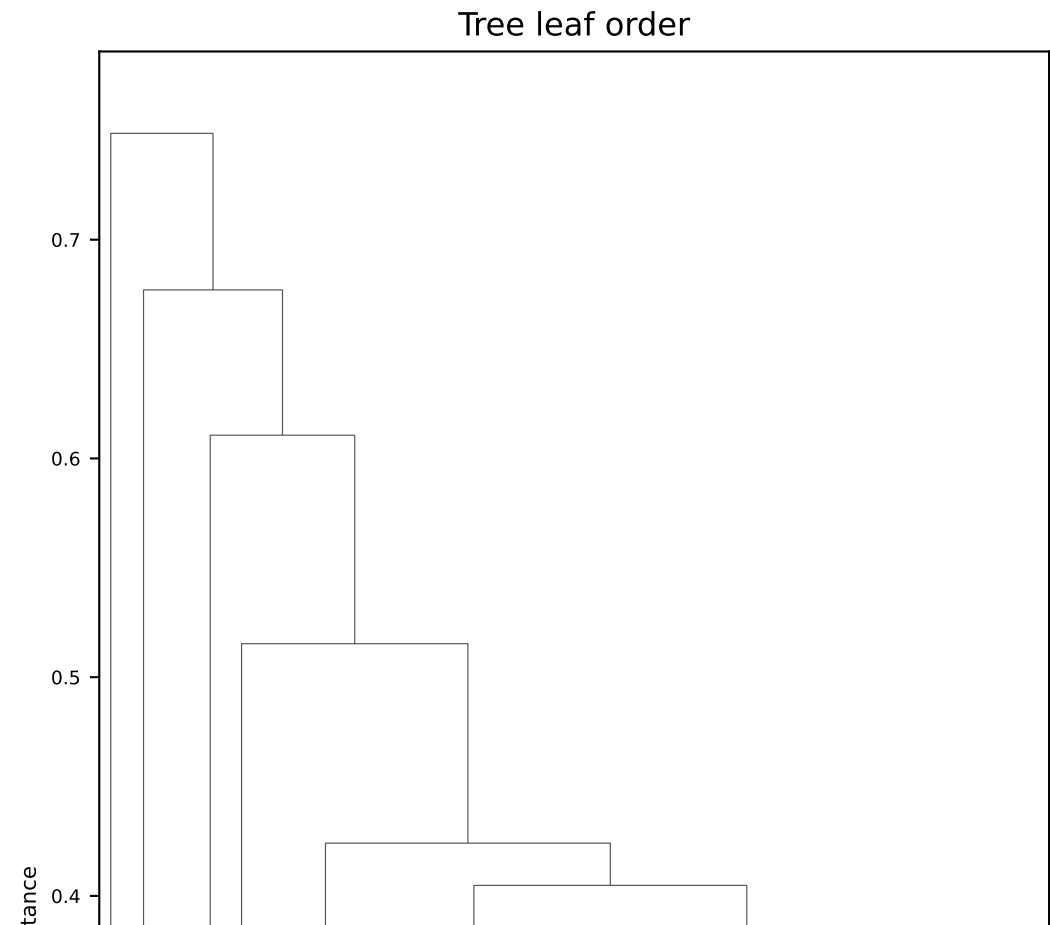
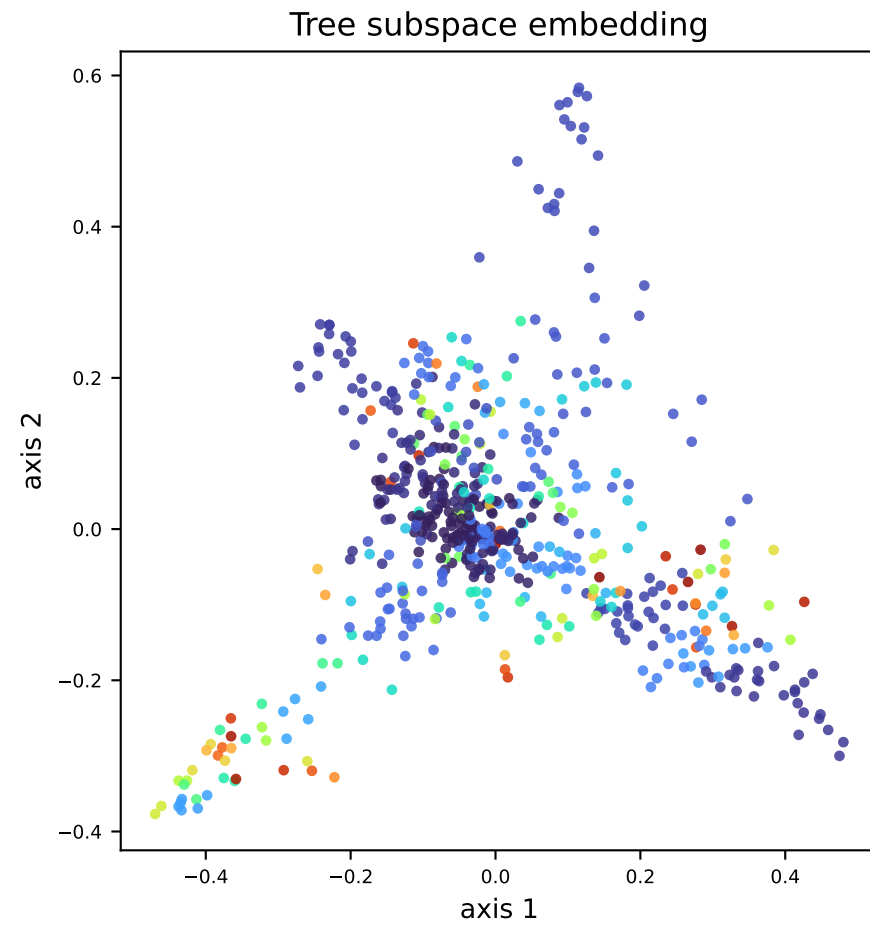
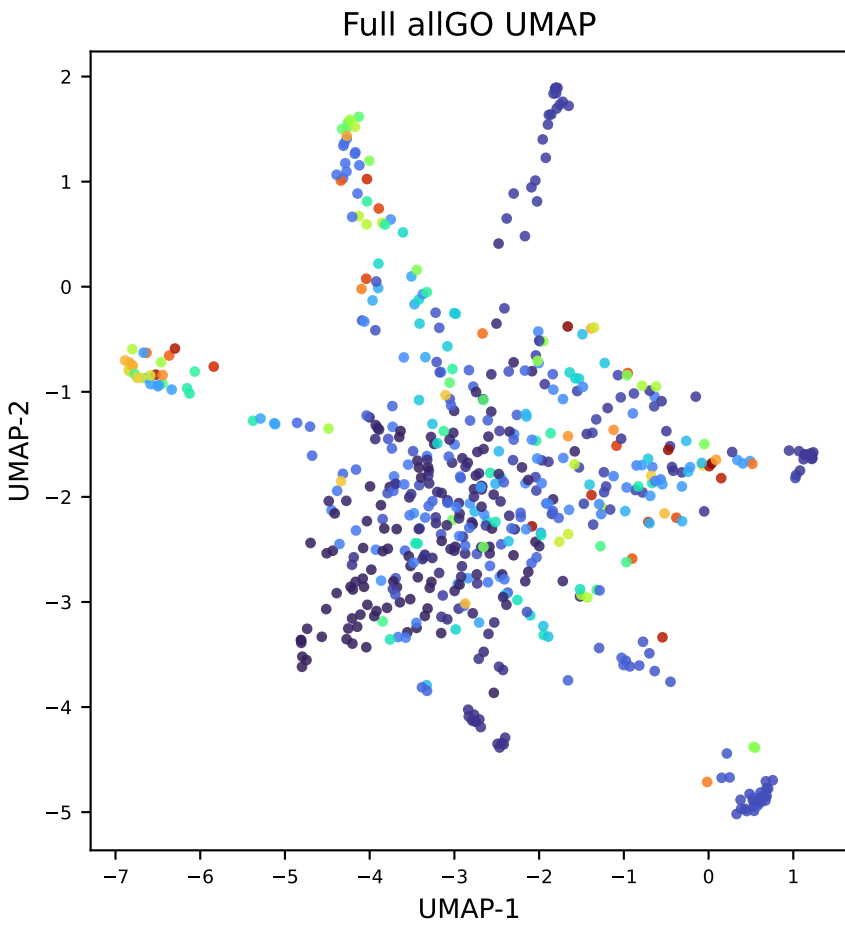
G0-BIC active: 871016.69
G0-BIC active/gene: 1446.87
G0 active params: 43779
G0 logLik: -103678.89

clusters: 203
largest cluster: 13 (0.022)
singletons: 68 (0.335 of clusters)
median cluster size: 2.0

coherent clusters: 55 (0.271)
median significant G0 terms: 0.0
min cluster q: 7.72e-12
median within TF-IDF cosine: 0.2268
weighted within TF-IDF cosine: 0.2480
block_energy_fraction: 0.1219
segmentation_bic: -657.8000



Rank 20: raw cosine subspace, binary, modes 02 06



raw_cosine_subspace__binary__adaptive_modes_02_06

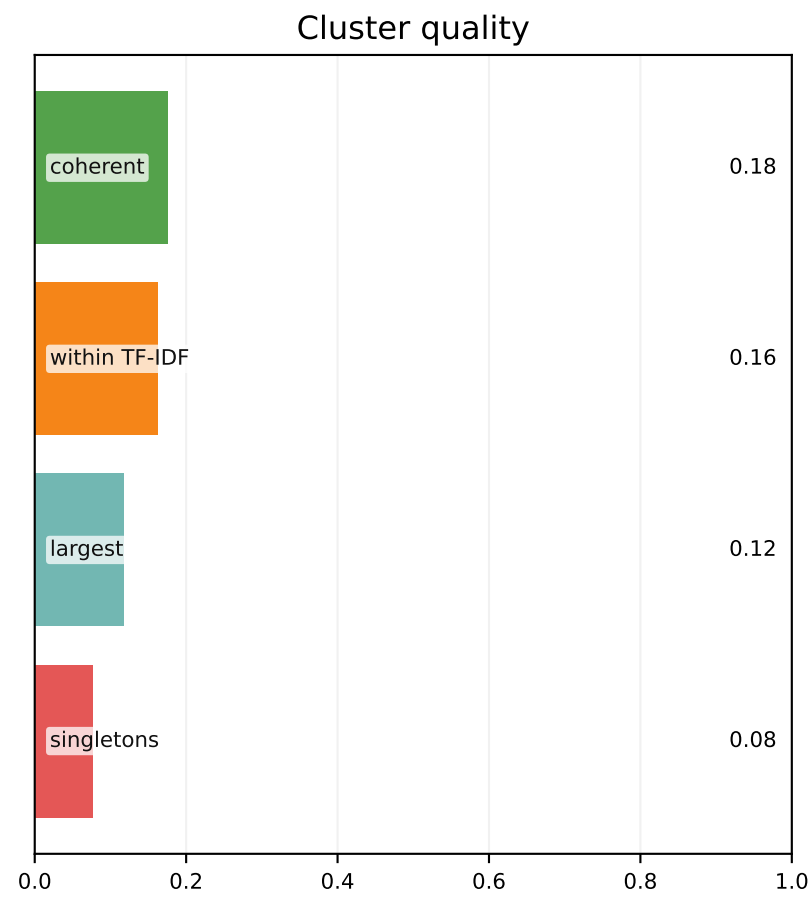
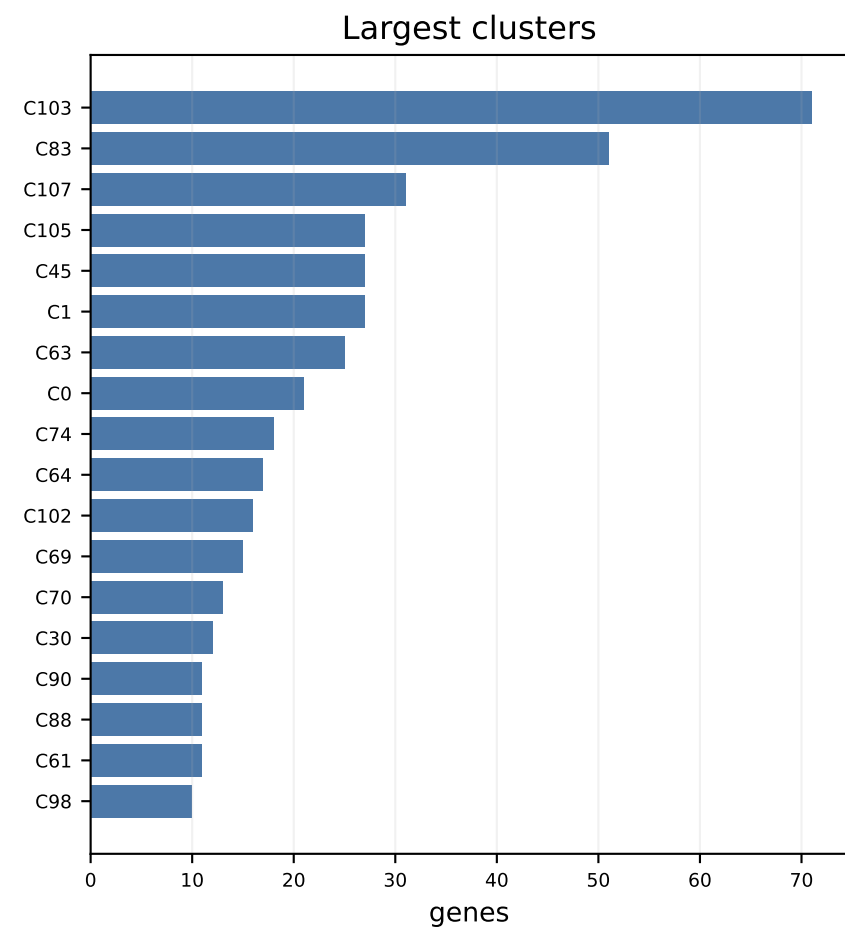
display rank: 20
raw G0-IC rank: 16
quality tier: quality_mixed

method family: raw_cosine_subspace
weighting: binary
block: adaptive_modes_02_06
modes: 2-6

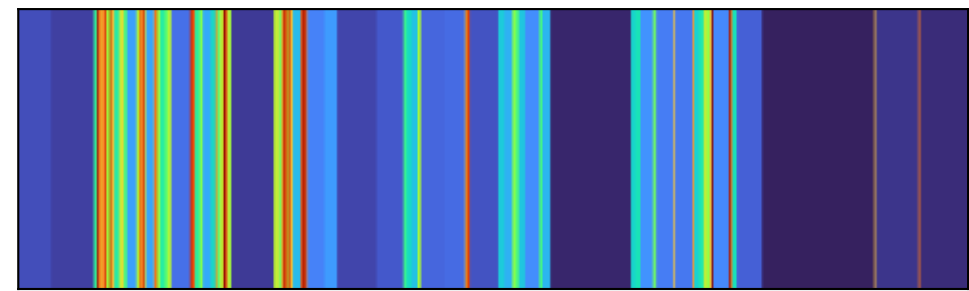
G0-BIC active: 888801.66
G0-BIC active/gene: 1476.41
G0 active params: 37578
G0 logLik: -159572.77

clusters: 108
largest cluster: 71 (0.118)
singletons: 46 (0.426 of clusters)
median cluster size: 2.0

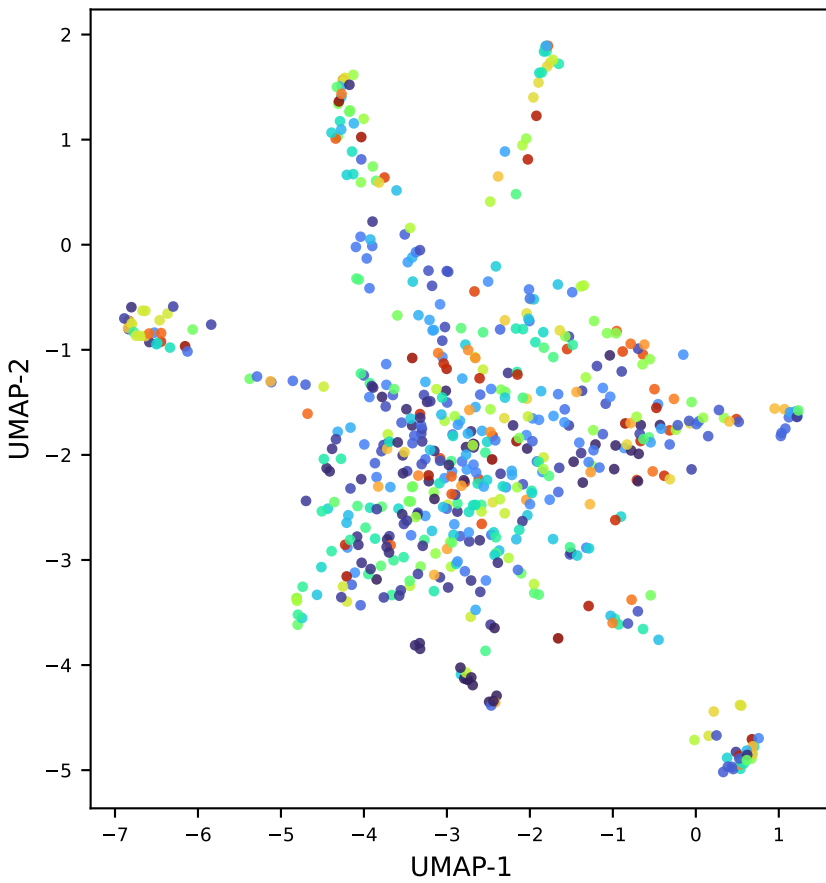
coherent clusters: 19 (0.176)
median significant G0 terms: 0.0
min cluster q: 8.51e-37
median within TF-IDF cosine: 0.2272
weighted within TF-IDF cosine: 0.1622
block_energy_fraction: 0.2318
segmentation_bic: -650.2908



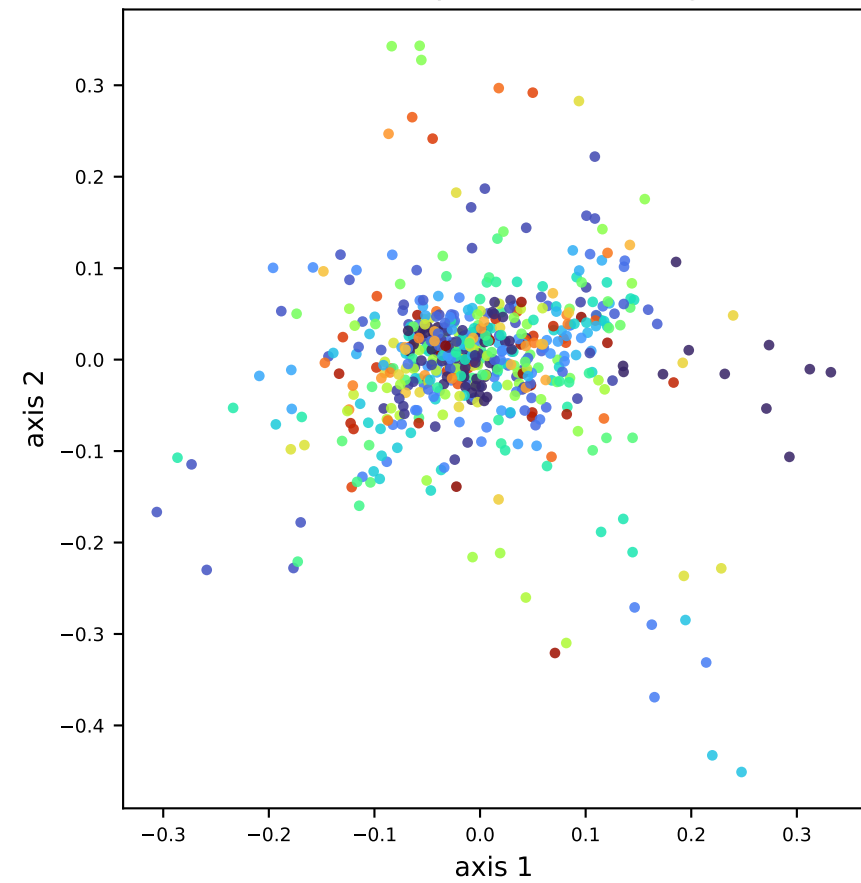
distance



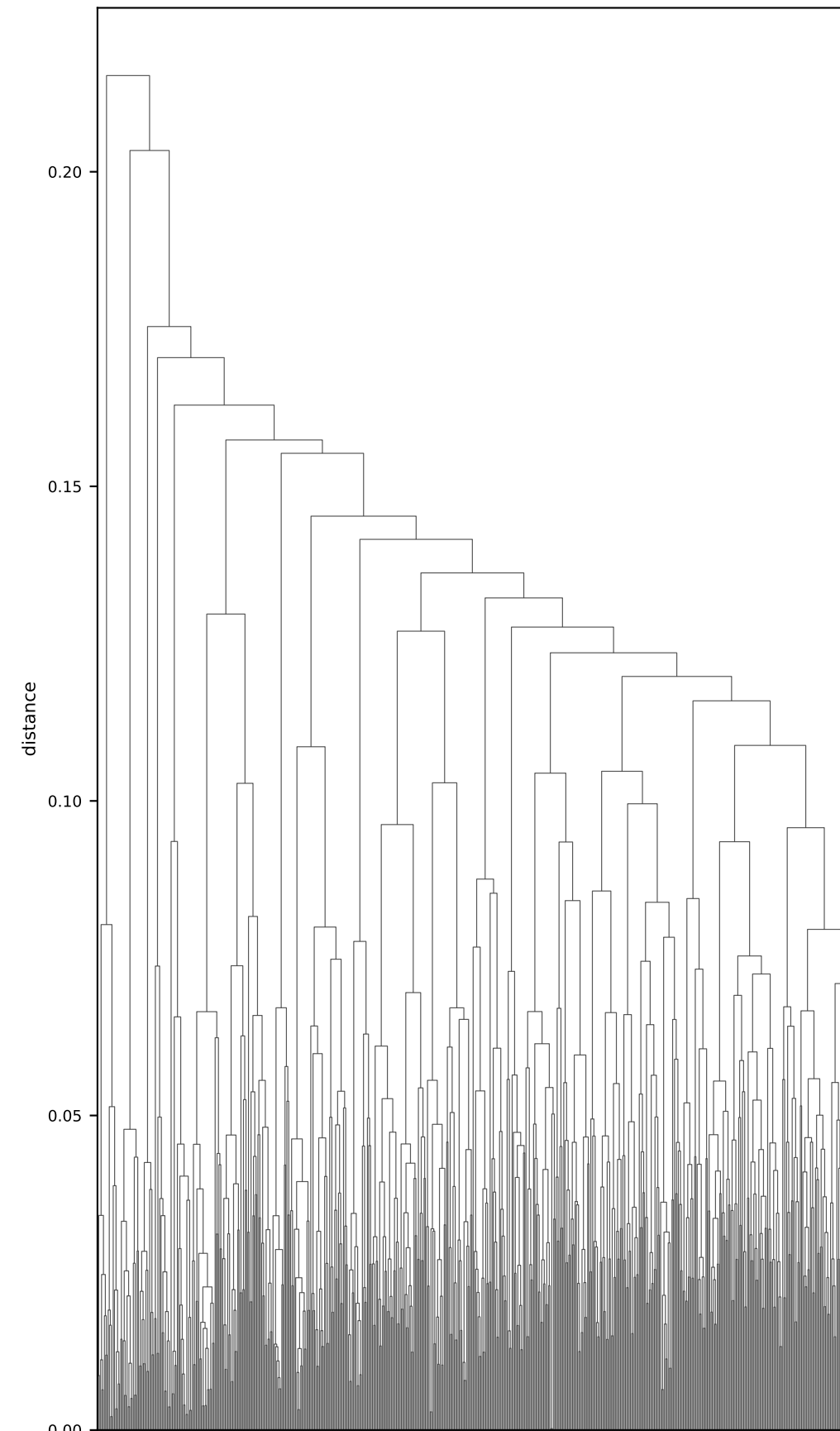
Full allGO UMAP



Tree subspace embedding



Tree leaf order



adaptive_diffusion_cosine_subspace_tfidf_adaptive_modes_11_15

```
display rank: 21
raw G0-IC rank: 19
quality tier: quality_mixed
```

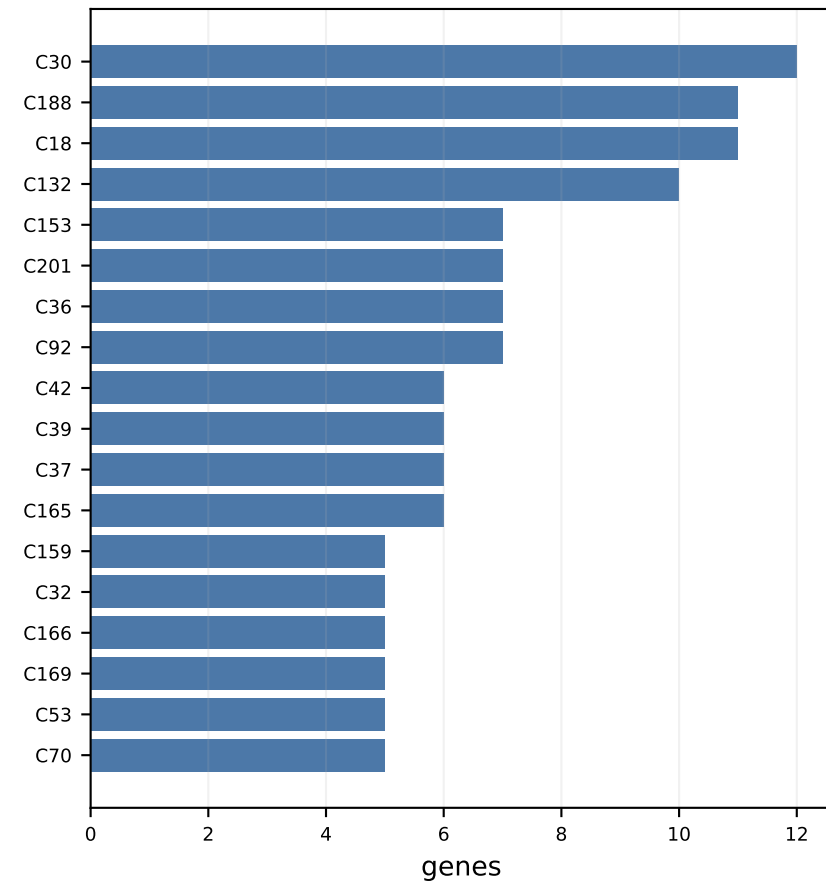
```
method family: adaptive_diffusion_cosine_subspace
weighting: tfidf
block: adaptive_modes_11_15
modes: 11-15
```

GO-BIC active: 1018637.38
GO-BIC active/gene: 1692.09
GO active params: 52353
GO logLik: -112501.32

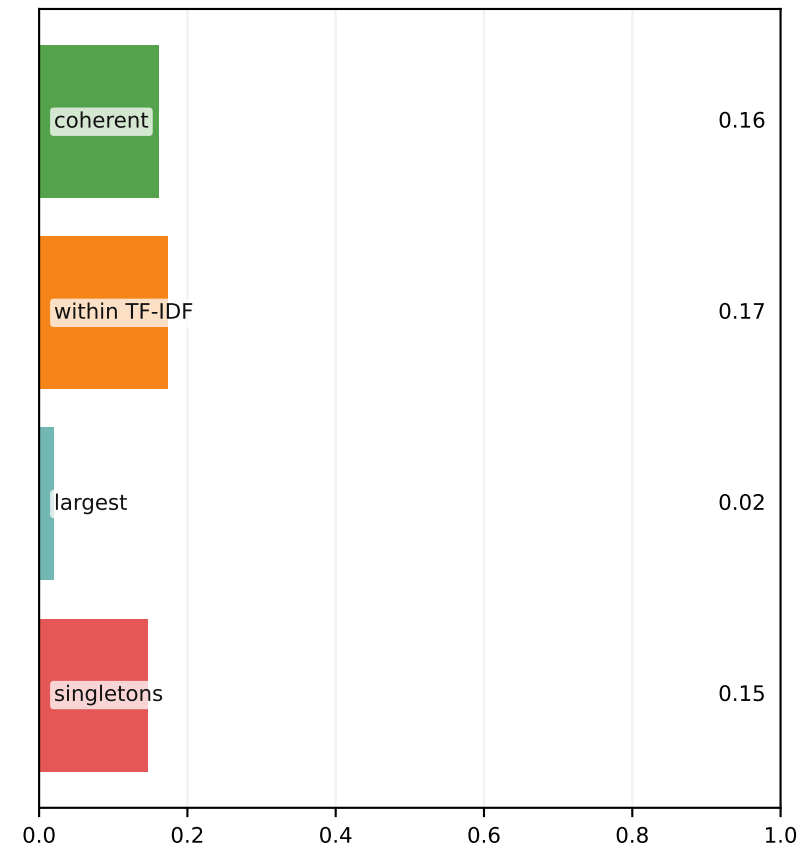
```
clusters: 236
largest cluster: 12 (0.020)
singletons: 88 (0.373 of clusters)
median cluster size: 2.0
```

```
coherent clusters: 38 (0.161)
median significant G0 terms: 0.0
min cluster q: 1.03e-12
median within TF-IDF cosine: 0.1361
weighted within TF-IDF cosine: 0.1738
block_energy_fraction: 0.0737
segmentation bic: -657.8000
```

Largest clusters



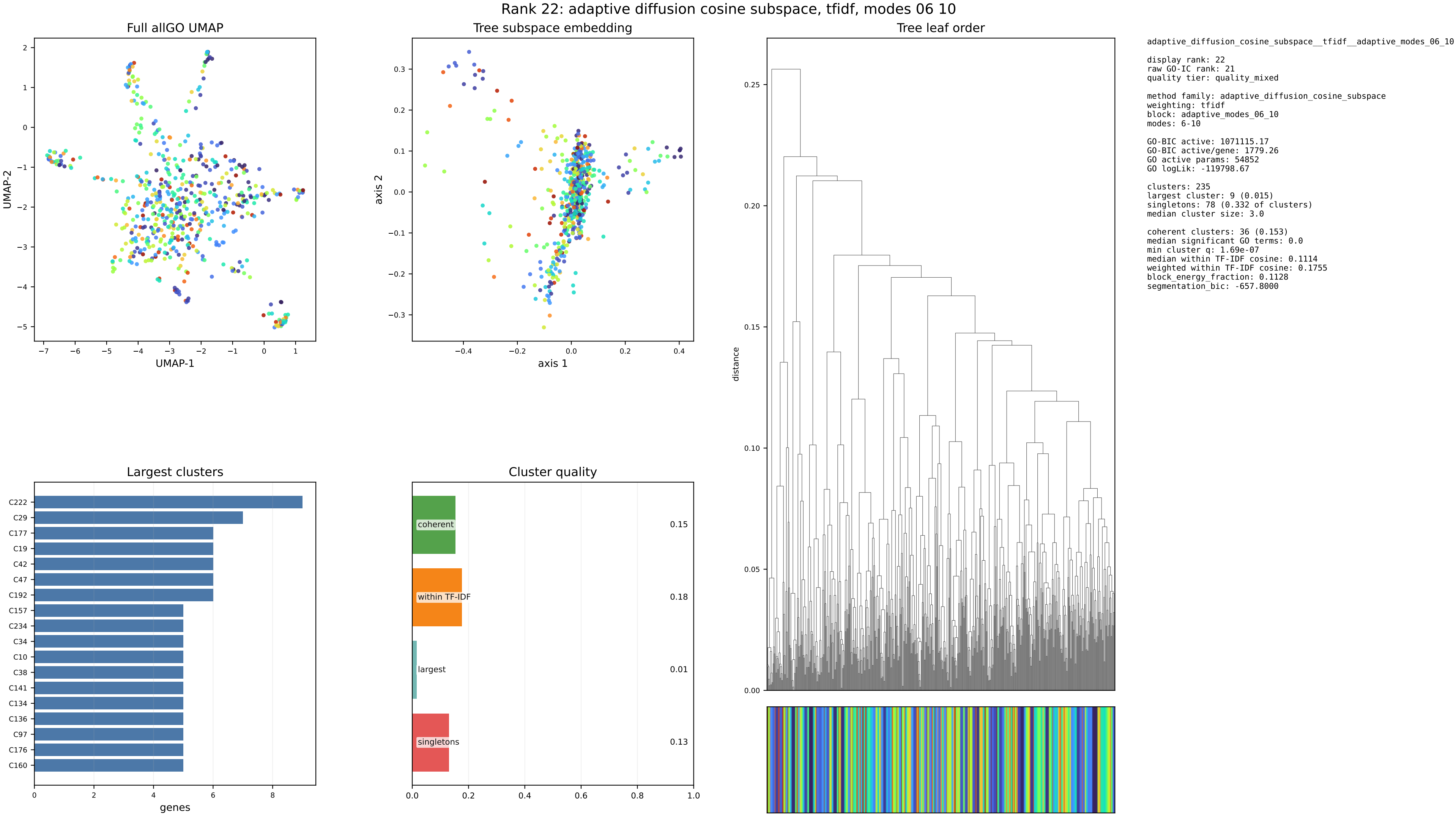
Cluster quality



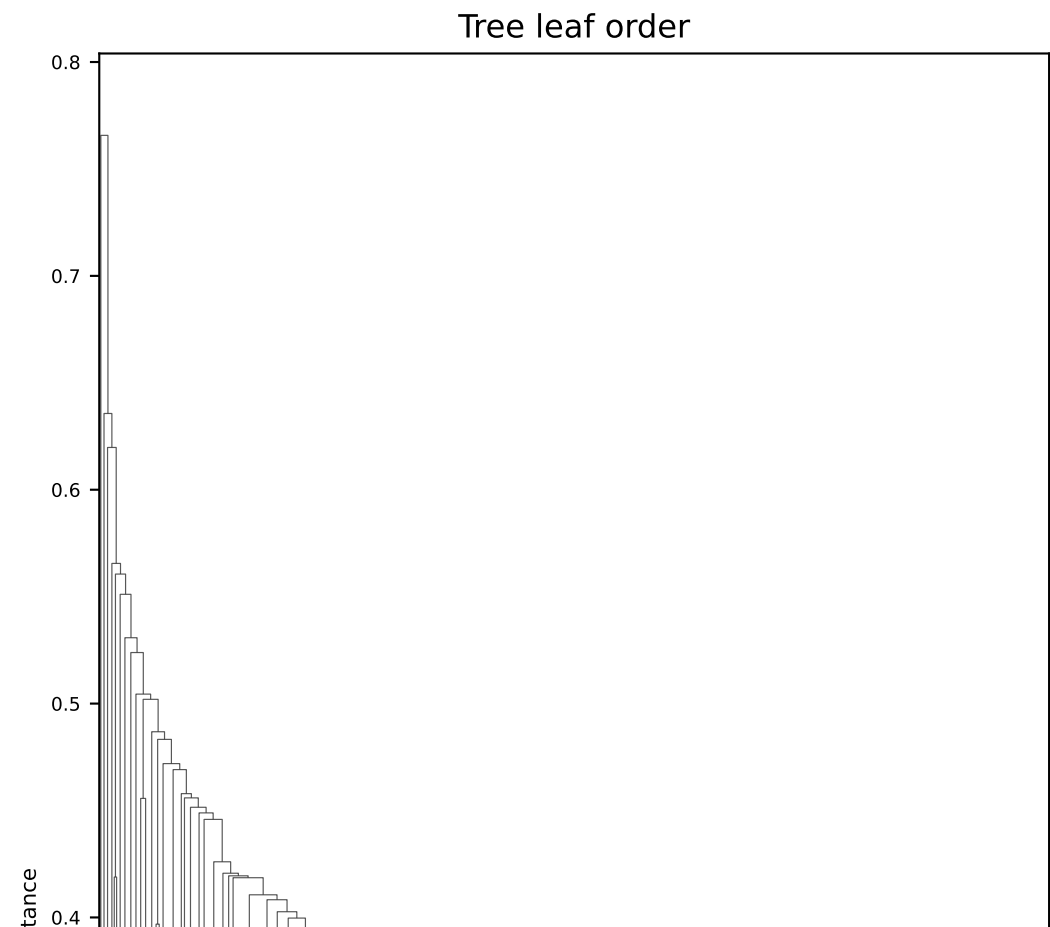
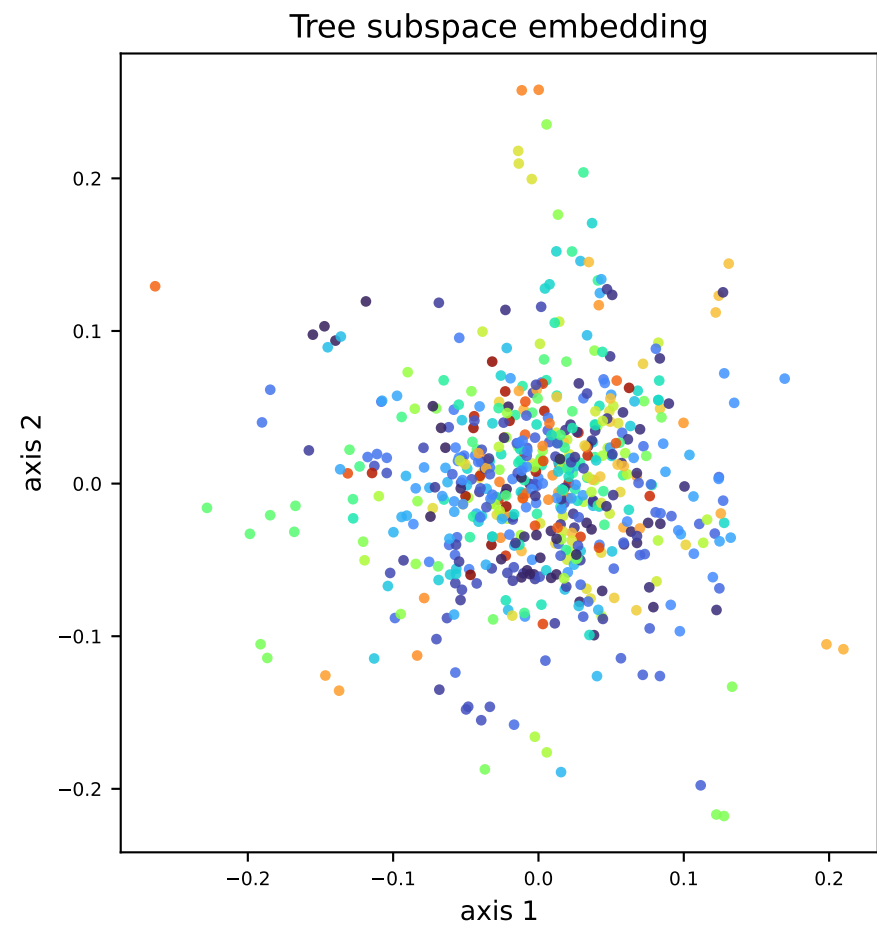
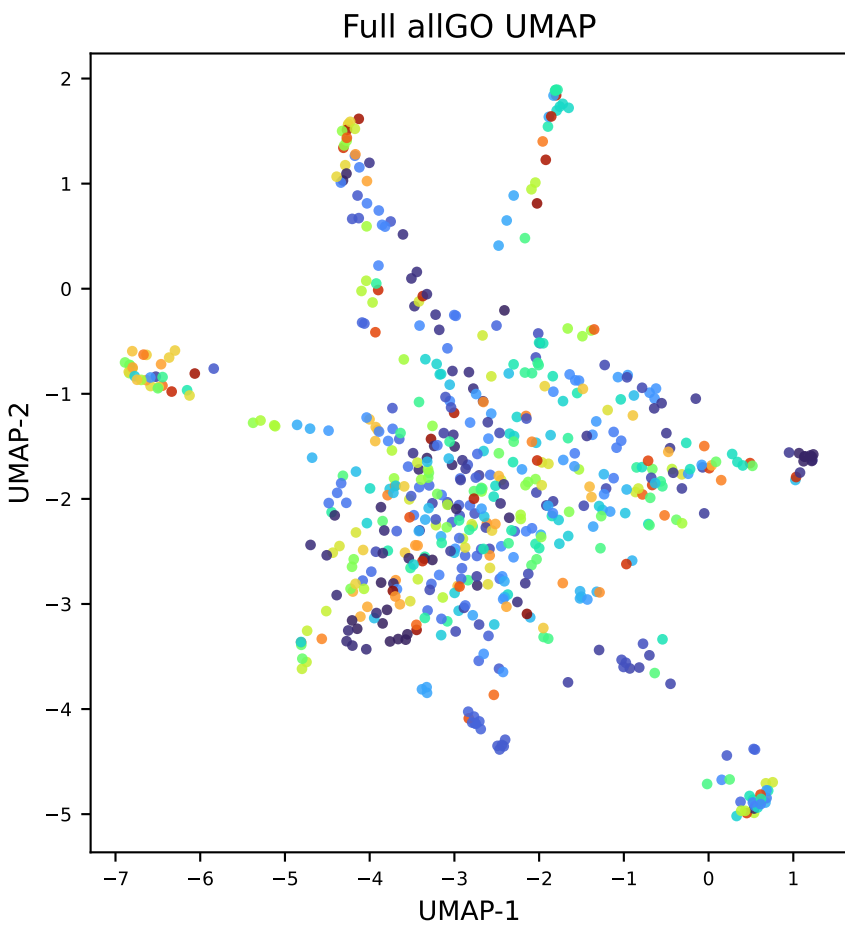
within TF-IDF

largest

singleton



Rank 23: raw cosine subspace, tfidf, modes 32 52



raw_cosine_subspace__tfidf__adaptive_modes_32_52

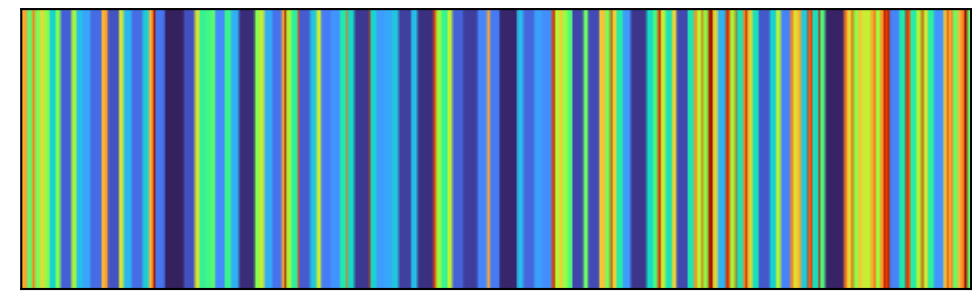
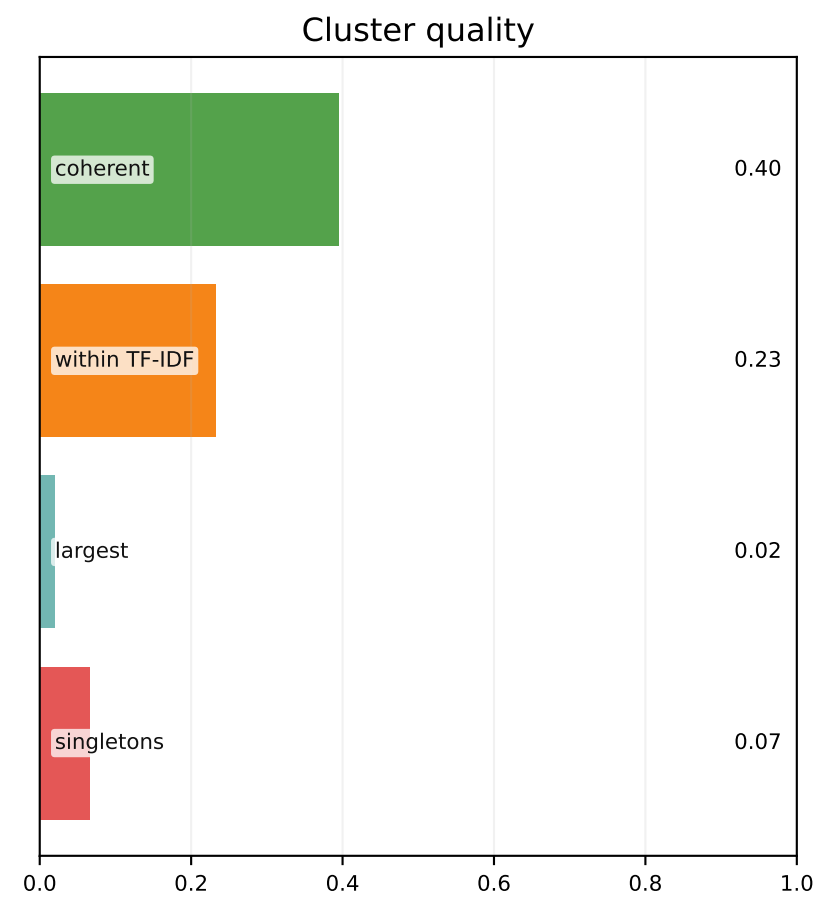
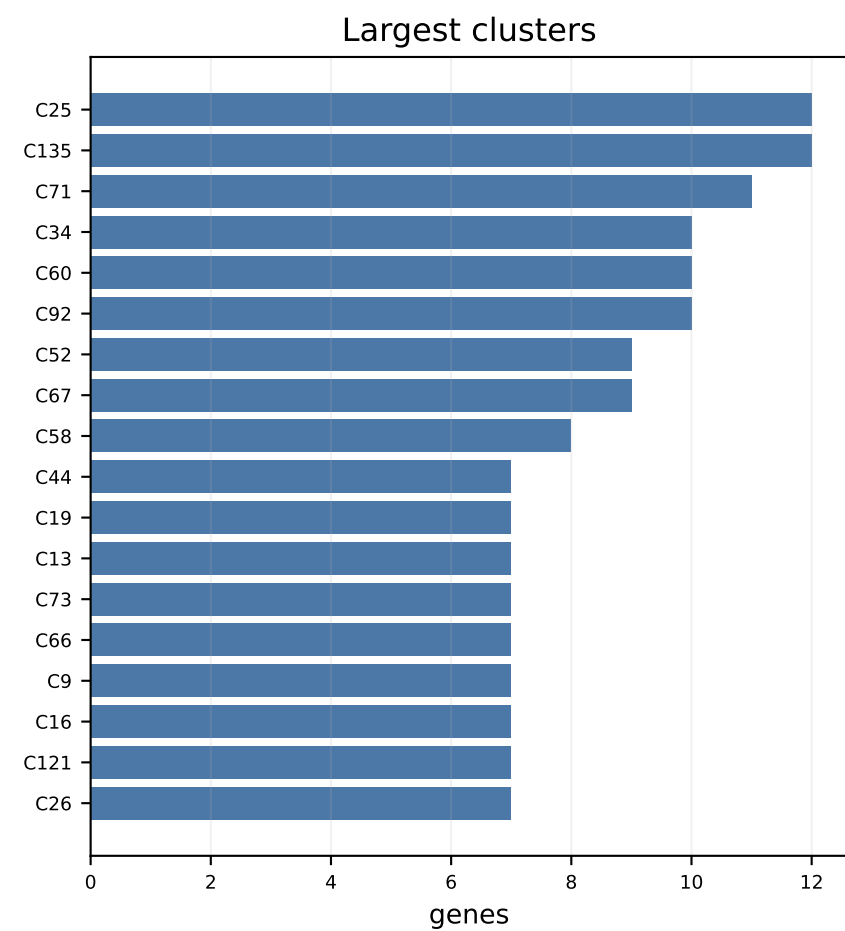
display rank: 23
raw G0-IC rank: 22
quality tier: quality_mixed

method family: raw_cosine_subspace
weighting: tfidf
block: adaptive_modes_32_52
modes: 32-52

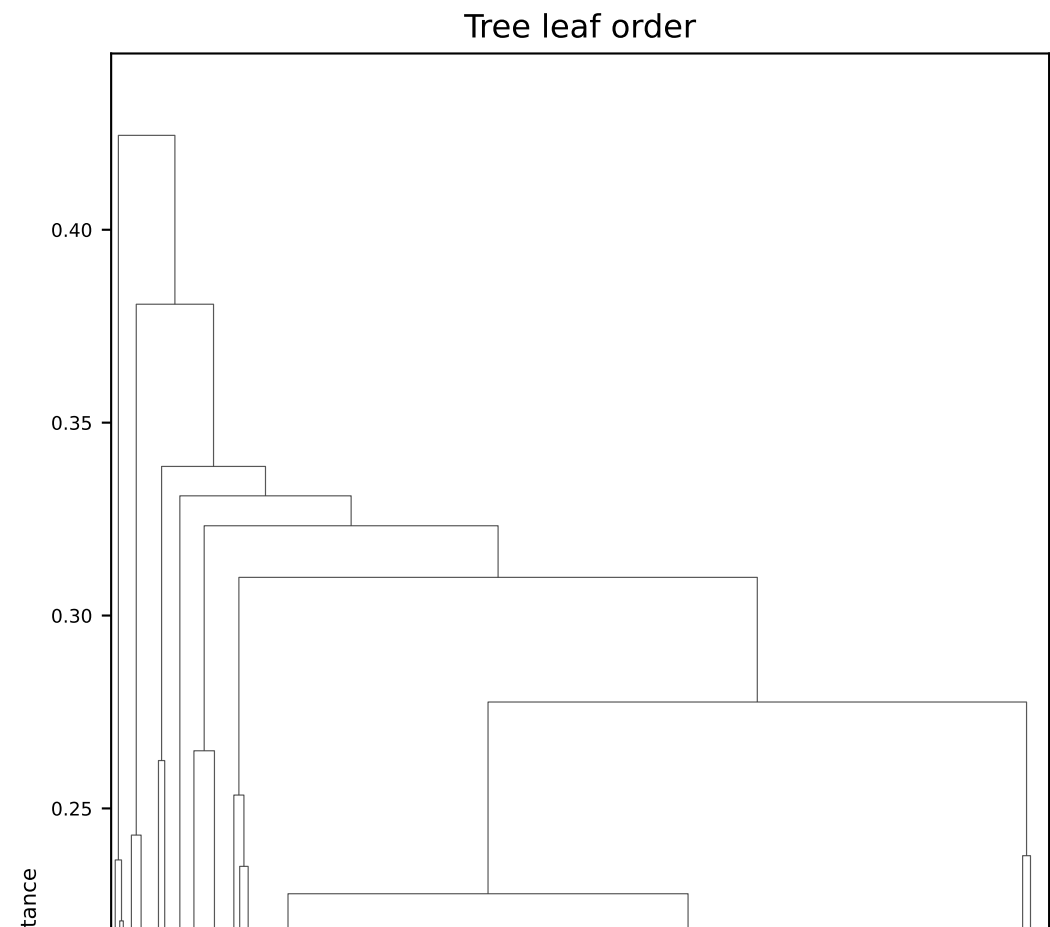
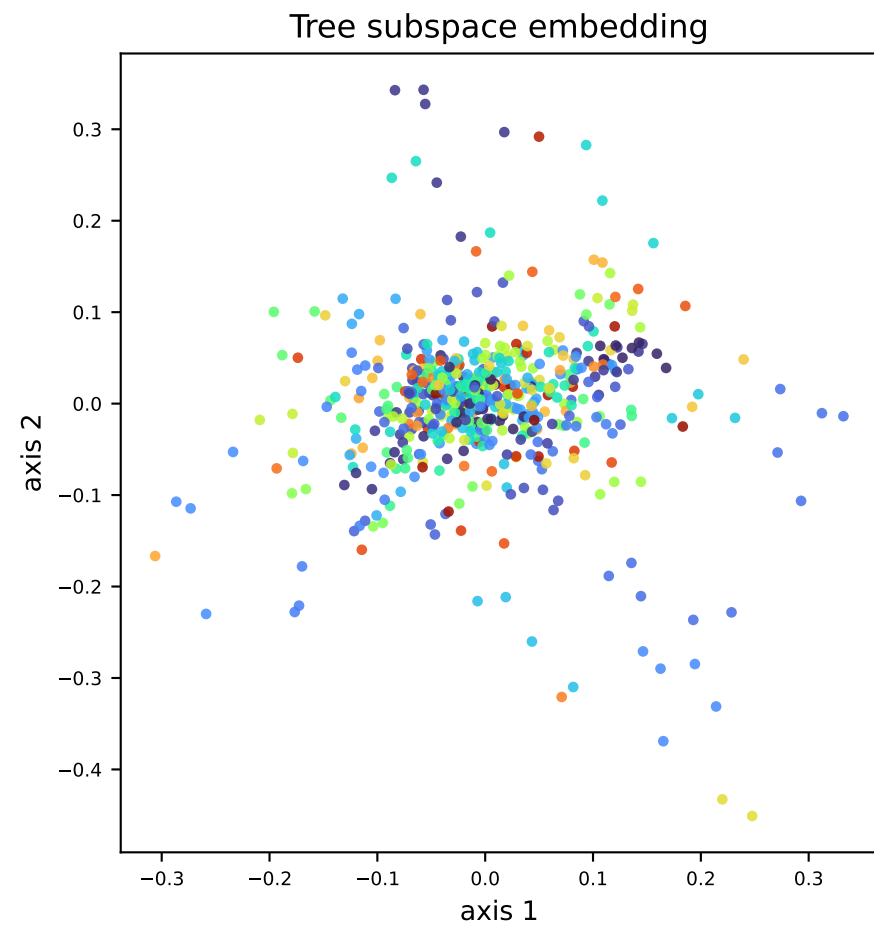
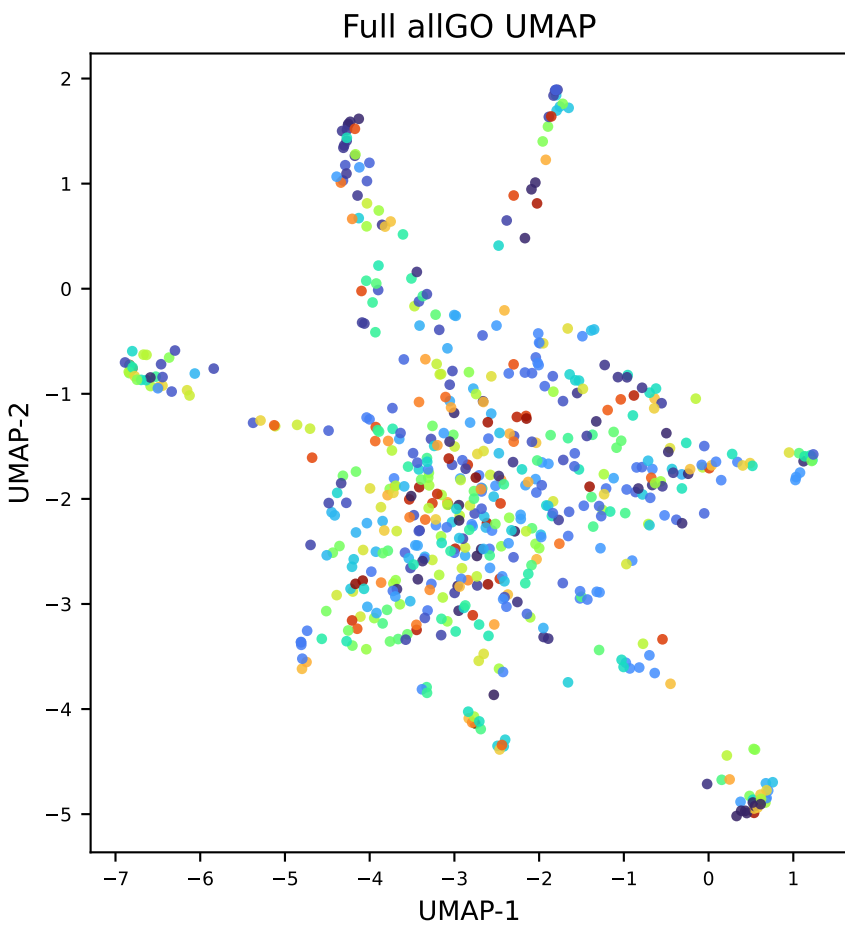
G0-BIC active: 1074047.89
G0-BIC active/gene: 1784.13
G0 active params: 52915
G0 logLik: -135946.81

clusters: 172
largest cluster: 12 (0.020)
singletons: 40 (0.233 of clusters)
median cluster size: 3.0

coherent clusters: 68 (0.395)
median significant G0 terms: 0.0
min cluster q: 1.40e-13
median within TF-IDF cosine: 0.1881
weighted within TF-IDF cosine: 0.2331
block_energy_fraction: 0.1670
segmentation_bic: -657.8000



Rank 24: raw cosine subspace, tfidf, modes 11 15



raw_cosine_subspace__tfidf__adaptive_modes_11_15

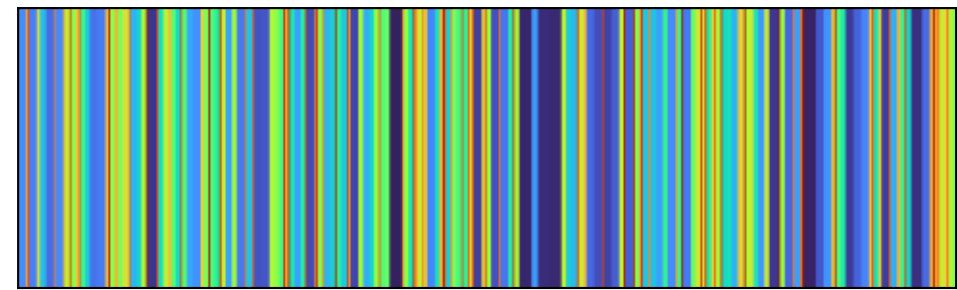
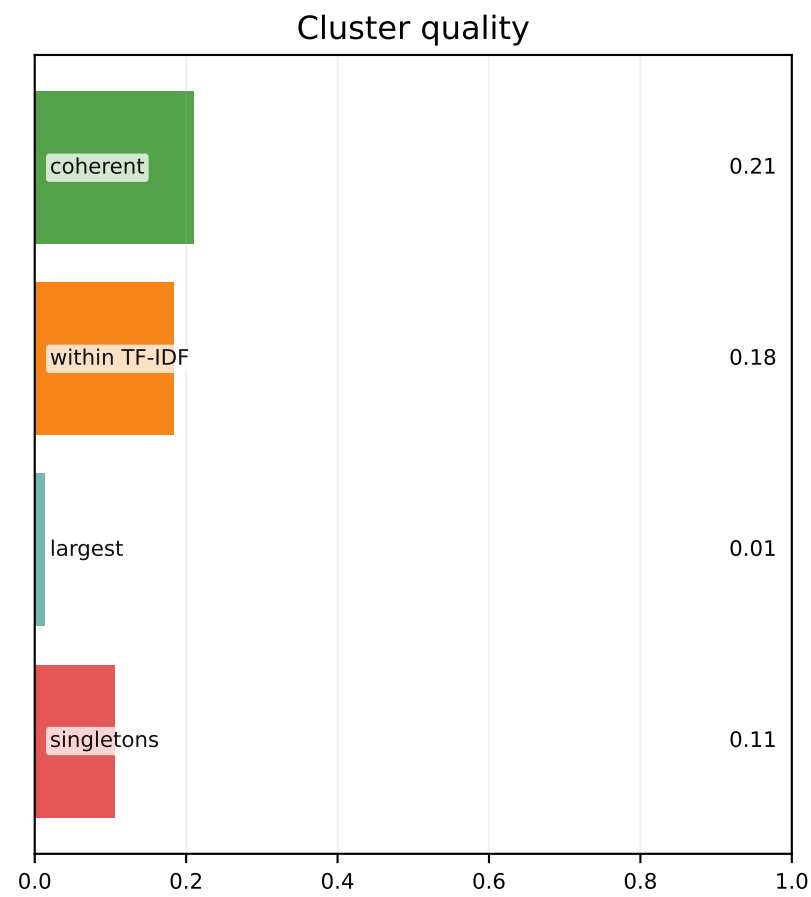
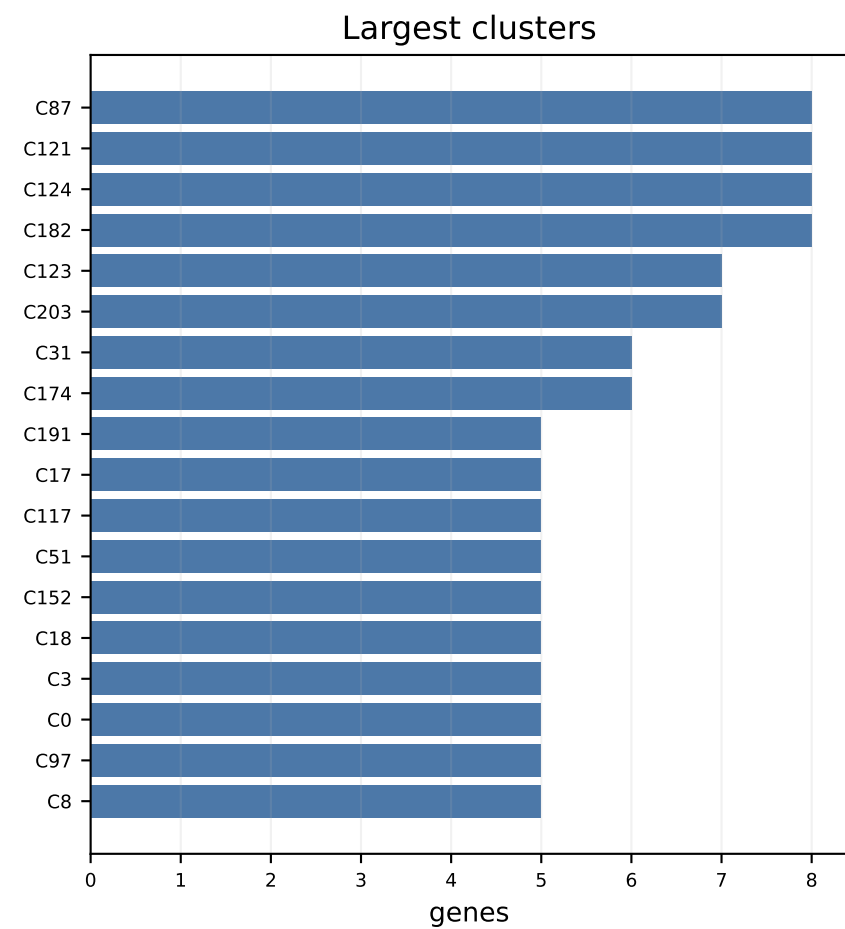
display rank: 24
raw G0-IC rank: 27
quality tier: quality_mixed

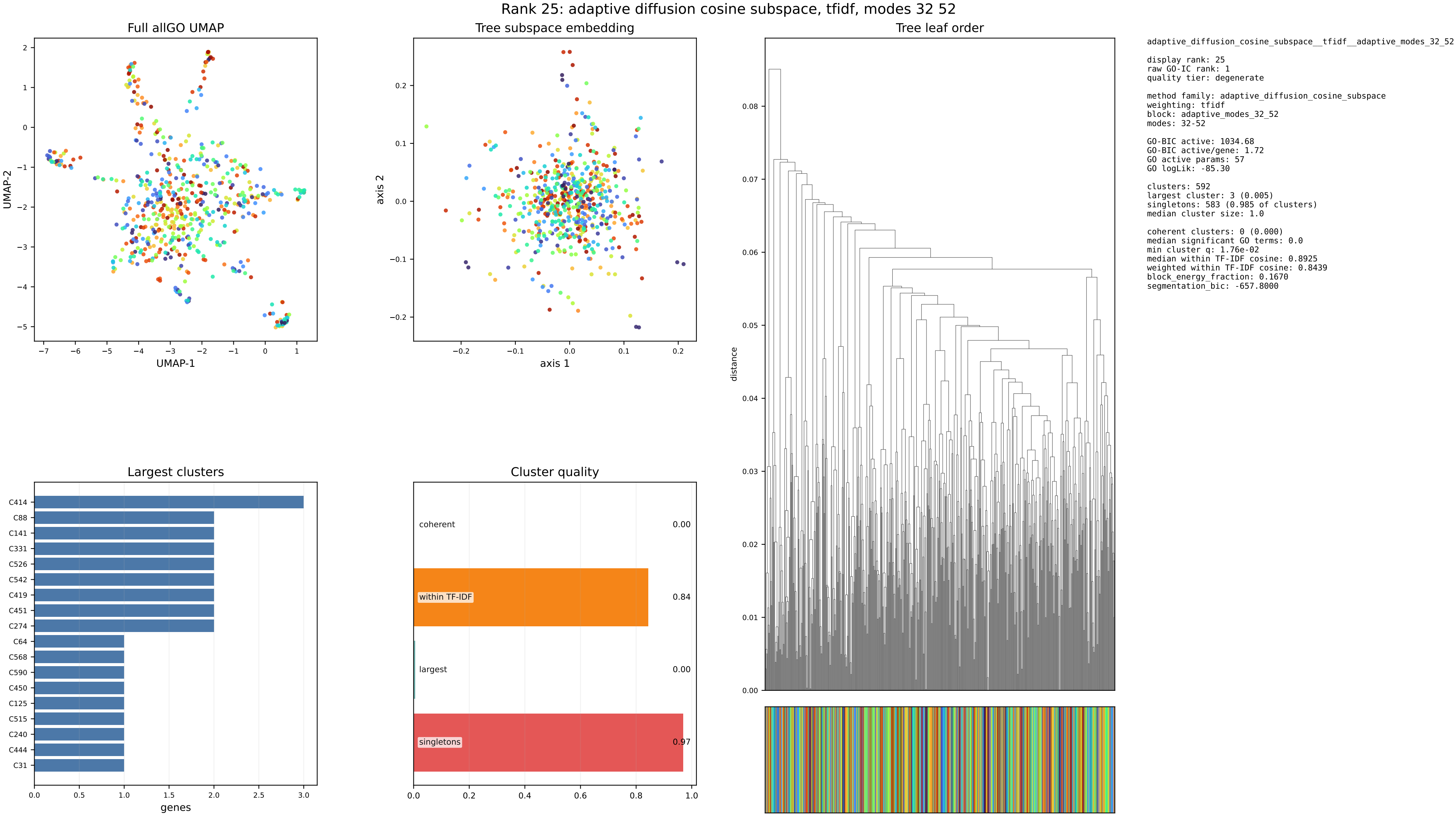
method family: raw_cosine_subspace
weighting: tfidf
block: adaptive_modes_11_15
modes: 11-15

G0-BIC active: 1183570.34
G0-BIC active/gene: 1966.06
G0 active params: 60033
G0 logLik: -136756.10

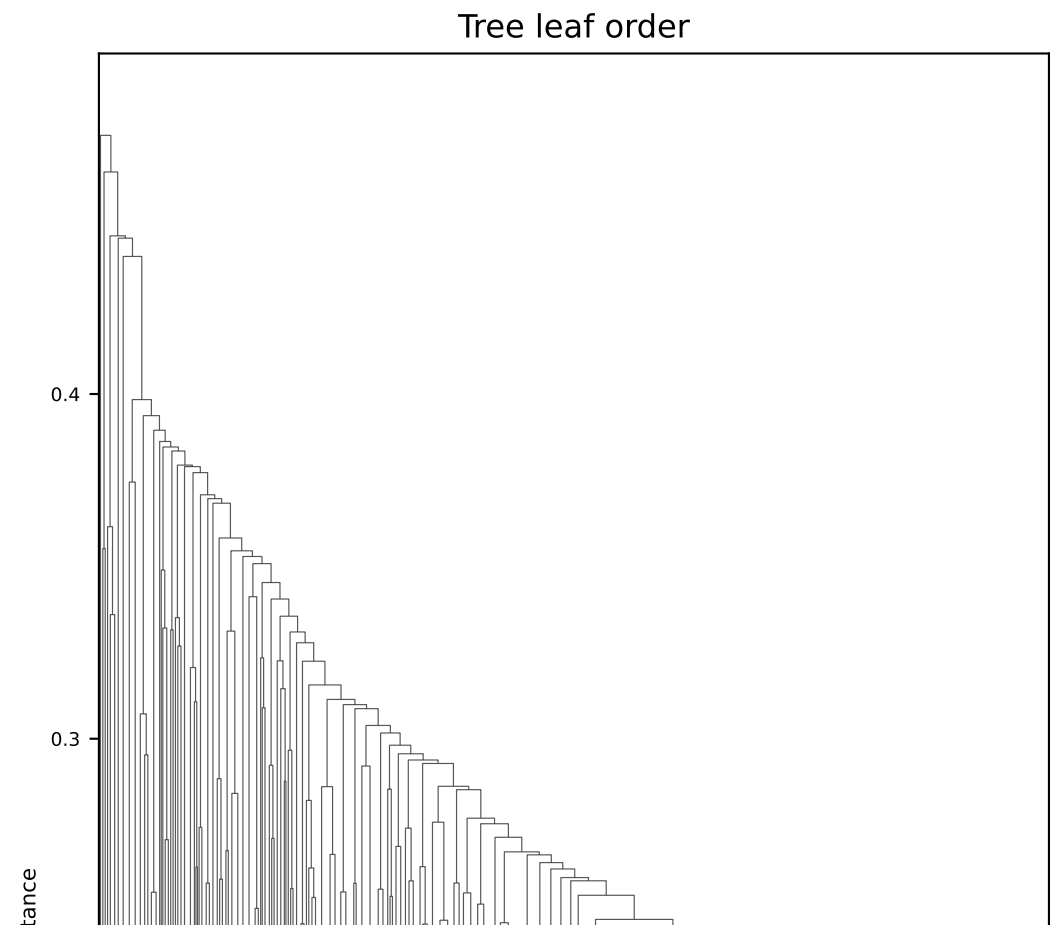
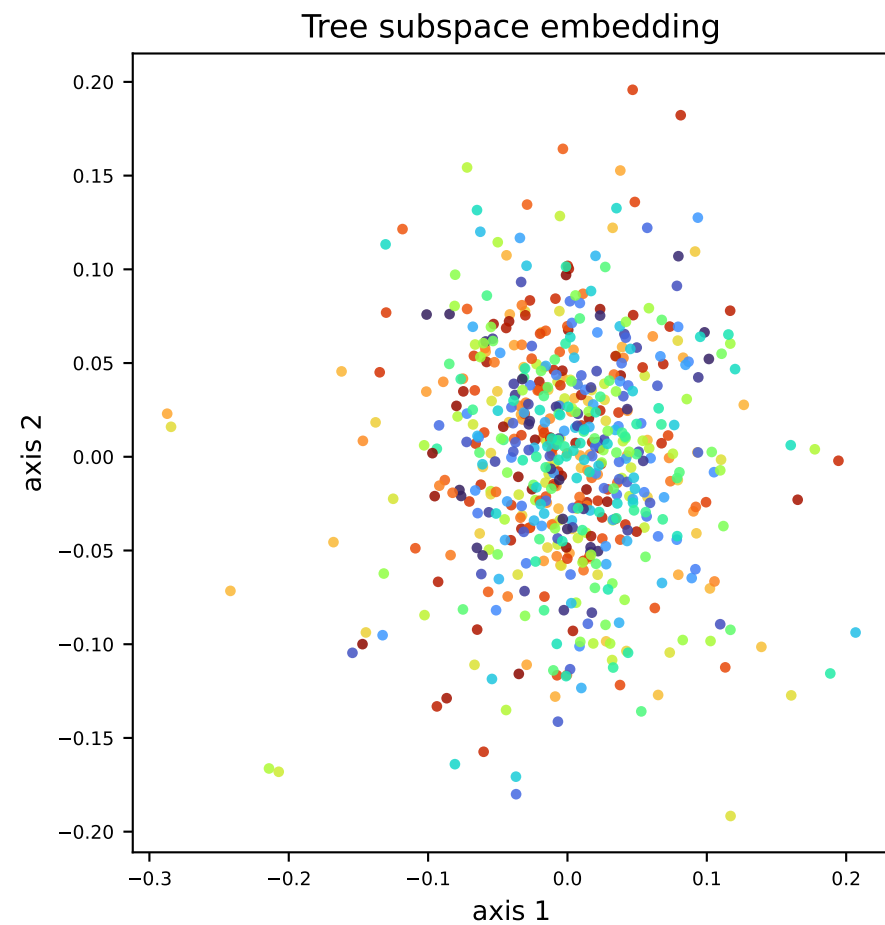
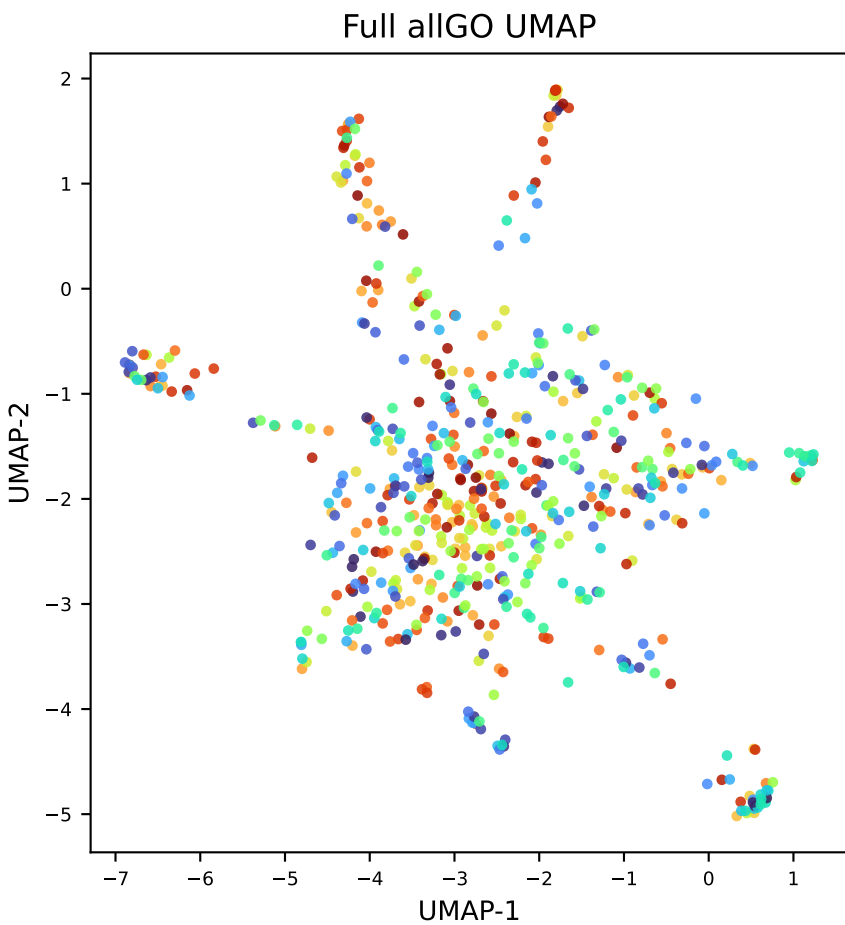
clusters: 214
largest cluster: 8 (0.013)
singletons: 64 (0.299 of clusters)
median cluster size: 3.0

coherent clusters: 45 (0.210)
median significant G0 terms: 0.0
min cluster q: 1.15e-08
median within TF-IDF cosine: 0.1231
weighted within TF-IDF cosine: 0.1832
block_energy_fraction: 0.0737
segmentation_bic: -657.8000





Rank 26: raw cosine subspace, binary, modes 41 55



```
raw_cosine_subspace__binary__adaptive_modes_41_55

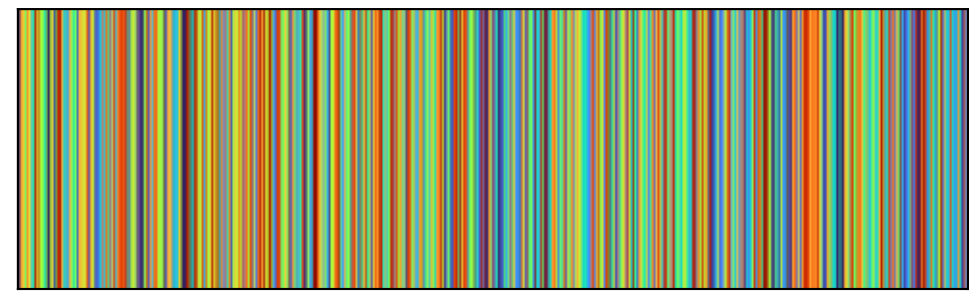
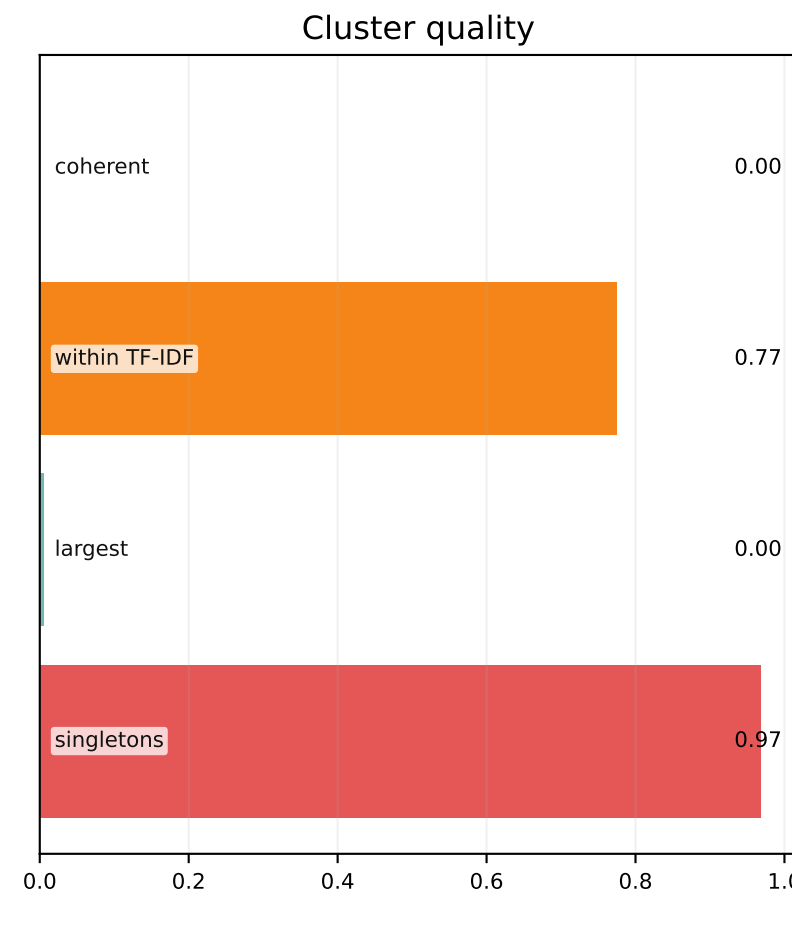
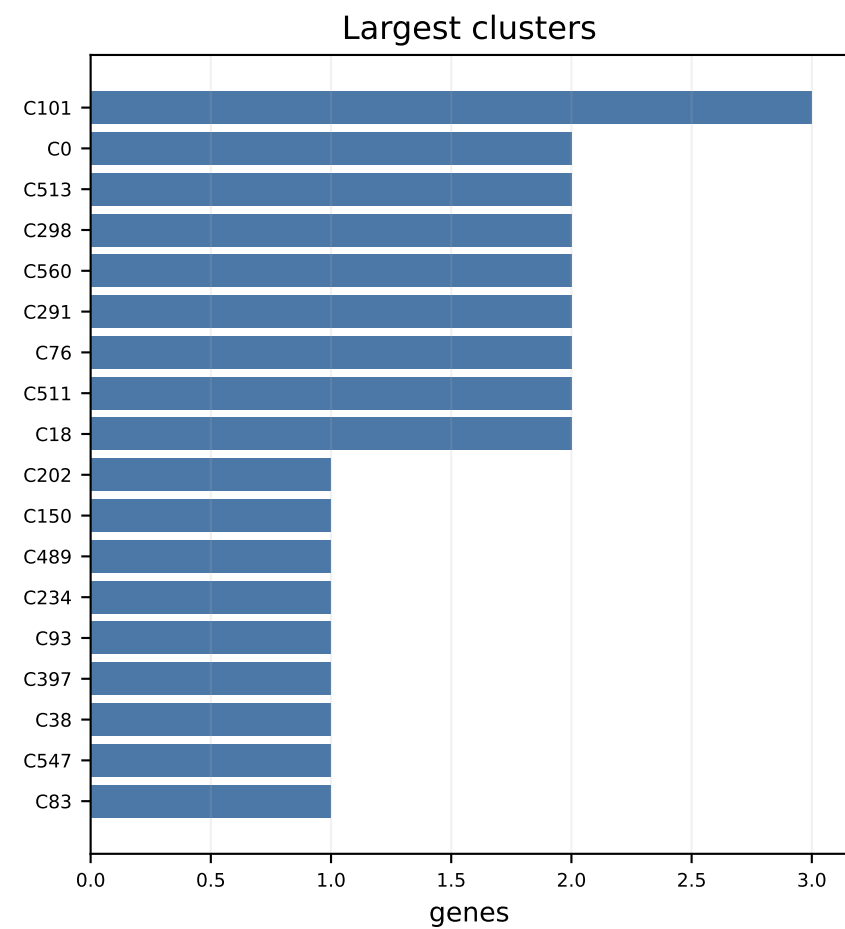
display rank: 26
raw G0-IC rank: 2
quality tier: degenerate

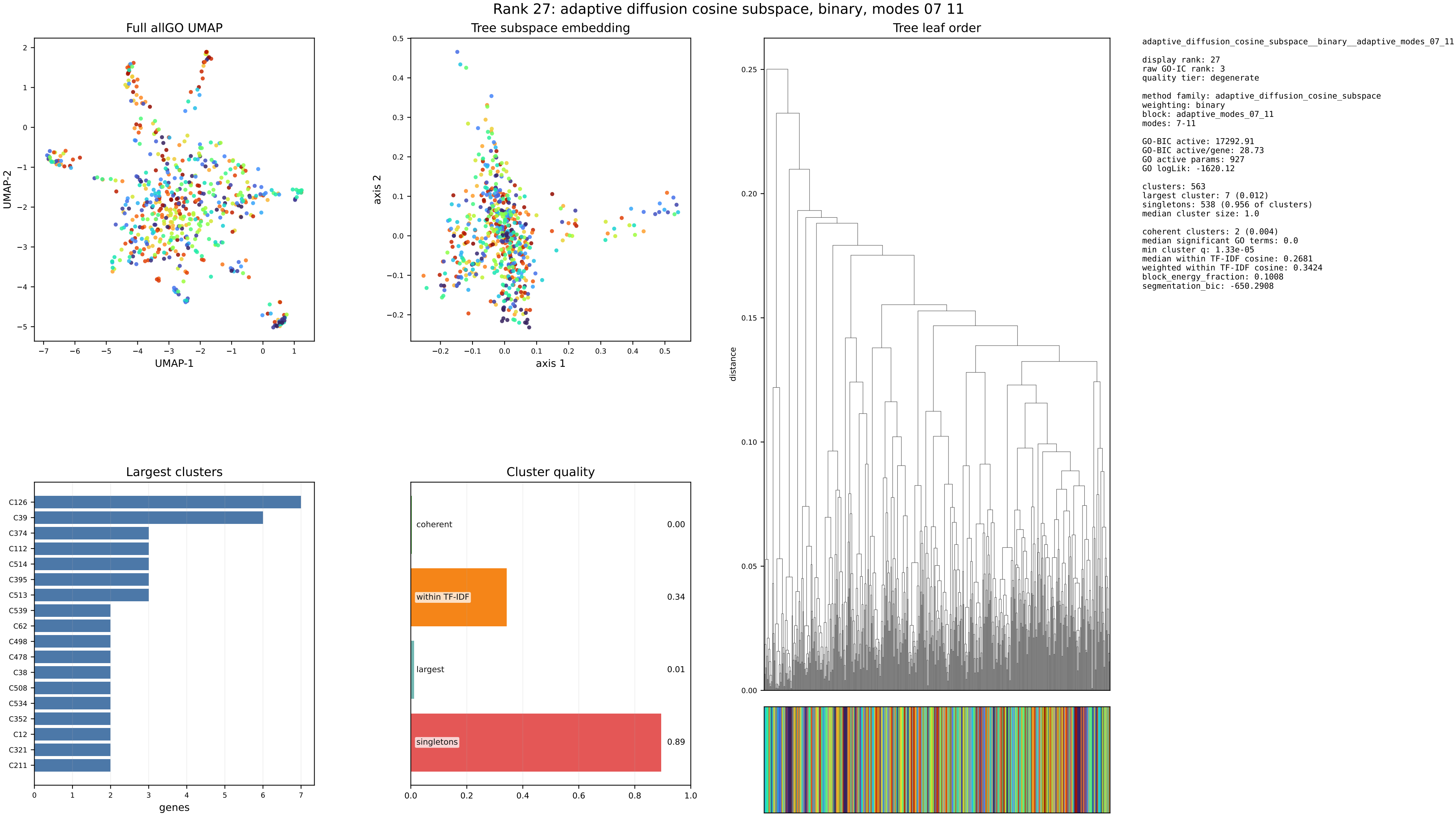
method family: raw_cosine_subspace
weighting: binary
block: adaptive_modes_41_55
modes: 41-55

G0-BIC active: 1059.93
G0-BIC active/gene: 1.76
G0 active params: 58
G0 logLik: -90.35

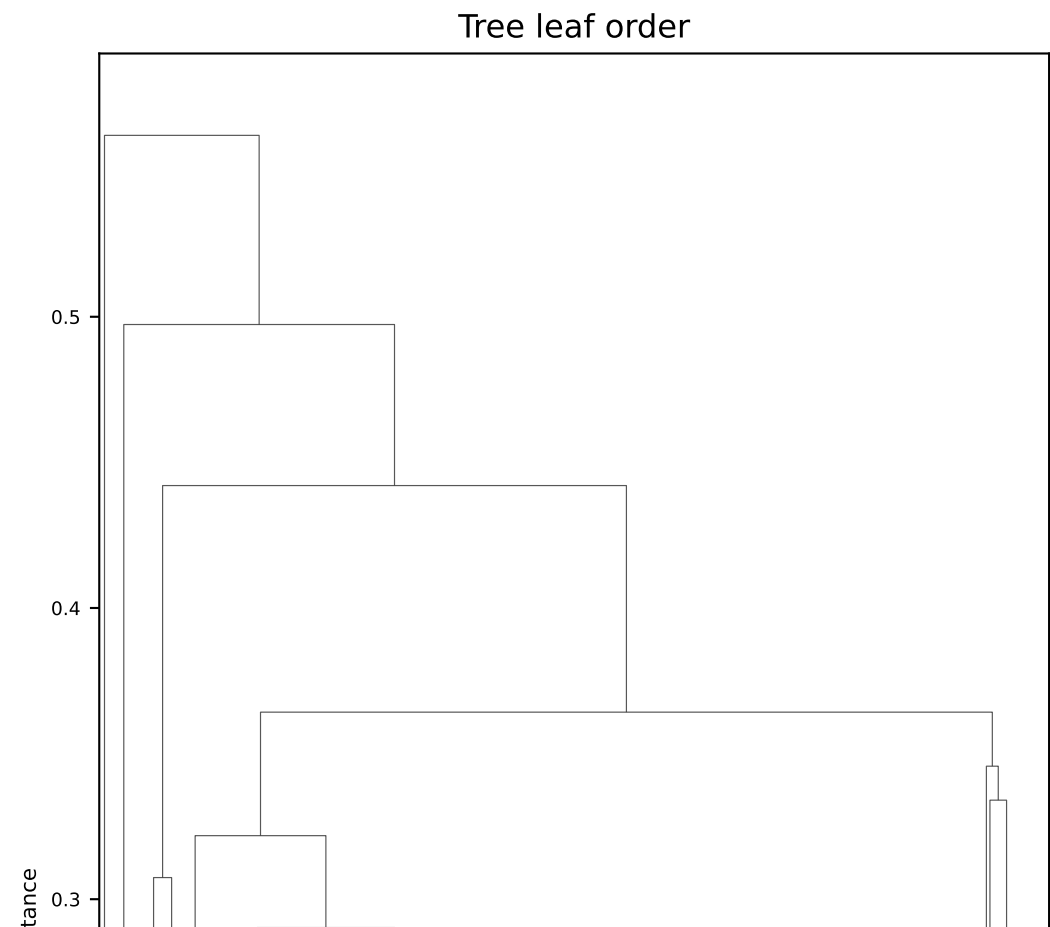
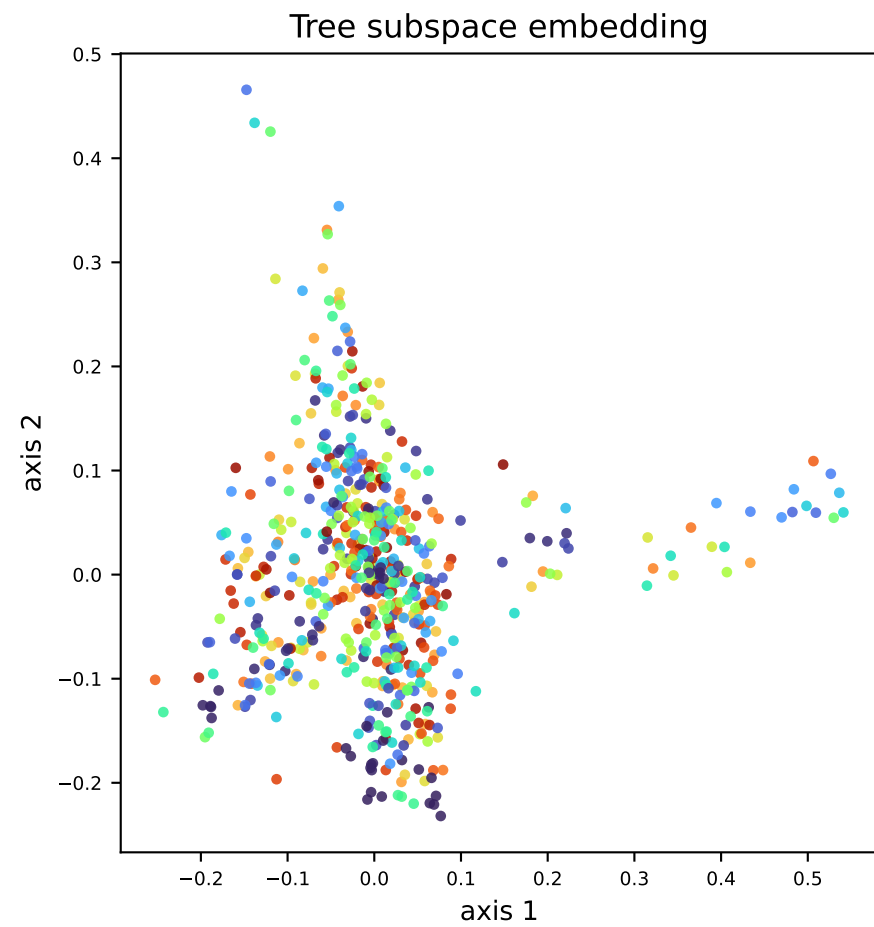
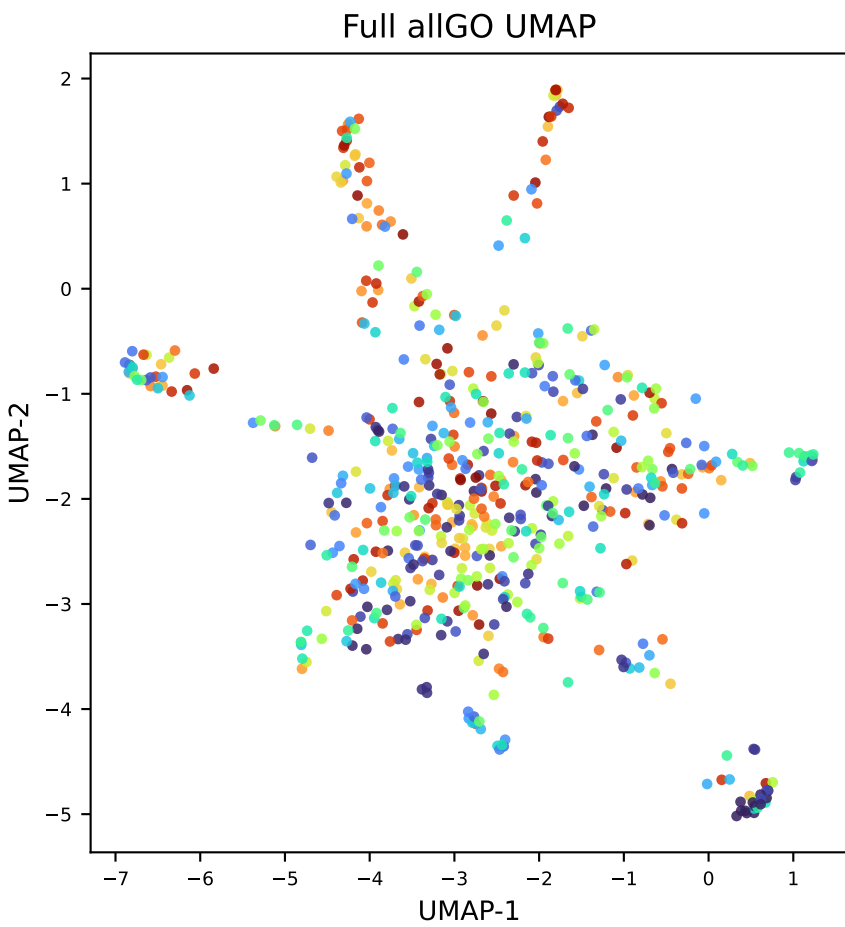
clusters: 592
largest cluster: 3 (0.005)
singletons: 583 (0.985 of clusters)
median cluster size: 1.0

coherent clusters: 0 (0.000)
median significant G0 terms: 0.0
min cluster q: 1.76e-02
median within TF-IDF cosine: 0.8925
weighted within TF-IDF cosine: 0.7742
block_energy_fraction: 0.0980
segmentation_bic: -650.2908
```





Rank 28: raw cosine subspace, binary, modes 07 11



raw_cosine_subspace__binary__adaptive_modes_07_11

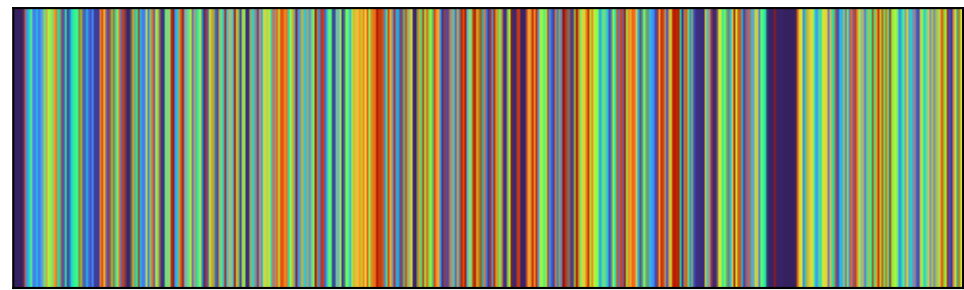
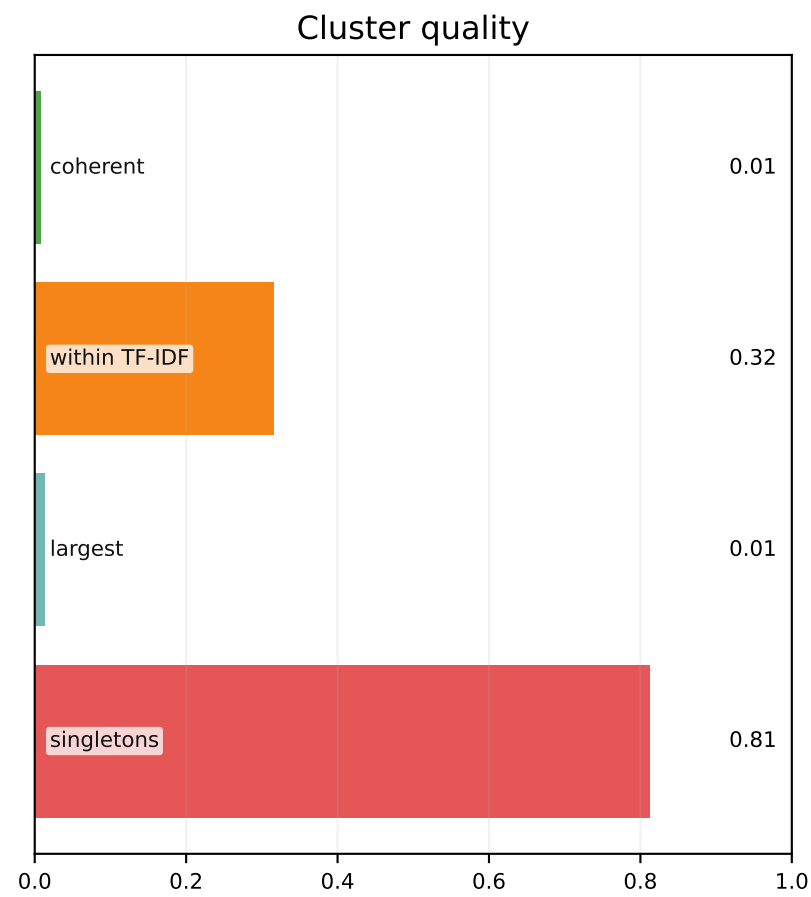
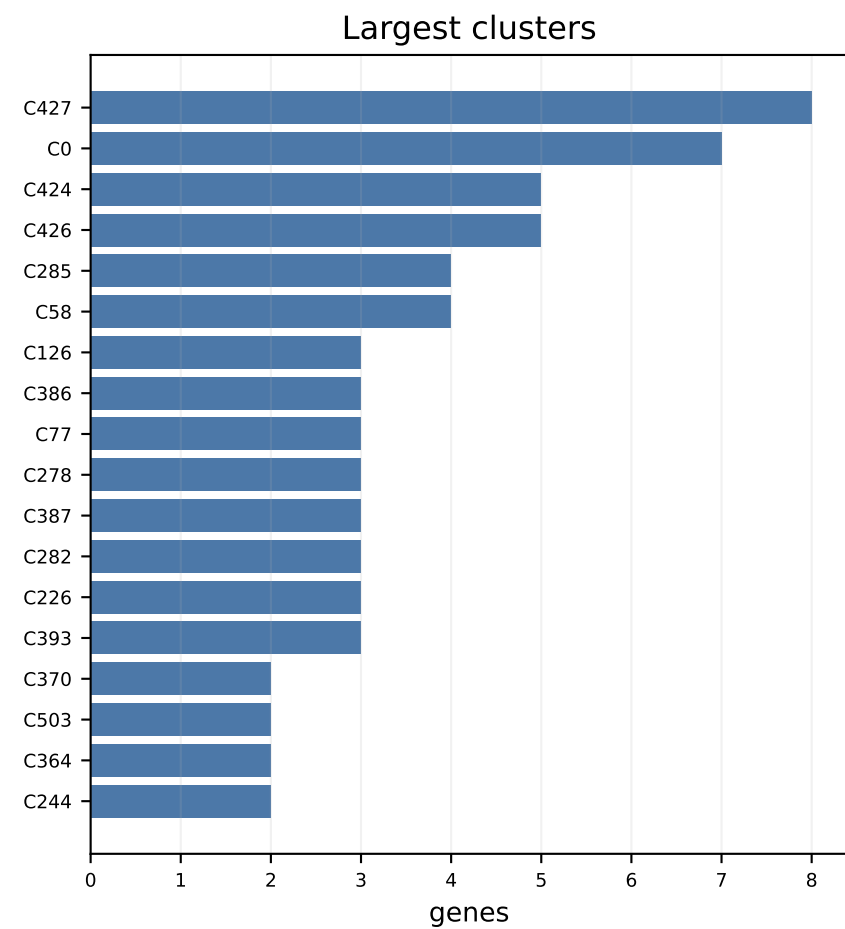
display rank: 28
raw G0-IC rank: 4
quality tier: degenerate

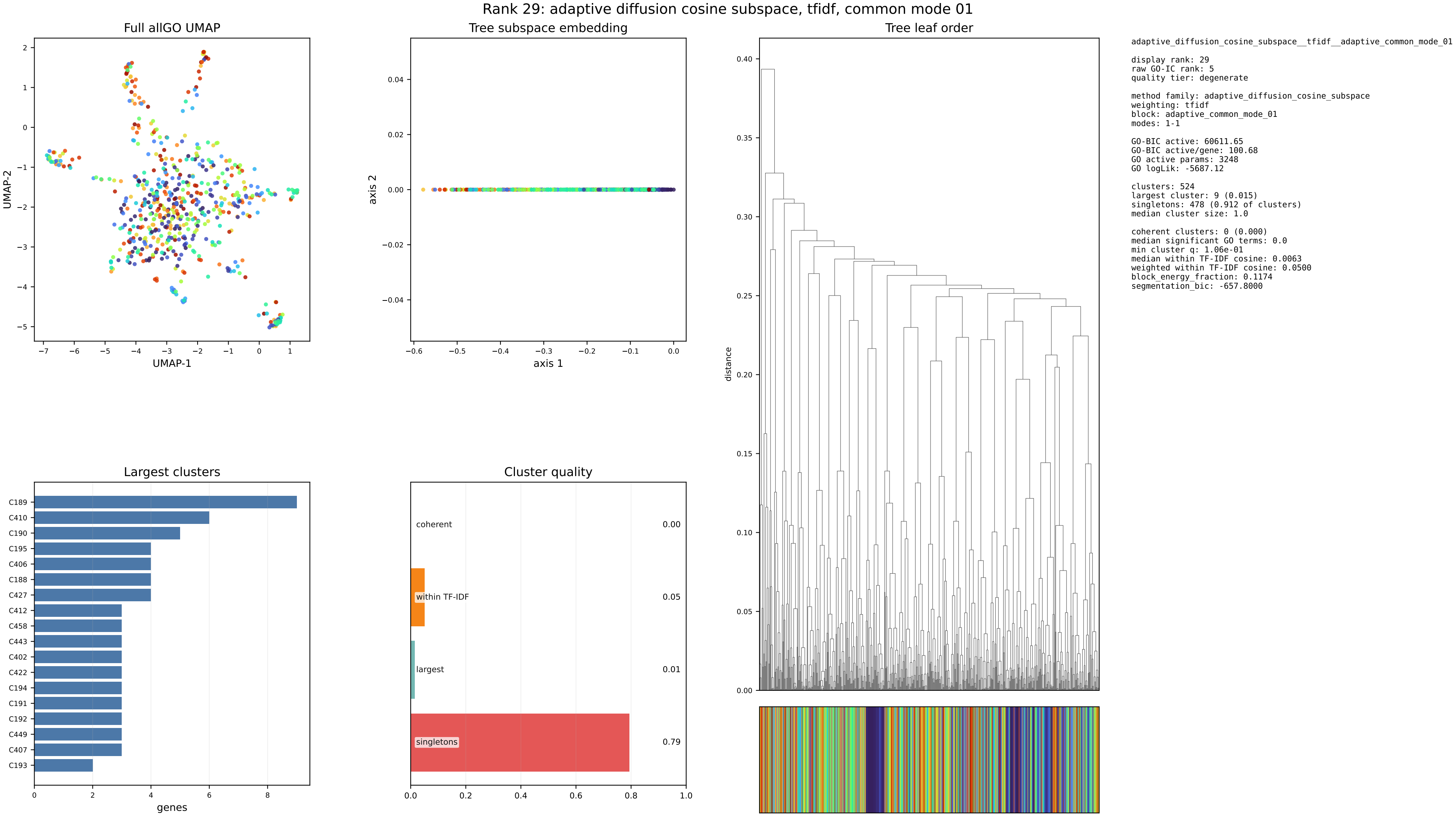
method family: raw_cosine_subspace
weighting: binary
block: adaptive_modes_07_11
modes: 7-11

G0-BIC active: 39834.82
G0-BIC active/gene: 66.17
G0 active params: 2112
G0 logLik: -3909.19

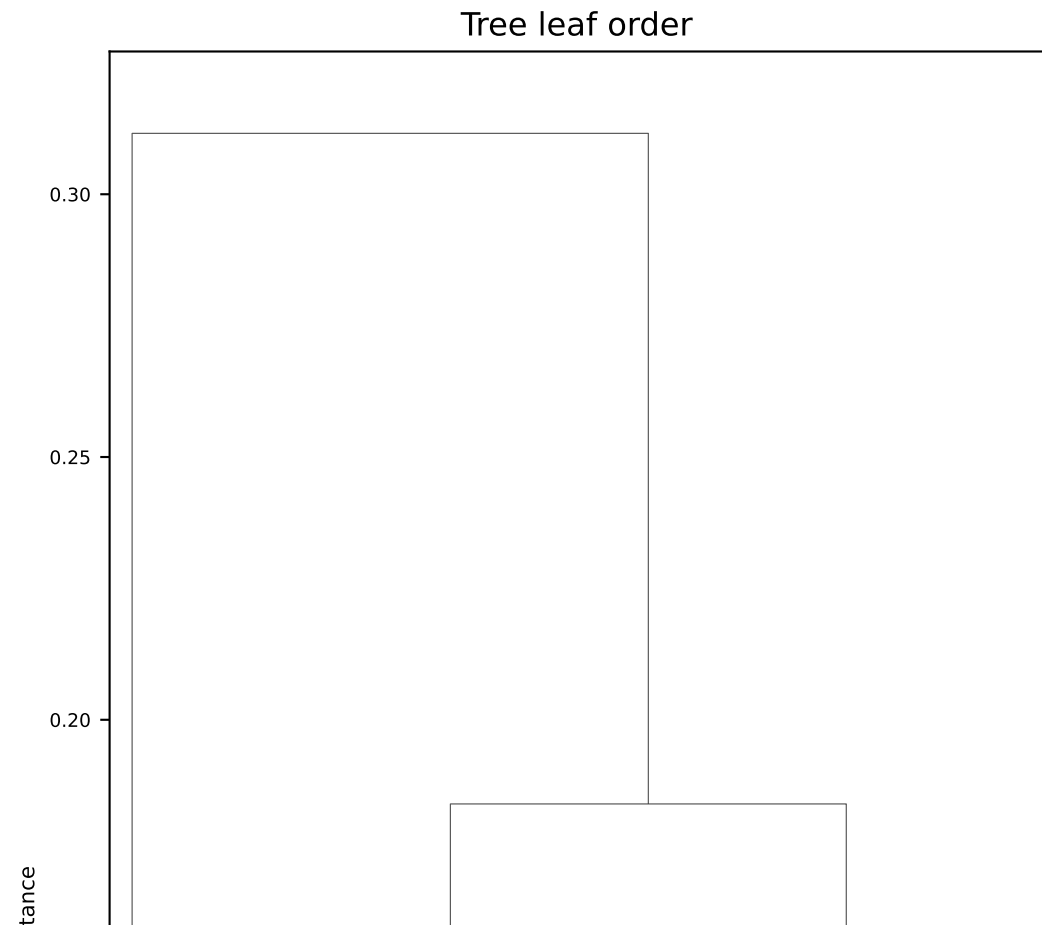
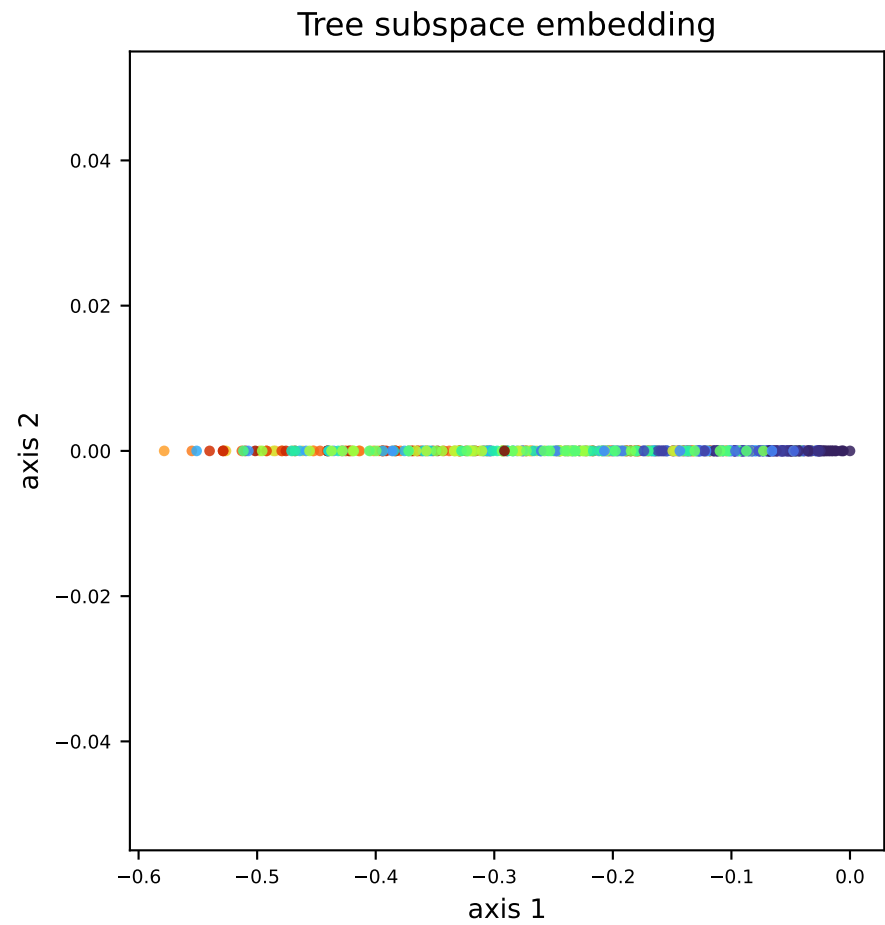
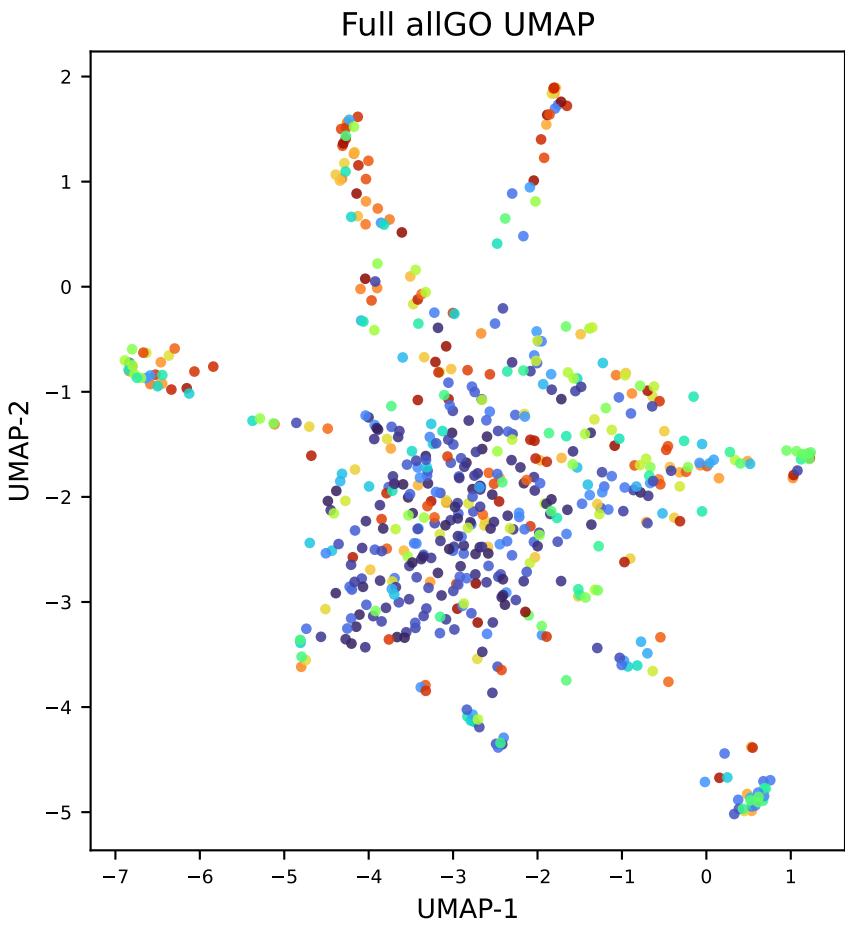
clusters: 531
largest cluster: 8 (0.013)
singletons: 489 (0.921 of clusters)
median cluster size: 1.0

coherent clusters: 4 (0.008)
median significant G0 terms: 0.0
min cluster q: 1.15e-08
median within TF-IDF cosine: 0.1485
weighted within TF-IDF cosine: 0.3162
block_energy_fraction: 0.1008
segmentation_bic: -650.2908





Rank 30: raw cosine subspace, tfidf, common mode 01



raw_cosine_subspace__tfidf__adaptive_common_mode_01

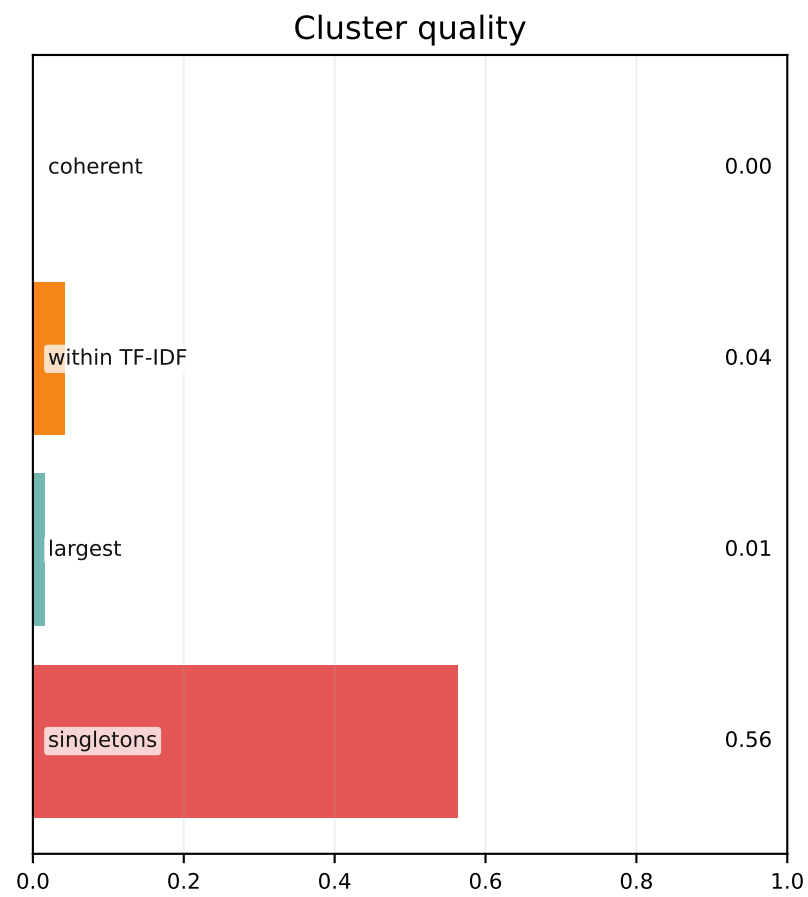
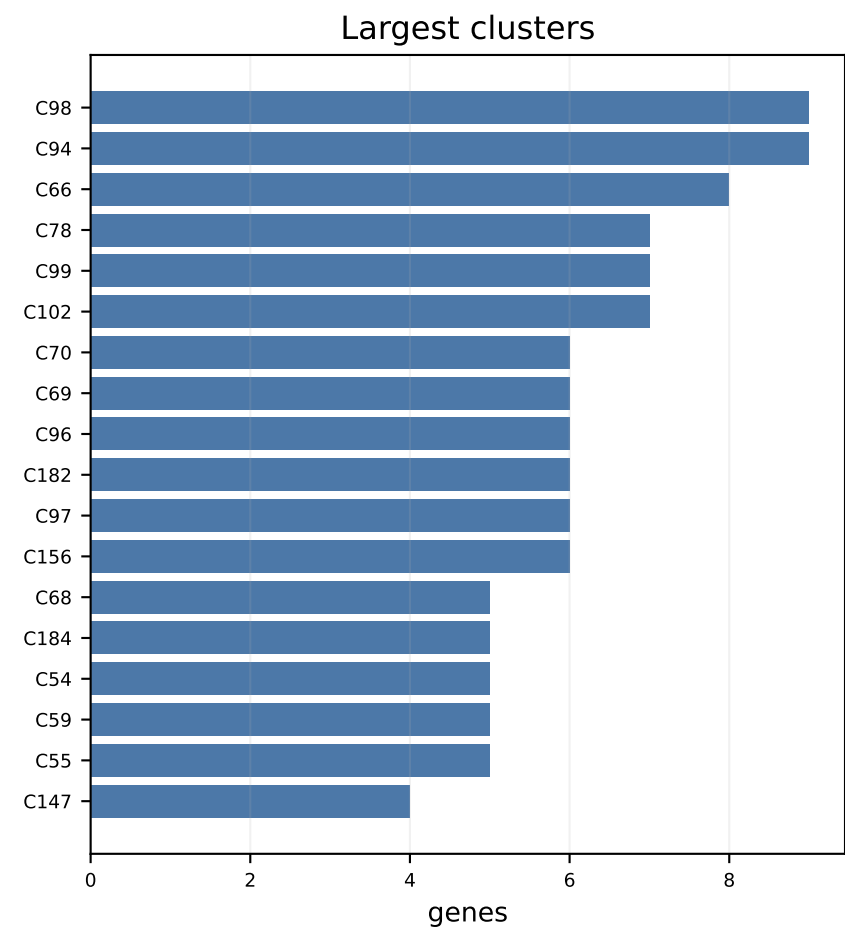
display rank: 30
raw G0-IC rank: 6
quality tier: degenerate

method family: raw_cosine_subspace
weighting: tfidf
block: adaptive_common_mode_01
modes: 1-1

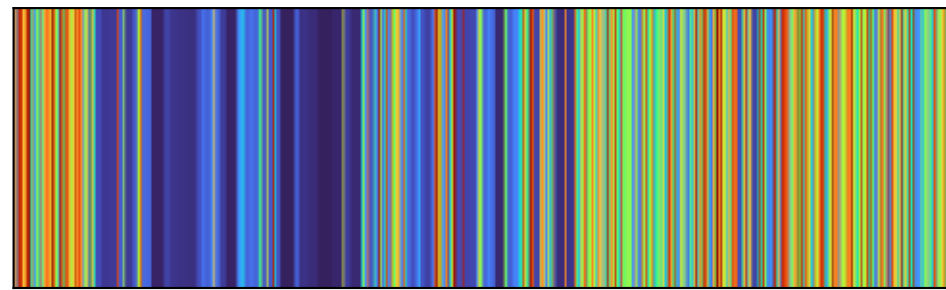
G0-BIC active: 262509.04
G0-BIC active/gene: 436.06
G0 active params: 13520
G0 logLik: -28777.66

clusters: 412
largest cluster: 9 (0.015)
singletons: 339 (0.823 of clusters)
median cluster size: 1.0

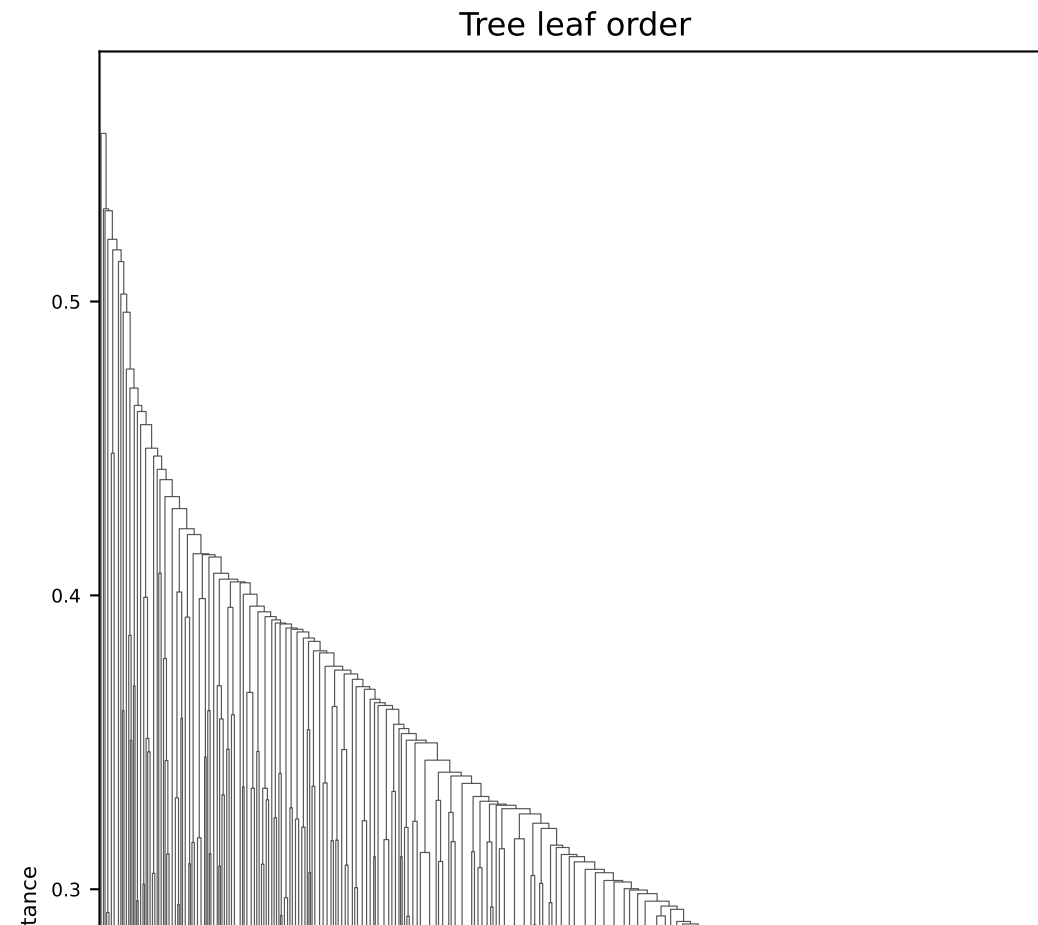
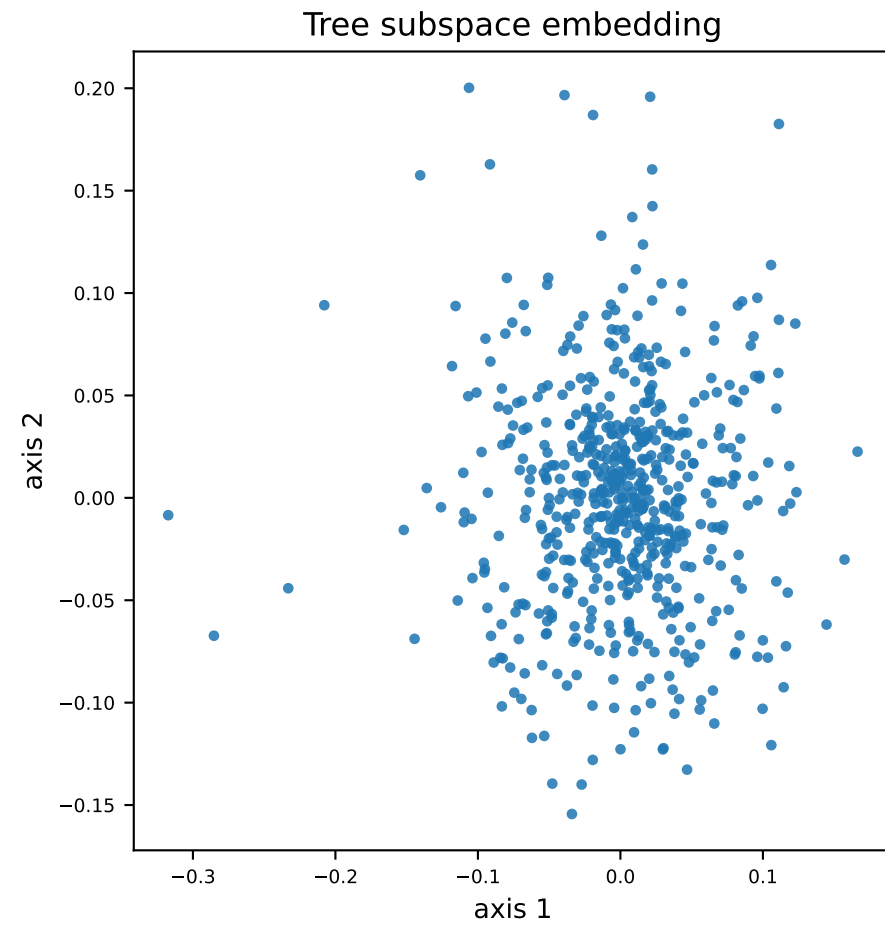
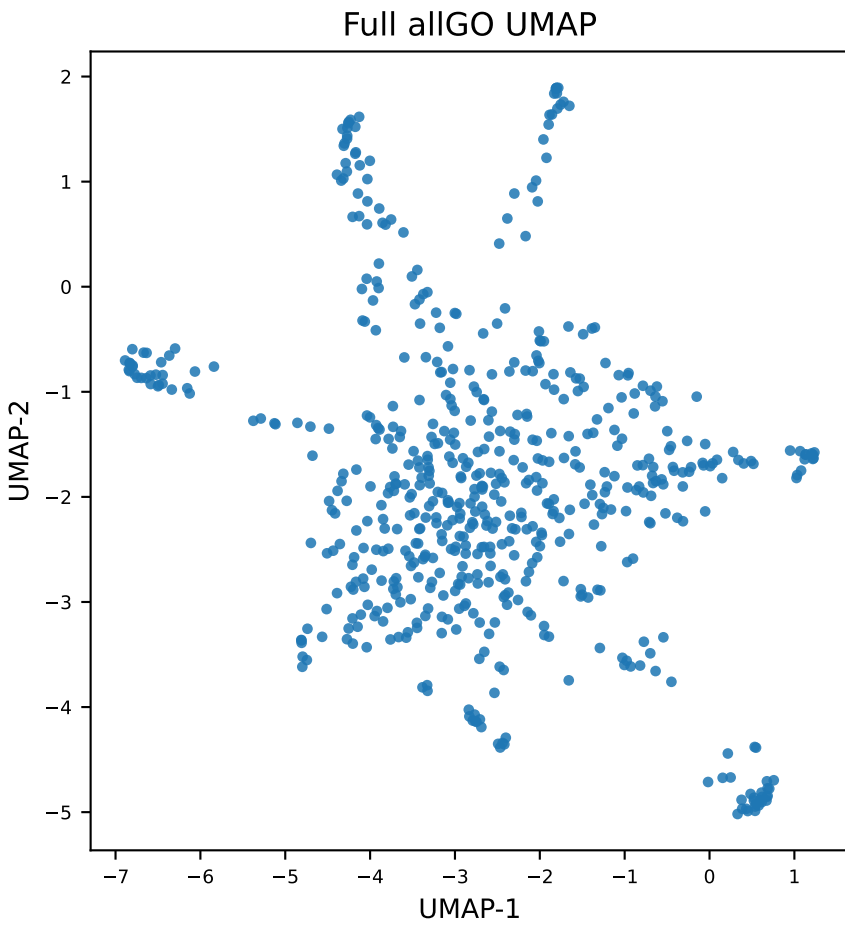
coherent clusters: 0 (0.000)
median significant G0 terms: 0.0
min cluster q: 1.06e-01
median within TF-IDF cosine: 0.0227
weighted within TF-IDF cosine: 0.0415
block_energy_fraction: 0.1174
segmentation_bic: -657.8000



distance



Rank 31: raw cosine subspace, binary, modes 56 80



raw_cosine_subspace__binary__adaptive_modes_56_80

display rank: 31
raw G0-IC rank: 12
quality tier: degenerate

method family: raw_cosine_subspace
weighting: binary
block: adaptive_modes_56_80
modes: 56-80

G0-BIC active: 866071.69
G0-BIC active/gene: 1438.66
G0 active params: 6368
G0 logLik: -384768.64

clusters: 1
largest cluster: 602 (1.000)
singletons: 0 (0.000 of clusters)
median cluster size: 602.0

coherent clusters: 0 (0.000)
median significant G0 terms: 0.0
min cluster q: 1.00e+00
median within TF-IDF cosine: 0.0369
weighted within TF-IDF cosine: 0.0369
block_energy_fraction: 0.1340
segmentation_bic: -650.2908

